

# CPYR Agent Communication



JULY 16

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CPYR  
V1.3



# CPYR Agent Streams

## Stream definition

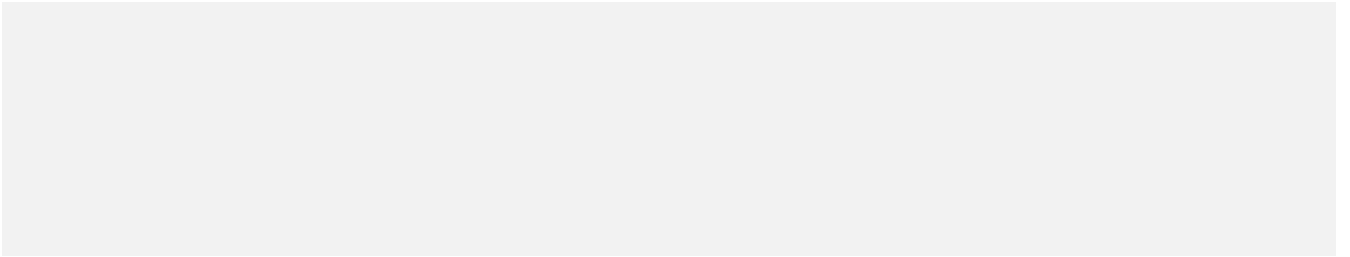
Communication on the Ethernet are defined by stream numbers at the 19<sup>th</sup> byte of the Header, This is a special tailored Header that the CPYR Agent uses in order to communicate with the CPYR Box.

CPYR Agents collect the data from different SW components allocates them to different Streams. To enable any future expansion stream version id has been given, and due to the limitation of the UDP frame size limitation 1490 Byte, stream chunk id was given in order to enable fragmentation of the messages.

Most important stream is the SW version info stream which preserves all the SW components which will be sent every 1 Sec.

**SW\_Version\_Info(0x102)- ST Number (72) - ST Version (1)**

| Signal Name                    | Description                                                                    | Startbit | Length [Bit] | Value Type | Factor | Offset | Minimum | Maximum  | Unit |
|--------------------------------|--------------------------------------------------------------------------------|----------|--------------|------------|--------|--------|---------|----------|------|
| CPYR agent_Verison_Variant     | Major release variant to main processor                                        | 0        | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| CPYR agent_Version_Major       | Major updates which will not support backwards compatibility (ie. SPI updates) | 8        | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| CPYR agent_Version_Minor       | Incremental updates                                                            | 16       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| CPYR agent_Version_Field       | Minor updates and re-releases. >100 decimal indicates a US release.            | 24       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| Algorunner_Verison_Variant     | Major release variant to Algo processor                                        | 32       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| Algorunner_Version_Major       | Major updates which will not support backwards compatibility (ie. SPI updates) | 40       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| Algorunner_Version_Minor       | Incremental updates                                                            | 48       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| Algorunner_Version_Field       | Minor updates and re-releases.                                                 | 56       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| ETSEL_Version_Major            | ETSEL Major Version                                                            | 64       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| ETSEL_Version_Minor            | ETSEL Minor Version                                                            | 72       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| ETSEL_Version_Revision         | ETSEL Revision Version                                                         | 80       | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| TAPS_Version                   | OCTAP Version                                                                  | 88       | 32           | IEEE Float | 1      | 0      | 0       | 3.40E+38 | N/A  |
| Tracker_Version_Major          | Tracker Major Version                                                          | 120      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| Tracker_Version_Minor          | Tracker Minor Version                                                          | 128      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| Tracker_Version_Revision       | Tracker Revision Version                                                       | 136      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| SPP_Version_Major              | SPP Major Version                                                              | 144      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| SPP_Version_Minor              | SPP Minor Version                                                              | 152      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| SPP_Version_Revision           | SPP Revision Version                                                           | 160      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| OTP_Version_Major              | OTP Major Version                                                              | 168      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| OTP_Version_Minor              | OTP Minor Version                                                              | 176      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| OTP_Version_Revision           | OTP Revision Version                                                           | 184      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| VISIONCO_SPI_API_Version       | VISIONCO SPI API Version                                                       | 192      | 32           | IEEE Float | 1      | 0      | 0       | 3.40E+38 | N/A  |
| ME_SW_Version                  | Vision Software Version                                                        | 224      | 32           | IEEE Float | 1      | 0      | 0       | 3.40E+38 | N/A  |
| MAINRADAR_SW_Release_Revision  | hardware version change                                                        | 256      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| MAINRADAR_SW_Promote_Revision  | milestone updates for a certain hardware                                       | 264      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| MAINRADAR_SW_Field_Revision    | stage change in a milestone                                                    | 272      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| MAINRADAR_SW_Field             | minor change in a stage                                                        | 280      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| MAINRADAR_PBL_Release_Revision | milestone updates                                                              | 288      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| MAINRADAR_PBL_Promote_Revision | stage change                                                                   | 296      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| MAINRADAR_PBL_Field_Revision   | minor updates                                                                  | 304      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| VSE_CS_Version_Major           | Curvature and Sideslip Estimation Algorithm Major Version                      | 312      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| VSE_CS_Version_Minor           | Curvature and Sideslip Estimation Algorithm Minor Version                      | 320      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| VSE_CS_Version_Field           | Curvature and Sideslip Estimation Algorithm Field Version                      | 328      | 8            | Unsigned   | 1      | 0      | 0       | 255      | N/A  |
| VSE_YRE_Version_Major          | Yaw Rate Estimation Algorithm Major                                            | 336      | 32           | Unsigned   | 1      | 0      | 0       | 2.15E+09 | N/A  |



## Table of Contents

|                                                       |           |
|-------------------------------------------------------|-----------|
| CPYR Agent Streams.....                               | 2         |
| <b>Stream definition.....</b>                         | <b>3</b>  |
| <b>Data Link and Network Layer .....</b>              | <b>8</b>  |
| <b>Transport Layer .....</b>                          | <b>8</b>  |
| <b>User Datagram Protocol (UDP).....</b>              | <b>8</b>  |
| <b>Application Layer .....</b>                        | <b>6</b>  |
| <b>Payload Header.....</b>                            | <b>6</b>  |
| <b>Ethernet Application Messages.....</b>             | <b>7</b>  |
| <b>CPYR AGENT Application Messages .....</b>          | <b>9</b>  |
| <b>System_Power_Status .....</b>                      | <b>9</b>  |
| <b>Electrical_Status .....</b>                        | <b>24</b> |
| <b>CFG_Parameters .....</b>                           | <b>26</b> |
| <b>Veh_Estimation_Compensation.....</b>               | <b>39</b> |
| <b>Active_Fault_Status .....</b>                      | <b>41</b> |
| <b>Latch_Fault_Status.....</b>                        | <b>52</b> |
| <b>History_Fault_Status.....</b>                      | <b>64</b> |
| <b>CPYR AGENT_SPI_Debug.....</b>                      | <b>75</b> |
| <b>Temperature.....</b>                               | <b>76</b> |
| <b>SOD_Signals.....</b>                               | <b>79</b> |
| <b>VisionCo SPI Messages.....</b>                     | <b>81</b> |
| <b>VISIONCO_APP_DIAG_STATUS_MSG (0xA1) .....</b>      | <b>81</b> |
| <b>VISIONCO_DYNAMIC_CAL_MSG (0xA2) .....</b>          | <b>82</b> |
| <b>VISIONCO_TAC_CAL_MSG (0xA3).....</b>               | <b>82</b> |
| <b>VISIONCO_VIS_ROAD_DATA_MSG (0xE0) .....</b>        | <b>82</b> |
| <b>VISIONCO_VIS_LIGHT_SENSOR_DATA_MSG (0xE1).....</b> | <b>82</b> |
| <b>VISIONCO_VIS_TRAFFIC_SIGNS_MSG (0xE2) .....</b>    | <b>82</b> |
| <b>VISIONCO_VIS_AEB_INFO_MSG (0xE3) .....</b>         | <b>83</b> |

|                                         |    |
|-----------------------------------------|----|
| VISIONCO_VIS_OBSTACLES_MSG (0xE4) ..... | 83 |
| VISIONCO_VIS_FAILSAFE_MSG (0xE5).....   | 83 |
| VISIONCO_VIS_SRP_MSG (0xE6) .....       | 83 |
| VISIONCO_DIAGNOSTIC_MSG (0x41) .....    | 83 |
| CPYR_AGENT_VEH_STATE_MSG (0x20).....    | 85 |
| VISIONCO_BOOT_STATUS_MSG (0x10).....    | 86 |
| Feature Stream Messages.....            | 88 |
| CPU_Load_PDU_signal.....                | 88 |
| _Execution_Time_Pdu .....               | 90 |

## Data Link and Network Layer

Instrumented ECUs with an installed Ethernet Adapter may be used for Ethernet communication.

The CPYR agent instrumentation shall communicate at BRR 100 Mbps Full-Duplex or 1Gbit at the (RJ45)

Inst\_Bus. The Ethernet CPYR agent network specification shall be defined as below:

- Internet Protocol version 4.
- Subnet mask for the network is 255.255.255.0.
- External Device has a fixed IP address of 10.2.0.10

A Category 5e or better cable must be used when connecting nodes with a maximum length of 100 m.

## Transport Layer

For the instrumentation messages defined in this document, The CPYR agent shall use the User Datagram Protocol Transport (UDP) for sends and receives. CPYR agent is responsible for adding the UDP header and send the packets.

The Source Port field identifies the sender of the message.

The Destination Port field identifies the receiver of the message. See Application Layer section for definition of allowed ports.

The Length field indicates the length of the datagram including the header bytes.

The Checksum field is used for error checking and includes the header and all data bytes.

The Data field shall contain the Application message.

## User Datagram Protocol (UDP)

An Ethernet device driver shall be implemented and provide an abstraction of the User Datagram Protocol (UDP).

UDP device driver configuration shall support setting the UDP broadcast port used for broadcasting UDP datagrams.

The UDP device driver shall allow transmission of broadcast datagrams after the driver is initialized.

An API service shall be implemented that allows the UDP server to send a datagram to the broadcast port. The server may reject service requests to send a broadcast datagram when the previous broadcast datagram has not been sent.

It shall be possible to re-initialize the UDP device driver. Re-initialization may be used when attempting to recover from a fault condition.



# Application Layer

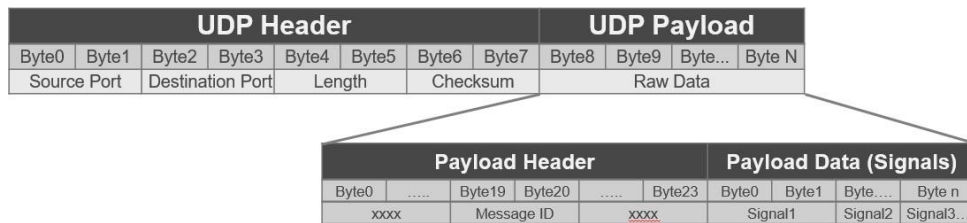
Communication from the CPYR agent application software is accomplished by using the

port. The table below specifies the Port address used.

| Port Name             | Port Address |
|-----------------------|--------------|
| CPYR agent Port       | 50014 C35E   |
| External Port(Server) | 50015 C35F   |

The content of the UDP data payload is likely to have two parts, header block and the data block.

The payload header block contains the message ID (Byte19 and Byte20) Stream Number and Stream Version and the data block contains the message signals as per the Inst\_Ethernet dbc.[3]



All data stored in the header shall be in little Endian format (i.e., Least Significant Byte (LSB) first).

All signals data in the messages shall be in little Endian format (i.e., Least Significant Byte (LSB) first).

## Payload Header

Application messages have a 24-byte header with the following format: Layer Byte... **Application messages have a 24-byte header with the following format:**

| Layer       | Byte Offset | Size (Bits) | Signal         | Description                                                                                                                                                                                                                                                               |
|-------------|-------------|-------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application | 0           | 16          | versionInfo    | Version of this header specification. The lower byte contains the size the message header. The upper byte contains the record version number.                                                                                                                             |
|             | 2           | 16          | sourceTxCnt    | Source Transmit Counter. Incremented on every transmission from the source Application.                                                                                                                                                                                   |
|             | 4           | 32          | sourceTxTime   | Source Transmit Time. Timestamp (ms) when packed was queued for transmission.                                                                                                                                                                                             |
|             | 8           | 8           | sourceInfo     | A constant identifying the sending application [0:63].                                                                                                                                                                                                                    |
|             | 9           | 8           | reservedSrc1   | Reserved for future use. Set to 0.                                                                                                                                                                                                                                        |
|             | 10          | 8           | reservedSrc2   | Reserved for future use. Set to 0.                                                                                                                                                                                                                                        |
|             | 11          | 8           | reservedSrc3   | Reserved for future use. Set to 0.                                                                                                                                                                                                                                        |
| Process     | 12          | 32          | streamIndex    | Stream-specific index used for data retrieval (e.g. Look Index for radar).                                                                                                                                                                                                |
|             | 16          | 16          | streamDataLen  | Stream data payload size (bytes). Size for each stream must be constant.                                                                                                                                                                                                  |
|             | 18          | 8           | streamTxCnt    | Stream Transmit Counter. Increments on each transmission of associated stream number.                                                                                                                                                                                     |
|             | 19          | 8           | streamNumber   | Stream number One source may transmit multiple, asynchronous streams.                                                                                                                                                                                                     |
|             | 20          | 8           | streamVersion  | Stream data structure format/version (how to interpret data).                                                                                                                                                                                                             |
|             | 21          | 8           | streamChunks   | A Stream may be transmitted from the process layer as M equal-size chunks that can be reconstructed on the receive side. Set to 0 if the transmission is not part of a series of chunks (normal mode). Set to M (>=2) to indicate stream is sent as a series of M chunks. |
|             | 22          | 8           | streamChunkIdx | Set to 0 when streamChuns is 0; otherwise, represents this chnk' s position in the reconstruction queue [0:(streamChunks-1)].                                                                                                                                             |

|  |    |   |              |                                    |
|--|----|---|--------------|------------------------------------|
|  | 23 | 8 | reservedStr3 | Reserved for future use. Set to 0. |
|--|----|---|--------------|------------------------------------|

## Ethernet Application Messages

The following table lists all the Ethernet application messages specified in this document.

| Message Name                  | Stream Number | DLC(Byte) | Source Node | Receiver Node | Update Rate (ms) | Description                                                   |
|-------------------------------|---------------|-----------|-------------|---------------|------------------|---------------------------------------------------------------|
| VISIONCO_BASE_DIAG_STATUS_MSG | 40            | 91        | AGEN<br>T   | External      | 55.5             | VisionCo4 base diagnostic status message.                     |
| VISIONCO_APP_DIAG_STATUS_MSG  | 41            | 52        | AGEN<br>T   | External      | 55.5             | VisionCo4 application diagnostic status message.              |
| VISIONCO_DYNAMIC_CAL_MSG      | 42            | 46        | AGEN<br>T   | External      | 55.5             | VisionCo4 Service Point Calibration (SPC) and AutoFix output. |

|                                    |    |      |       |          |           |                                                                                                                                                                                                                                                          |
|------------------------------------|----|------|-------|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VISIONCO_TAC_CAL_MSG               | 43 | 45   | AGENT | External | 55.5      | VisionCo4 Target Auto-Calibration (TAC) output.                                                                                                                                                                                                          |
| VISIONCO_VIS_ROAD_DATA_MSG         | 44 | 524  | AGENT | External | 55.5      | VisionCo4 road detection data message.                                                                                                                                                                                                                   |
| VISIONCO_VIS_LIGHT_SENSOR_DATA_MSG | 45 | 844  | AGENT | External | 55.5      | VisionCo4 AHBC function data message.                                                                                                                                                                                                                    |
| VISIONCO_VIS_TRAFFIC_SIGNS_MSG     | 46 | 524  | AGENT | External | 55.5      | Traffic signs as detected by vision subsystem.                                                                                                                                                                                                           |
| VISIONCO_VIS_AEB_INFO_MSG          | 47 | 36   | AGENT | External | 55.5      | AEB decisions ( for braking or warning) based on vision.                                                                                                                                                                                                 |
| VISIONCO_VIS_OBSTACLES_MSG         | 48 | 3320 | AGENT | External | 55.5      | Vision objects as detected by vision subsystem.                                                                                                                                                                                                          |
| VISIONCO_VIS_FAILSAFE_MSG          | 49 | 34   | AGENT | External | 55.5      | Vision failsafes                                                                                                                                                                                                                                         |
| VISIONCO_VIS_SRP_MSG               | 50 | 37   | AGENT | External | 55.5      | Suspension Road Preview                                                                                                                                                                                                                                  |
| VISIONCO_DIAGNOSTIC_MSG            | 51 | 920  | AGENT | External | 55.5      | The application diagnostics protocol is a message that contains information about the current status of the application (internal states, diagnostics, errors, etc.). The message is being sent from VisionCo at the end of every image processing loop. |
| VISIONCO_BOOT_STATUS_MSG           | 52 | 44   | AGENT | External | Boot Time | Boot Status message                                                                                                                                                                                                                                      |
| CPYR_AGENT_VEH_STATE_MSG           | 53 | 65   | AGENT | External | 15        | CPYR agent vehicle state message.                                                                                                                                                                                                                        |
| VISIONCO_TIMESYNC_MSG (0xE8)       | 54 |      |       |          |           |                                                                                                                                                                                                                                                          |
| VISIONCO_SAFETY_MSG (0xE7)         | 55 |      |       |          |           |                                                                                                                                                                                                                                                          |
| Init_Params_Vision (0x92)          | 56 |      |       |          |           |                                                                                                                                                                                                                                                          |
| Intrinsic Calibration (0x84)       | 57 |      |       |          |           |                                                                                                                                                                                                                                                          |
| Init_IPB (0x78)                    | 58 |      |       |          |           |                                                                                                                                                                                                                                                          |
| Init_PCW (0x79)                    | 59 |      |       |          |           |                                                                                                                                                                                                                                                          |
| BT_TPP_SPP                         | 60 | 48   |       |          |           | OBJECTSENSE Signals Received from Algorunner                                                                                                                                                                                                             |
| BT_TPP_TAD                         | 61 | 460  |       |          |           | OBJECTSENSE Signals Received from Algorunner                                                                                                                                                                                                             |
| BT_TPP_AutoHitch                   | 62 | 64   |       |          |           | OBJECTSENSE Signals Received from Algorunner                                                                                                                                                                                                             |
| BT_TPP_DriveHistory                | 63 | 40   |       |          |           | OBJECTSENSE Signals Received from Algorunner                                                                                                                                                                                                             |
| BT_TPP_Localization_Results        | 64 | 32   |       |          |           | OBJECTSENSE Signals Received from Algorunner                                                                                                                                                                                                             |
| System_Power_Status                | 70 | 125  | AGENT | External | 5         | Reports the power status that happen at the 5 ms rate                                                                                                                                                                                                    |
| Electrical_Status                  | 71 | 24   | AGENT | External | 25        | Reports the status of SW & LEDs and the RAW Data of ADC Inputs                                                                                                                                                                                           |
| _SW_Version_Info                   | 72 | 22   | AGENT | External | 1000      | Reports SW version numbers                                                                                                                                                                                                                               |
| CFG_Parameters                     | 73 | 148  | AGENT | External | 50        | Reports CFG Parameters                                                                                                                                                                                                                                   |
| Veh_Estimation_Compensation        | 74 | 19   | AGENT | External | 10        | Reports information used for resim                                                                                                                                                                                                                       |
| Active_Fault_Status                | 75 | 24   | AGENT | External | 50        | Active fault status bytes 0 to 31                                                                                                                                                                                                                        |
| Latched_Fault_Status               | 76 | 24   | AGENT | External | 50        | Latched fault status bytes 0 to 31                                                                                                                                                                                                                       |
| History_Fault_Status               | 77 | 24   | AGENT | External | 50        | History fault status bytes 0 to 31                                                                                                                                                                                                                       |
| CPYR_AGENT_SPI_Debug               | 78 | 8    | AGENT | External | 1000      | Contains debug information from the SPI driver.                                                                                                                                                                                                          |
| _Temperature                       | 79 | 10   | AGENT | External | 100       | Message with the CPYR_AGENT sensors temperatures: imager, MIPS, VMP, Imager, VisionCo4, IMAGE2P, Algorunner and thermal status                                                                                                                           |
| OBJECTSENSE_Signals                | 80 | 43   | AGENT | External | 25        | Reports information used for HIL testing                                                                                                                                                                                                                 |

|     |                     |     |    |       |          |    |                                            |
|-----|---------------------|-----|----|-------|----------|----|--------------------------------------------|
| 81  | SOD_Signals         | 81  | 25 | AGENT | External | 50 | Reports information used for HIL testing   |
| 82  | Reserved            |     |    |       |          |    |                                            |
| ... | Reserved            |     |    |       |          |    |                                            |
| 99  | Reserved            |     |    |       |          |    |                                            |
| 100 | cpu core throughput | 100 |    |       |          |    | cpu core throughput metric                 |
| 101 | Module              | 101 |    |       |          |    | Status of tasks periodicity and scheduling |
| 110 | AFL                 | 110 |    |       |          | 40 | feature debug streams                      |
| 111 | LKS                 | 111 |    |       |          | 40 | feature debug streams                      |
| 112 | L25                 | 112 |    |       |          | 40 | feature debug streams                      |
| 113 | TSR, WWA, IACC      | 113 |    |       |          | 20 | feature debug streams                      |
| 114 | TA_Veh              | 114 |    |       |          | 40 | feature debug streams                      |
| 115 | FSM                 | 115 |    |       |          | 40 | feature debug streams                      |
| 116 | ACC                 | 116 |    |       |          | 40 | feature debug streams                      |
| 117 | HA                  | 117 |    |       |          | 40 | feature debug streams                      |
| 118 | CHA                 | 118 |    |       |          | 40 | feature debug streams                      |

## CPYR AGENT Application Messages

### System\_Power\_Status

-System\_Power\_Status(0x100)-STNumber(70)-STVersion(1) **This message will be used to test functionalities.**

Message System\_Power\_Status (0x100) shall be implemented as follows:

| SignalName                         | State Description                                                          | StartBit | Size | Sign | Factor | Offset | Min | Max  |
|------------------------------------|----------------------------------------------------------------------------|----------|------|------|--------|--------|-----|------|
| Rear_USSs_lout                     | Current representative of sensed voltage of Rear USS                       | 0        | 32   | @1+  | 1      | 0 0    | 0   |      |
| Heater_lout                        | Current representative of sensed voltage of Heater                         | 32       | 32   | @1+  | 1      | 0 0    | 0   | 0    |
| RVC_lout                           | Current representative of sensed voltage of RVC                            | 64       | 32   | @1+  | 1      | 0 0    | 0   | 2000 |
| POC_LDO1_FVC_Current_SENSE_Voltage | Scaled engineering value of FVC LDO1 regulator current sense voltage       | 96       | 32   | @1+  | 1      | 0 0    | 0   | 25   |
| POC_LDO1_RVC_Current_SENSE_Voltage | Scaled engineering value of RVC LDO1 regulator current sense voltage       | 128      | 32   | @1+  | 1      | 0 0    | 0   | 25   |
| POC_LDO2_RH_Current_SENSE_Voltage  | Scaled engineering value of RH Mirror LDO2 regulator current sense voltage | 160      | 32   | @1+  | 1      | 0 0    | 0   | 25   |
| POC_LDO2_LH_Current_SENSE_Voltage  | Scaled engineering value of LH Mirror LDO2 regulator current sense voltage | 192      | 32   | @1+  | 1      | 0 0    | 0   | 25   |
| POC_LDO3_FWC_Current_SENSE_Voltage | Scaled engineering value of FWC LDO3 regulator current sense voltage       | 224      | 32   | @1+  | 1      | 0 0    | 0   | 25   |
| LKS_SW_Voltage                     | Lane Keeping Switch Voltage                                                | 256      | 32   | @1+  | 1      | 0 0    | 0   | 25   |

|                                        |                                                                                                                                                         |      |    |     |   |   |   |       |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|-----|---|---|---|-------|
| APA_SW_Voltage_AN1                     | Active Park Assist Switch Voltage from ADC AN1                                                                                                          | 288  | 32 | @1+ | 1 | 0 | 0 | 25    |
| APA_SW_Voltage_AN24                    | Active Park Assist Switch Voltage from ADC AN24                                                                                                         | 320  | 32 | @1+ | 1 | 0 | 0 | 25    |
| BPA_SW_Voltage                         | Base Park Aid Switch Voltage                                                                                                                            | 352  | 32 | @1+ | 1 | 0 | 0 | 25    |
| View_SW_Voltage                        | Surround View Switch Voltage                                                                                                                            | 384  | 32 | @1+ | 1 | 0 | 0 | 25    |
| L_BLIS_OUT_Voltage                     | Left BLIS_OUT LED Voltage                                                                                                                               | 416  | 32 | @1+ | 1 | 0 | 0 | 25    |
| R_BLIS_OUT_Voltage                     | Right BLIS_OUT LED Voltage                                                                                                                              | 448  | 32 | @1+ | 1 | 0 | 0 | 25    |
| PARK_AID_OUT_Voltage                   | PARK AID_OUT LED Voltage                                                                                                                                | 480  | 32 | @1+ | 1 | 0 | 0 | 25    |
| SIDERADAR_RR_Current_SENSE_Voltage     | Scaled engineering value of RR SIDERADARs current sense voltage.                                                                                        | 512  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| SIDERADAR_LR_Current_SENSE_Voltage     | Scaled engineering value of LR SIDERADARs current sense voltage.                                                                                        | 544  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| SIDERADAR_LF_Current_SENSE_Voltage     | Scaled engineering value of LF SIDERADARs current sense voltage.                                                                                        | 576  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| SIDERADAR_RF_Current_SENSE_Voltage     | Scaled engineering value of RF SIDERADARs current sense voltage.                                                                                        | 608  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| FLR_Current_SENSE_Voltage              | Scaled engineering value of Forward Looking Radar current sense voltage.                                                                                | 640  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| Heater_Current_SENSE_Voltage           | Scaled engineering value of Heater current sense voltage.                                                                                               | 672  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| SIDERADAR_RR_PWR_Voltage               | SIDERADAR RR power Voltage                                                                                                                              | 704  | 32 | @1+ | 1 | 0 | 0 | 25    |
| SIDERADAR_LR_PWR_Voltage               | SIDERADAR LR power Voltage                                                                                                                              | 736  | 32 | @1+ | 1 | 0 | 0 | 25    |
| ALGORUNNER_3V3_Voltage                 | ALGORUNNER 3V3 Voltage Supply                                                                                                                           | 768  | 32 | @1+ | 1 | 0 | 0 | 6.6   |
| ALGORUNNER_1V8_Voltage                 | ALGORUNNER 1V8 Voltage Supply                                                                                                                           | 800  | 32 | @1+ | 1 | 0 | 0 | 1.8   |
| ALGORUNNER_1V0_Voltage                 | ALGORUNNER 1V0 Voltage Supply                                                                                                                           | 832  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| ALGORUNNER_1V35_Voltage                | ALGORUNNER 1V35 Voltage Supply                                                                                                                          | 864  | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| VISIONCO_3V3_Voltage                   | VISIONCO 3V3 Voltage                                                                                                                                    | 896  | 32 | @1+ | 1 | 0 | 0 | 6.6   |
| VISIONCO_1V0_Voltage                   | VISIONCO_1V0_Voltage                                                                                                                                    | 928  | 32 | @1+ | 1 | 0 | 0 | 6.6   |
| VISIONCO_1V1_Voltage                   | VISIONCO 1V1 Voltage                                                                                                                                    | 960  | 32 | @1+ | 1 | 0 | 0 | 6.6   |
| VISIONCO_1V8_Voltage                   | VISIONCO 1V8 Voltage                                                                                                                                    | 992  | 32 | @1+ | 1 | 0 | 0 | 6.6   |
| SIDERADAR_LF_PWR_Voltage               | SIDERADAR LF Power Voltage                                                                                                                              | 1024 | 32 | @1+ | 1 | 0 | 0 | 25    |
| SIDERADAR_RF_PWR_Voltage               | SIDERADAR RF Power Voltage                                                                                                                              | 1056 | 32 | @1+ | 1 | 0 | 0 | 25    |
| FLR_PWR_Voltage                        | FLR Power Voltage                                                                                                                                       | 1088 | 32 | @1+ | 1 | 0 | 0 | 25    |
| HEATER_PWR_Voltage                     | HEATER Power Voltage                                                                                                                                    | 1120 | 32 | @1+ | 1 | 0 | 0 | 25    |
| HSD_R_ULTRASONIC_Current_SENSE_Voltage | Scaled engineering value of HSD Rear ULTRASONIC current sense voltage.                                                                                  | 1152 | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| HSD_F_ULTRASONIC_Current_SENSE_Voltage | Scaled engineering value of HSD Front ULTRASONIC current sense voltage.                                                                                 | 1184 | 32 | @1+ | 1 | 0 | 0 | 3.3   |
| NTSC_ENCODER_P_Voltage                 | NTSC ENCODER P Voltage                                                                                                                                  | 1216 | 32 | @1+ | 1 | 0 | 0 | 25    |
| NTSC_ENCODER_N_Voltage                 | NTSC ENCODER N Voltage                                                                                                                                  | 1248 | 32 | @1+ | 1 | 0 | 0 | 25    |
| VBattU                                 | VBattU                                                                                                                                                  | 1280 | 32 | @1+ | 1 | 0 | 0 | 25.5  |
| K_3V3_Decay_Thresh                     | #N/A                                                                                                                                                    | 1312 | 32 | @1+ | 1 | 0 | 0 | 65535 |
| k_Switches_Open_Max_Volt_Level         | k_Switches_Open_Max_Volt_Level<br><br>Max voltage that can be available at the ADC if switch is not pressed/Open assuming worst case vin=19v is 2.428 V | 1344 | 32 | @1+ | 1 | 0 | 0 | 19    |
| k_LED_DRIVER_ON_Volt_Level             | erence Voltage of LED ON Voltage                                                                                                                        | 1376 | 32 | @1+ | 1 | 0 | 0 | 25    |
| k_LED_DRIVER_min_batt_volt             | erence to Minimum Battery Voltage                                                                                                                       | 1408 | 32 | @1+ | 1 | 0 | 0 | 25    |

|                                    |                                                                                                                                                                                          |      |    |     |   |   |   |       |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|-----|---|---|---|-------|
| VISIONCO4_off_wait_ms              | Running timer of VISIONCO4_OFF state.                                                                                                                                                    | 1440 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| VISIONCO4_core_dump_wait_ms        | Running timer of VISIONCO4_core_dump state.                                                                                                                                              | 1456 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| IMAGE2P_off_wait_ms                | IMAGE2P_OFF state timeout timer                                                                                                                                                          | 1472 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| ALGORUNNER_off_wait_ms             | ALGORUNNER_OFF state timeout timer                                                                                                                                                       | 1488 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| MAINRADAR_off_wait_ms              | MAINRADAR_OFF state timeout timer                                                                                                                                                        | 1504 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| SIDERADAR_RFRONT_off_wait_ms       | SIDERADAR_RFRONT_OFF state timeout timer                                                                                                                                                 | 1520 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| SIDERADAR_RREAR_off_wait_ms        | SIDERADAR_RREAR_OFF state timeout timer                                                                                                                                                  | 1536 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| SIDERADAR_LFRONTInitTimeout        | The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_LFRONT_Init.                                                                 | 1552 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| SIDERADAR_LFRONT_off_wait_ms       | SIDERADAR_LFRONT_OFF state timeout timer                                                                                                                                                 | 1568 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| SIDERADAR_LREAR_off_wait_ms        | SIDERADAR_LREAR_OFF state timeout timer                                                                                                                                                  | 1584 | 16 | @1+ | 1 | 0 | 0 | 1000  |
| Front_USSs_lout                    | Current representative of sensed voltage of Front USS                                                                                                                                    | 1600 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| SIDERADAR_LR_lout                  | Current representative of sensed voltage of LR SIDERADAR                                                                                                                                 | 1616 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| SIDERADAR_LF_lout                  | Current representative of sensed voltage of LF SIDERADAR                                                                                                                                 | 1632 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| SIDERADAR_RR_lout                  | Current representative of sensed voltage of RR SIDERADAR                                                                                                                                 | 1648 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| SIDERADAR_RF_lout                  | Current representative of sensed voltage of RF SIDERADAR                                                                                                                                 | 1664 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| MAINRADAR_lout                     | Current representative of sensed voltage of MAINRADAR                                                                                                                                    | 1680 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| LH_MIRROR_lout                     | Current representative of sensed voltage of LH_MIRROR                                                                                                                                    | 1696 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| FVC_lout                           | Current representative of sensed voltage of FVC                                                                                                                                          | 1712 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| RH_MIRROR_lout                     | Current representative of sensed voltage of RH_MIRROR                                                                                                                                    | 1728 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| FWC_MIRROR_lout                    | Current representative of sensed voltage of FWC_MIRROR                                                                                                                                   | 1744 | 16 | @1+ | 1 | 0 | 0 | 2000  |
| K_Max_Time_In_VISIONCO4_Init       | Maximum time to remain in VISIONCO4_Init mode before resetting VISIONCO4 by going to VISIONCO4_OFF.                                                                                      | 1760 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Min_Time_In_VISIONCO4_Off_ms     | Minimum time in which the VISIONCO4 must remain off.                                                                                                                                     | 1776 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_VISIONCO4_Core_Dump  | Maximum time to remain in VISIONCO4_Core_Dump mode before resetting VISIONCO4 by going to VISIONCO4_OFF.<br>The default value is set to 60 seconds to conform with the BPP spec from ME. | 1792 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_VISIONCO4_SOFT_RESET | #N/A                                                                                                                                                                                     | 1808 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_IMAGE2P_Retry_Counter_Max        | Maximum number of retrys for initialization the IMAGE2P.                                                                                                                                 | 1824 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_IMAGE2P_Init         | Maximum time to remain in IMAGE2P_Init mode before resetting IMAGE2P by going to IMAGE2P_OFF.                                                                                            | 1840 | 16 | @1+ | 1 | 0 | 0 | 65535 |

|                                |                                                              |      |    |     |   |   |   |       |
|--------------------------------|--------------------------------------------------------------|------|----|-----|---|---|---|-------|
| K_Min_Time_In_IMAGE2P_Off_ms   | Minimum time in which the IMAGE2P must remain off.           | 1856 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_ALGORUNNER_Retry_Counter_Max | Maximum number of retries for initialization the ALGORUNNER. | 1872 | 16 | @1+ | 1 | 0 | 0 | 65535 |

|                                       |                                                                                                                                                                |      |    |     |   |   |   |       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|-----|---|---|---|-------|
| K_Max_Time_In_ALGORUNNER_Init         | Maximum time to remain in ALGORUNNER_Init mode before resetting ALGORUNNER by going to ALGORUNNER_OFF.<br>Currently it's set to 60000ms to satisfy DV testing. | 1888 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Min_Time_In_ALGORUNNER_Off_ms       | Minimum time in which the ALGORUNNER must remain off.                                                                                                          | 1904 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_MAINRADAR_Retry_Counter_Max         | Maximum number of retrys for initialization the MAINRADAR.                                                                                                     | 1920 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_MAINRADAR_Init          | Maximum time to remain in MAINRADAR_Init mode before resetting MAINRADAR by going to MAINRADAR_OFF.                                                            | 1936 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Min_Time_In_MAINRADAR_Off_ms        | Minimum time in which the MAINRADAR must remain off.                                                                                                           | 1952 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_SIDERADAR_RFRONT_Retry_Counter_Max  | Maximum number of retrys for initialization the SIDERADAR_RFRONT.                                                                                              | 1968 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_SIDERADAR_RFRONT_Init   | Maximum time to remain in SIDERADAR_RFRONT_Init mode before resetting SIDERADAR_RFRONT by going to SIDERADAR_RFRONT_OFF.                                       | 1984 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Min_Time_In_SIDERADAR_RFRONT_Off_ms | Minimum time in which the SIDERADAR_RFRONT must remain off.                                                                                                    | 2000 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_SIDERADAR_RREAR_Retry_Counter_Max   | Maximum number of retrys for initialization the SIDERADAR_RREAR.                                                                                               | 2016 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_SIDERADAR_RREAR_Init    | Maximum time to remain in SIDERADAR_RREAR_Init mode before resetting SIDERADAR_RREAR by going to SIDERADAR_RREAR_OFF.                                          | 2032 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Min_Time_In_SIDERADAR_RREAR_Off_ms  | Minimum time in which the SIDERADAR_RREAR must remain off.                                                                                                     | 2048 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_SIDERADAR_LFRONT_Retry_Counter_Max  | Maximum number of retrys for initialization the SIDERADAR_LFRONT.                                                                                              | 2064 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_SIDERADAR_LFRONT_Init   | Maximum time to remain in SIDERADAR_LFRONT_Init mode before resetting SIDERADAR_LFRONT by going to SIDERADAR_LFRONT_OFF.                                       | 2080 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Min_Time_In_SIDERADAR_LFRONT_Off_ms | Minimum time in which the SIDERADAR_LFRONT must remain off.                                                                                                    | 2096 | 16 | @1+ | 1 | 0 | 0 | 1     |
| K_SIDERADAR_LREAR_Retry_Counter_Max   | Maximum number of retrys for initialization the SIDERADAR_LREAR.                                                                                               | 2112 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Max_Time_In_SIDERADAR_LREAR_Init    | Maximum time to remain in SIDERADAR_LREAR_Init mode before resetting SIDERADAR_LREAR by going to SIDERADAR_LREAR_OFF.                                          | 2128 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| K_Min_Time_In_SIDERADAR_LREAR_Off_ms  | Minimum time in which the SIDERADAR_LREAR must remain off.                                                                                                     | 2144 | 16 | @1+ | 1 | 0 | 0 | 65535 |
| k_Therm_A2D_Hi                        | Limits on highest temp value for the Thermistor out of range DTC.<br>Note: Thermistor counts to temp have inverse relationship.                                | 2160 | 16 | @1+ | 1 | 0 | 0 | 4096  |
| k_Therm_A2D_Lo                        | Limits on lowest temp value for the Thermistor out of range DTC.<br>Note: Thermistor counts to temp have inverse relationship.                                 | 2176 | 16 | @1+ | 1 | 0 | 0 | 4096  |
| k_battery_supply_time_constant        | Time constant for calculating filtered battery supply "VBattU" from UnFilterVBattU                                                                             | 2192 | 16 | @1+ | 1 | 0 | 1 | 1000  |



|                              |                                 |      |    |     |   |   |   |       |
|------------------------------|---------------------------------|------|----|-----|---|---|---|-------|
| SIDERADAR_RREAR_Init_wait_ms | Right Rear SIDERADAR Init timer | 2208 | 16 | @1+ | 1 | 0 | 0 | 65535 |
|                              |                                 |      |    |     |   |   |   |       |

|                                               |                                                                                  |      |    |     |   |   |     |       |
|-----------------------------------------------|----------------------------------------------------------------------------------|------|----|-----|---|---|-----|-------|
| SIDERADAR_LFRONT_Init_wait_ms                 | From SIDERADAR Init Timer                                                        | 2224 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| SIDERADAR_RFRONT_Off_wait_ms                  | SIDERADAR_RFRONT_OFF state timeout timer                                         | 2240 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| SIDERADAR_RFRONT_Init_wait_ms                 | From SIDERADAR Init Timer                                                        | 2256 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| MAINRADAR_Off_wait_ms                         | MAINRADAR_OFF state timeout timer                                                | 2272 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| MAINRADAR_Init_wait_ms                        | MAINRADAR Init Timer                                                             | 2288 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| VISIONCO4_Off_wait_ms                         | Running timer of VISIONCO4_OFF state.                                            | 2304 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| VISIONCO4_Init_wait_ms                        | VISIONCO4 Init timer                                                             | 2320 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| ALGORUNNER_Off_wait_ms                        | ALGORUNNER_OFF state timeout timer                                               | 2336 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| ALGORUNNER_Init_Retry_Counter                 | Counter of the retrys for initialization the ALGORUNNER.                         | 2352 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| ALGORUNNER_Init_wait_ms                       | ALGORUNNER INIT timeout                                                          | 2368 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| VISIONCO4_Init_Retry_Counter                  | Counter for the retrys of VISIONCO4 initialization.                              | 2384 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| VISIONCO4_Core_Dump_wait_ms                   | VisionCo core dump wait timer                                                    | 2400 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| VISIONCO4_Core_Dump_Complete_wait_ms          | VisionCo core dump complete timer                                                | 2416 | 16 | @1+ | 1 | 0 | 0   | 65535 |
| VISIONCO4_Init_Retry_Counter                  | Counter for the retrys of VISIONCO4 initialization.                              | 2432 | 8  | @1+ | 1 | 0 | 0   | 255   |
| IMAGE2P_Init_Retry_Counter                    | Counter of the retrys for initialization the IMAGE2P.                            | 2440 | 8  | @1+ | 1 | 0 | 0   | 255   |
| ALGORUNNER_Init_Retry_Counter                 | Counter of the retrys for initialization the ALGORUNNER.                         | 2448 | 8  | @1+ | 1 | 0 | 0   | 255   |
| MAINRADAR_Init_Retry_Counter                  | Counter of the retrys for initialization the MAINRADAR.                          | 2456 | 8  | @1+ | 1 | 0 | 0   | 255   |
| SIDERADAR_RFRONT_Init_Retry_Counter           | Counter of the retrys for initialization the SIDERADAR_RFRONT.                   | 2464 | 8  | @1+ | 1 | 0 | 0   | 255   |
| SIDERADAR_RREAR_Init_Retry_Counter            | Counter of the retrys for initialization the SIDERADAR_RREAR.                    | 2472 | 8  | @1+ | 1 | 0 | 0   | 255   |
| SIDERADAR_LFRONT_Init_Retry_Counter           | Counter of the retrys for initialization the SIDERADAR_LFRONT.                   | 2480 | 8  | @1+ | 1 | 0 | 0   | 255   |
| SIDERADAR_LREAR_Init_Retry_Counter            | Counter of the retrys for initialization the SIDERADAR_LREAR.                    | 2488 | 8  | @1+ | 1 | 0 | 0   | 255   |
| Bypass_ADC_conversion                         | Defines which ADC channel shall be Bypassed in ADC conversion                    | 2496 | 8  | @1+ | 1 | 0 | 0   | 0     |
| CPYR_agentOnBdT                               | CPYR agent on Board Temperature                                                  | 2504 | 8  | @1+ | 1 | 0 | -40 | 125   |
| VisionCo4OnBdT                                | VisionCo4 on Board Temperature                                                   | 2512 | 8  | @1+ | 1 | 0 | -40 | 125   |
| IMAGE2POnBdT                                  | IMAGE2P on Board Temperature                                                     | 2520 | 8  | @1+ | 1 | 0 | -40 | 125   |
| AlgorunnerOnBdT                               | Algorunner on Board Temperature                                                  | 2528 | 8  | @1+ | 1 | 0 | -40 | 125   |
| CPYR_agent_Rx_Valid_VisionCo4_SPI_Msgs        | EyqQ4 to CPYR agent Valid SPI flags Ture<br>False                                | 2536 | 8  | @1+ | 1 | 0 | 0   | 0     |
| CPYR_agent_Rx_Valid_ALGORUNNER_Ethernet_Msgs  | Algorunner to CPYR agent Ethernet messages flag:<br>true<br>false                | 2544 | 8  | @1+ | 1 | 0 | 0   | 0     |
| CPYR_agent_Rx_Valid_SIDERADAR_RFRONT_CAN_Msgs | Internal PMM variable set/cleared according to the status reported from the E2E. | 2552 | 8  | @1+ | 1 | 0 | 0   | 0     |
| CPYR_agent_Rx_Valid_SIDERADAR_RREAR_CAN_Msgs  | Right Rear SIDERADAR to CPYR agent can messages flag TRUE<br>FALSE               | 2560 | 8  | @1+ | 1 | 0 | 0   | 0     |

|                                               |                                                                       |      |   |     |   |   |   |   |
|-----------------------------------------------|-----------------------------------------------------------------------|------|---|-----|---|---|---|---|
| CPYR agent_Rx_Valid_SIDERADAR_LFRONT_CAN_Msgs | left front SIDERADAR to CPYR AGENT can<br>messages Vaild flag<br>TRUE | 2568 | 8 | @1+ | 1 | 0 | 0 | 0 |
|-----------------------------------------------|-----------------------------------------------------------------------|------|---|-----|---|---|---|---|

|                                                        |                                                                                                                               |      |   |     |   |   |   |     |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------|---|-----|---|---|---|-----|
|                                                        | FALSE                                                                                                                         |      |   |     |   |   |   |     |
| CPYR agent_Rx_Valid_SIDERADAR_LREAR_CAN_Msgs           | Left read SIDERADAR to CPYR agent valid can messages flag<br>TRUE<br>FALSE                                                    | 2576 | 8 | @1+ | 1 | 0 | 0 | 0   |
| VisionCo4_SPI_Rolling_Count_Err                        | VisionCo4 SPI to CPYR agent rolling counter error flag<br>TRUE<br>FALSE                                                       | 2584 | 8 | @1+ | 1 | 0 | 0 | 0   |
| ALGORUNNER_ETHERNET_Rolling_Count_Err                  | this flag is set/clear when the number of attempts for communicate with the Algo runner through the Ethernet<br>TRUE<br>FALSE | 2592 | 8 | @1+ | 1 | 0 | 0 | 0   |
| VisionCo4_SPI_Lost_Comm                                | This flag is set/clear regarding the status of the the communication with the VisionCo4<br>TRUE<br>FALSE                      | 2600 | 8 | @1+ | 1 | 0 | 0 | 0   |
| ALGORUNNER_ETHERNET_Lost_Comm                          | This flag is set/clear regarding the Ethernet connection status between the CPYR agent and the Algo runner<br>TRUE<br>FALSE   | 2608 | 8 | @1+ | 1 | 0 | 0 | 0   |
| Com_with_MAINRADAR_Is_Dropped                          | This flag is set/clear regarding the communication with the MAINRADAR active or dropped<br>TRUE<br>FALSE                      | 2616 | 8 | @1+ | 1 | 0 | 0 | 0   |
| Com_with_SIDERADAR_RFRONT_Is_Dropped                   | This flag is set/clear regarding the communication with the front SIDERADAR active or dropped<br>TRUE<br>FALSE                | 2624 | 8 | @1+ | 1 | 0 | 0 | 0   |
| Com_with_SIDERADAR_RREAR_Is_Dropped                    | This flag is set/clear regarding the communication with the right rear SIDERADAR active or dropped<br>TRUE<br>FALSE           | 2632 | 8 | @1+ | 1 | 0 | 0 | 0   |
| Com_with_SIDERADAR_LFRONT_Is_Dropped                   | This flag is set/clear regarding the communication with the left front SIDERADAR active or dropped<br>TRUE<br>FALSE           | 2640 | 8 | @1+ | 1 | 0 | 0 | 0   |
| Com_with_SIDERADAR_LREAR_Is_Dropped                    | This flag is set/clear regarding the communication with the left rear SIDERADAR active or dropped<br>TRUE<br>FALSE            | 2648 | 8 | @1+ | 1 | 0 | 0 | 0   |
| k_Max_Time_CPYR agent_Thermal_Shutdown_ms              | 0                                                                                                                             | 2656 | 8 | @1+ | 1 | 0 | 0 | 500 |
| k_Max_Time_IMAGE2P_Off_IMAGE2P_Thermal_Shutdown_ms     | 0                                                                                                                             | 2664 | 8 | @1+ | 1 | 0 | 0 | 500 |
| k_Max_Time_IMAGE2P_CAMs_VBattState_Not_In_NORM_VOLT_ms | 0                                                                                                                             | 2672 | 8 | @1+ | 1 | 0 | 0 | 500 |
| k_Max_Time_USSs_VBattState_Not_In_NORM_VOLT_ms         | 0                                                                                                                             | 2680 | 8 | @1+ | 1 | 0 | 0 | 500 |
| k_Max_Time_USSs_Power_Failure_ms                       | 0                                                                                                                             | 2688 | 8 | @1+ | 1 | 0 | 0 | 500 |
| k_Max_Time_CAMs_Power_Failure_ms                       | 0                                                                                                                             | 2696 | 8 | @1+ | 1 | 0 | 0 | 500 |
| K_VisionCo4_Retry_Counter_Max                          | Maximum number of retrys for initialization the VisionCo4.                                                                    | 2704 | 8 | @1+ | 1 | 0 | 0 | 255 |
| K_XCP_Force_VISIONCO_Core_Dump_Failure                 | if the flag<br>K_XCP_Force_VISIONCO_Core_Dump_Failure = true then VISIONCO shall be in Core dump                              | 2712 | 8 | @1+ | 1 | 0 | 0 | 1   |

|                                  |                                                                                             |      |   |     |   |   |      |     |
|----------------------------------|---------------------------------------------------------------------------------------------|------|---|-----|---|---|------|-----|
|                                  | state.                                                                                      |      |   |     |   |   |      |     |
| k_VISIONCO4_cycle_counter        | #N/A                                                                                        | 2720 | 8 | @1+ | 1 | 0 | 1    | 255 |
| k_ALGORUNNER_cycle_counter       | #N/A                                                                                        | 2728 | 8 | @1+ | 1 | 0 | 1    | 255 |
| k_MAINRADAR_cycle_counter        | #N/A                                                                                        | 2736 | 8 | @1+ | 1 | 0 | 1    | 255 |
| k_SIDERADAR_RFRONT_cycle_counter | #N/A                                                                                        | 2744 | 8 | @1+ | 1 | 0 | 1    | 255 |
| k_SIDERADAR_RREAR_cycle_counter  | #N/A                                                                                        | 2752 | 8 | @1+ | 1 | 0 | 1    | 255 |
| k_SIDERADAR_LFRONT_cycle_counter | #N/A                                                                                        | 2760 | 8 | @1+ | 1 | 0 | 1    | 255 |
| k_SIDERADAR_LREAR_cycle_counter  | #N/A                                                                                        | 2768 | 8 | @1+ | 1 | 0 | 1    | 255 |
| K_ThermRecoverDelta              | Current representative of sensed voltage of Rear USS                                        | 2776 | 8 | @1+ | 1 | 0 | 0    | 255 |
| K_ThermRecoverDelta              | Defines the amount of temp needed to cool down from over temp recovery                      | 2784 | 8 | @1+ | 1 | 0 | 0    | 255 |
| K_ThermRecoverDelta              | Defines the amount of temp needed to cool down from over temp recovery                      | 2792 | 8 | @1+ | 1 | 0 | 0    | 255 |
| k_OverTempEnterThresh            | Thermistor Upper temp threshold of Normal temp band. Min range -50C to +150C                | 2800 | 8 | @1+ | 1 | 0 | -50  | 150 |
| k_TR_OverTempEnterThresh         | Thermistor Upper temp threshold of Normal temp band. Min range -50C to +150C                | 2808 | 8 | @1+ | 1 | 0 | -50  | 150 |
| k_VisionCoOverTempEnterThresh    | VisionCo MIPS, VMP upper temp threshold of Normal temperature band. Min range -50C to +150C | 2816 | 8 | @1+ | 1 | 0 | -50  | 150 |
| k_IMAGE2P_OverTempEnterThresh    | Thermistor Upper temp threshold of Normal temp band. Min range -50C to +150C                | 2824 | 8 | @1+ | 1 | 0 | -50  | 150 |
| k_ImagerOverTempEnterThresh      | Imager Upper temp threshold of Normal temp band. Min range -50C to +150C                    | 2832 | 8 | @1+ | 1 | 0 | -50  | 150 |
| k_ThermTemp_Inc                  | Thermistor temperature out of range increment filter                                        | 2840 | 8 | @1+ | 1 | 0 | -128 | 127 |
| k_ThermTemp_Dec                  | Thermistor temperature out of range decrement filter                                        | 2848 | 8 | @1+ | 1 | 0 | -128 | 127 |
| k_ThermRange_Inc                 | Thermistor out of range increment filter                                                    | 2856 | 8 | @1+ | 1 | 0 | -128 | 127 |
| k_ThermRange_Dec                 | Thermistor out of range decrement filter                                                    | 2864 | 8 | @1+ | 1 | 0 | -128 | 127 |
| k_Switches_Pressed_Volt_Level    | k_Switches_Pressed_Volt_Level                                                               | 2872 | 8 | @1+ | 1 | 0 | 1    | 25  |
| k_Switches_Vmin                  | k_Switches_Vmin                                                                             | 2880 | 8 | @1+ | 1 | 0 | 10   | 25  |
| k_Switches_Vmax                  | k_Switches_Vmax                                                                             | 2888 | 8 | @1+ | 1 | 0 | 6    | 25  |
| k_Switches_Debounce_Counts       | k_Switches_Debounce_Counts                                                                  | 2896 | 8 | @1+ | 1 | 0 | 0    | 200 |
| k_LED_Driver_debounce_counts     | erence Debounce time for LED driver                                                         | 2904 | 8 | @1+ | 1 | 0 | 0    | 100 |
| k_BootOverTemp_Inc               | k_BootOverTemp_Inc                                                                          | 2912 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_BootOverTemp_Dec               | k_BootOverTemp_Dec                                                                          | 2920 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_BIST_Inc          | k_VISIONCOBoot_BIST_Inc                                                                     | 2928 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_BIST_Dec          | 0                                                                                           | 2936 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_CS_Inc            | k_VISIONCOBoot_CS_Inc                                                                       | 2944 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_CS_Dec            | k_VISIONCOBoot_CS_Dec                                                                       | 2952 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_DDR_Inc           | k_VISIONCOBoot_DDR_Inc                                                                      | 2960 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_DDR_Dec           | k_VISIONCOBoot_DDR_Dec                                                                      | 2968 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_LoadFlash_Inc     | k_VISIONCOBoot_LoadFlash_Inc                                                                | 2976 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_LoadFlash_Dec     | k_VISIONCOBoot_LoadFlash_Dec                                                                | 2984 | 8 | @1+ | 1 | 0 | 0    | 127 |
| k_VISIONCOBoot_AppCS_Inc         | k_VISIONCOBoot_AppCS_Inc                                                                    | 2992 | 8 | @1+ | 1 | 0 | 0    | 127 |

|                                   |                                                                                                                      |      |   |     |   |   |   |     |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|------|---|-----|---|---|---|-----|
| k_VISIONCOBoot_AppCS_Dec          | k_VISIONCOBoot_AppCS_Dec                                                                                             | 3000 | 8 | @1+ | 1 | 0 | 0 | 127 |
| k_VISIONCOBoot_Cal_CS_Inc         | k_VISIONCOBoot_Cal_CS_Inc                                                                                            | 3008 | 8 | @1+ | 1 | 0 | 0 | 127 |
| k_VISIONCOBoot_Cal_CS_Dec         | k_VISIONCOBoot_Cal_CS_Dec                                                                                            | 3016 | 8 | @1+ | 1 | 0 | 0 | 127 |
| k_VISIONCOBoot_MEST_CS_Inc        | k_VISIONCOBoot_MEST_CS_Inc                                                                                           | 3024 | 8 | @1+ | 1 | 0 | 0 | 127 |
| k_VISIONCOBoot_MEST_CS_Dec        | k_VISIONCOBoot_MEST_CS_Dec                                                                                           | 3032 | 8 | @1+ | 1 | 0 | 0 | 127 |
| k_LIN_Bus_Fault_Inc_Cnt           | k_LIN_Bus_Fault_Inc this is the Increment step size counter for LIN Rear Camera Bus Signal / Message Failure fault   | 3040 | 8 | @1+ | 1 | 0 | 0 | 255 |
| k_LIN_Bus_Fault_De_Cnt            | k_LIN_Bus_Fault_Dec this is the decrement step size counter for LIN Rear Camera Bus Signal / Message Failure fault   | 3048 | 8 | @1+ | 1 | 0 | 0 | 255 |
| k_LIN_RearCamera_Fault_Inc_Cnt    | #N/A                                                                                                                 | 3056 | 8 | @1+ | 1 | 0 | 0 | 255 |
| k_LIN_RearCamera_Fault_Dec_Cnt    | #N/A                                                                                                                 | 3064 | 8 | @1+ | 1 | 0 | 0 | 255 |
| k_LIN_R_Cam_Nocfg_Fault_Inc_Cnt   | #N/A                                                                                                                 | 3072 | 8 | @1+ | 1 | 0 | 0 | 255 |
| k_LIN_R_Cam_Nocfg_Fault_Dec_Cnt   | #N/A                                                                                                                 | 3080 | 8 | @1+ | 1 | 0 | 0 | 255 |
| k_SPI_Rolling_Count_Delta         | it is the acceptable value of missing SPI MSGs between CPYR agent & VISIONCO4 and vice versa.                        | 3088 | 8 | @1+ | 1 | 0 | 0 | 255 |
| k_Ethernet_Rolling_Count_Delta    | it is the acceptable value of missing Ethernet MSGs between CPYR agent & ALGORUNNER and vice versa.                  | 3096 | 8 | @1+ | 1 | 0 | 0 | 255 |
| SIDERADAR_LREAR_Power_Mode_State  | Left Rear SIDERADAR power mode status: Init<br>Norm<br>Standby<br>off                                                | 3104 | 8 | @1+ | 1 | 0 | 0 | 255 |
| SIDERADAR_RREAR_Power_Mode_State  | Right Rear SIDERADAR power mode status                                                                               | 3112 | 8 | @1+ | 1 | 0 | 0 | 255 |
| SIDERADAR_LFRONT_Power_Mode_State | Left Front SIDERADAR Power Mode status :<br>Init, Norm, Stand by, OFF                                                | 3120 | 8 | @1+ | 1 | 0 | 0 | 255 |
| SIDERADAR_RFRONT_Power_Mode_State | Front SIDERADAR power mode status                                                                                    | 3128 | 8 | @1+ | 1 | 0 | 0 | 255 |
| MAINRADAR_Power_Mode_State        | MAINRADAR_Power_Mode_State                                                                                           | 3136 | 8 | @1+ | 1 | 0 | 0 | 255 |
| IMAGE2P_Power_Mode_State          | IMAGE2P Power Mode Status                                                                                            | 3144 | 8 | @1+ | 1 | 0 | 0 | 255 |
| ALGORUNNER_Power_Mode_State       | ALGORUNNER Power Mode Status                                                                                         | 3152 | 8 | @1+ | 1 | 0 | 0 | 255 |
| VISIONCO4_Power_Mode_State        | VisionCo Power mode state var                                                                                        | 3160 | 8 | @1+ | 1 | 0 | 0 | 255 |
| CPYR agentApplicationFilePresent  | The primary boot loader detected a loadable/correct CPYR agent application files.                                    | 3168 | 1 | @1+ | 1 | 0 | 0 | 1   |
| ProgrammingMsgReceived            | TRUE if a CAN programming message is received while CPYR agent is running in PBL_INIT, FALSE otherwise.              | 3169 | 1 | @1+ | 1 | 0 | 0 | 1   |
| VisionCo4InitTimeout              | The CPYR agent doesn't receive a correctly formatted application SPI messages in K_Max_Time_In_VISIONCO4_Init.       | 3170 | 1 | @1+ | 1 | 0 | 0 | 1   |
| IMAGE2PInitTimeout                | The CPYR agent doesn't receive a correctly formatted application Ethernet messages in K_Max_Time_In_IMAGE2P_Init.    | 3171 | 1 | @1+ | 1 | 0 | 0 | 1   |
| ALGORUNNERInitTimeout             | The CPYR agent doesn't receive a correctly formatted application Ethernet messages in K_Max_Time_In_ALGORUNNER_Init. | 3172 | 1 | @1+ | 1 | 0 | 0 | 1   |
| MAINRADARInitTimeout              | The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_MAINRADAR_Init.    | 3173 | 1 | @1+ | 1 | 0 | 0 | 1   |

|                             |                                                                                                                           |      |   |     |   |   |   |   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|------|---|-----|---|---|---|---|
| SIDERADAR_RFRONTInitTimeout | The CPYR agent doesn' t receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_RFRONT_Init. | 3174 | 1 | @1+ | 1 | 0 | 0 | 1 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|------|---|-----|---|---|---|---|

|                                     |                                                                                                                                                 |      |   |     |   |   |   |   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------|---|-----|---|---|---|---|
| SIDERADAR_RREARInitTimeout          | The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_RREAR_Init.                         | 3175 | 1 | @1+ | 1 | 0 | 0 | 1 |
| SIDERADAR_LREARInitTimeout          | The CPYR agent doesn't receive a correctly formatted application CAN-FD messages in K_Max_Time_In_SIDERADAR_LREAR_Init.                         | 3176 | 1 | @1+ | 1 | 0 | 0 | 1 |
| LKA_SW_Status                       | LKA_SW_Status<br><br>1 (PRESSED).<br>0 (NOT_PRESSED).                                                                                           | 3177 | 1 | @1+ | 1 | 0 | 0 | 1 |
| View_SW_Status                      | View_SW_Status<br><br>1 (PRESSED or ACTIVE).<br>0 (NOT_PRESSED or INACTIVE).                                                                    | 3178 | 1 | @1+ | 1 | 0 | 0 | 1 |
| BPA_SW_Status                       | BPA_SW_Status<br><br>1 (PRESSED or ACTIVE).<br>0 (NOT_PRESSED or INACTIVE).                                                                     | 3179 | 1 | @1+ | 1 | 0 | 0 | 1 |
| APA_SW_Status                       | APA_SW_Status<br><br>1 (PRESSED or ACTIVE).<br>0 (NOT_PRESSED or INACTIVE).                                                                     | 3180 | 1 | @1+ | 1 | 0 | 0 | 1 |
| VisionCo4_SPI_E2E_PROFILE_ERR       | #N/A                                                                                                                                            | 3181 | 1 | @1+ | 1 | 0 | 0 | 0 |
| VisionCo4_ERROUT                    | 0                                                                                                                                               | 3182 | 1 | @1+ | 1 | 0 | 0 | 0 |
| TR_OTA_Ongoing                      | Internal PMM flag set/cleared according to the status reported from the OTA_Mgr.                                                                | 3183 | 1 | @1+ | 1 | 0 | 0 | 0 |
| TR_OTA_Diag_Failure                 | Internal PMM flag set/cleared according to the status reported from the OTA diagnostics.                                                        | 3184 | 1 | @1+ | 1 | 0 | 0 | 0 |
| VisionCo4_OTA_Ongoing               | Internal PMM flag set/cleared according to the status reported from the OTA_Mgr.                                                                | 3185 | 1 | @1+ | 1 | 0 | 0 | 0 |
| VisionCo4_OTA_Diag_Failure          | Internal PMM flag set/cleared according to the status reported from the OTA diagnostics.                                                        | 3186 | 1 | @1+ | 1 | 0 | 0 | 0 |
| K_Enable_Lost_Comm_VISIONCO4_Reset  | If TRUE, then lost of communication from the VISIONCO4 while in VISIONCO4_NORM or VISIONCO4_OFF states will cause a reset of the VISIONCO4.     | 3187 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_VISIONCO4_Diag_Shutdown    | Set to TRUE so that VISIONCO4 does not reset if this flag is TRUE.                                                                              | 3188 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Enable_Lost_Comm_IMAGE2P_Reset    | If TRUE, then lost of communication from the IMAGE2P while in IMAGE2P_NORM or IMAGE2P_OFF states will cause a reset of the IMAGE2P.             | 3189 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_IMAGE2P_Diag_Shutdown      | Set to TRUE so that IMAGE2P does not reset if this flag is TRUE.                                                                                | 3190 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Enable_Lost_Comm_ALGORUNNER_Reset | If TRUE, then lost of communication from the ALGORUNNER while in ALGORUNNER_NORM or ALGORUNNER_OFF states will cause a reset of the ALGORUNNER. | 3191 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_ALGORUNNER_Diag_Shutdown   | Set to TRUE so that ALGORUNNER does not reset if this flag is TRUE.                                                                             | 3192 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Enable_Lost_Comm_MAINRADAR_Reset  | If TRUE, then lost of communication from the MAINRADAR while in MAINRADAR_NORM or MAINRADAR_OFF states will cause a reset of the MAINRADAR.     | 3193 | 1 | @1+ | 1 | 0 | 0 | 1 |



|                                           |                                                                    |      |   |     |   |   |   |   |
|-------------------------------------------|--------------------------------------------------------------------|------|---|-----|---|---|---|---|
| K_Ignore_MAINRADAR_Diag_Shutdown          | Set to TRUE so that MAINRADAR does not reset if this flag is TRUE. | 3194 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Enable_Lost_Comm_SIDERADAR_RFRONT_Reset | If TRUE, then lost of communication                                | 3195 | 1 | @1+ | 1 | 0 | 0 | 1 |

|                                           |                                                                                                                                                                         |      |   |     |   |   |   |   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|-----|---|---|---|---|
|                                           | from the SIDERADAR_RFRONT while in SIDERADAR_RFRONT_NORM or SIDERADAR_RFRONT_OFF states will cause a reset of the SIDERADAR_RFRONT.                                     |      |   |     |   |   |   |   |
| K_Ignore_SIDERADAR_RFRONT_Diag_Shutdown   | Set to TRUE so that SIDERADAR_RFRONT does not reset if this flag is TRUE.                                                                                               | 3196 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Enable_Loss_Comm_SIDERADAR_RREAR_Reset  | If TRUE, then lost of communication from the SIDERADAR_RREAR while in SIDERADAR_RREAR_NORM or SIDERADAR_RREAR_OFF states will cause a reset of the SIDERADAR_RREAR.     | 3197 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_SIDERADAR_RREAR_Diag_Shutdown    | Set to TRUE so that SIDERADAR_RREAR does not reset if this flag is TRUE.                                                                                                | 3198 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Enable_Lost_Comm_SIDERADAR_LFRONT_Reset | If TRUE, then lost of communication from the SIDERADAR_LFRONT while in SIDERADAR_LFRONT_NORM or SIDERADAR_LFRONT_OFF states will cause a reset of the SIDERADAR_LFRONT. | 3199 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_SIDERADAR_LFRONT_Diag_Shutdown   | Set to TRUE so that SIDERADAR_LFRONT does not reset if this flag is TRUE.                                                                                               | 3200 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Enable_Lost_Comm_SIDERADAR_LREAR_Reset  | If TRUE, then lost of communication from the SIDERADAR_LREAR while in SIDERADAR_LREAR_NORM or SIDERADAR_LREAR_OFF states will cause a reset of the SIDERADAR_LREAR.     | 3201 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_SIDERADAR_LREAR_Diag_Shutdown    | Set to TRUE so that SIDERADAR_LREAR does not reset if this flag is TRUE.                                                                                                | 3202 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_ULTRASONIC_FRONT_Diag_Shutdown   | Set to TRUE so that ULTRASONIC_FRONT does not reset if this flag is TRUE.                                                                                               | 3203 | 1 | @1+ | 1 | 0 | 0 | 1 |
| K_Ignore_ULTRASONIC_REAR_Diag_Shutdown    | Set to TRUE so that ULTRASONIC_REAR does not reset if this flag is TRUE.                                                                                                | 3204 | 1 | @1+ | 1 | 0 | 0 | 1 |
| k_Ignore_RH_MIRR_CAM_Diag_Shutdown        | Set to TRUE so that RH_MIRR_CAM does not reset due to diag reasons                                                                                                      | 3205 | 1 | @1+ | 1 | 0 | 0 | 1 |
| k_Ignore_LH_MIRR_CAM_Diag_Shutdown        | Set to TRUE so that LH_MIRR_CAM does not reset due to diag reasons.                                                                                                     | 3206 | 1 | @1+ | 1 | 0 | 0 | 1 |
| k_Ignore_FVC_Diag_Shutdown                | Set to TRUE so that FVC CAM does not reset due to diag reasons.                                                                                                         | 3207 | 1 | @1+ | 1 | 0 | 0 | 1 |
| k_Ignore_RVC_Diag_Shutdown                | Set to TRUE so that RVC CAM does not reset due to diag reasons.                                                                                                         | 3208 | 1 | @1+ | 1 | 0 | 0 | 1 |
| FVC_Power_Mode_State                      | FCV power mode status:<br>on<br>off                                                                                                                                     | 3209 | 1 | @1+ | 1 | 0 | 0 | 1 |
| SIDERADAR_RREAR_Reset_Request             | Right Rear Radar Reset Request                                                                                                                                          | 3210 | 1 | @1+ | 1 | 0 | 0 | 1 |
| RVC_Power_Mode_State                      | RVC power mode status:<br>on<br>off                                                                                                                                     | 3211 | 1 | @1+ | 1 | 0 | 0 | 1 |
| LH_MIRR_CAM_Power_Mode_State              | Left hand mirror cam power mode status :<br>on<br>off                                                                                                                   | 3212 | 1 | @1+ | 1 | 0 | 0 | 1 |
| RH_MIRR_CAM_Power_Mode_State              | Right Hand Mirror cam power mode status:<br>on<br>off                                                                                                                   | 3213 | 1 | @1+ | 1 | 0 | 0 | 1 |

|                             |                                                 |      |   |     |   |   |   |   |
|-----------------------------|-------------------------------------------------|------|---|-----|---|---|---|---|
| REAR_USSs_Power_Mode_State  | Rear Ultrasonic power mode status:<br>on<br>off | 3214 | 1 | @1+ | 1 | 0 | 0 | 1 |
| FRONT_USSs_Power_Mode_State | Front ultrasonics power mode status<br>on       | 3215 | 1 | @1+ | 1 | 0 | 0 | 1 |

|                                |                                                                      |      |   |     |   |   |   |   |
|--------------------------------|----------------------------------------------------------------------|------|---|-----|---|---|---|---|
|                                | off                                                                  |      |   |     |   |   |   |   |
| SIDERADAR_RFRONT_Reset_Request | Front SIDERADAR Reset Request                                        | 3216 | 1 | @1+ | 1 | 0 | 0 | 1 |
| SIDERADAR_RFRONTInitTimeout    | SIDERADAR Init Sequence time out timer flag                          | 3217 | 1 | @1+ | 1 | 0 | 0 | 1 |
| MAINRADAR_Reset_Request        | MAINRADAR Reset Request                                              | 3218 | 1 | @1+ | 1 | 0 | 0 | 1 |
| ALGORUNNER_Reset_Request       | ALGORUNNER reset request                                             | 3219 | 1 | @1+ | 1 | 0 | 0 | 1 |
| VISIONCO4SoftResetTimeout      | Internal PMM flag set/cleared according to Soft reset Time Out event | 3220 | 1 | @1+ | 1 | 0 | 0 | 1 |

## Electrical\_Status

Electrical\_Status(0x101)-STNumber(71)-STVersion(1) This message provides the data of all input signals.

Message Electrical\_Status (0x101) shall be implemented as follows:

| SignalName                             | State Description  | StartBit | Size | Sign | Offset | Factor | Min | Max |
|----------------------------------------|--------------------|----------|------|------|--------|--------|-----|-----|
| LKS_SW_Voltage                         | ADC Physical value | 0        | 32   | @1+  | 1      | 0      | 0   | 25  |
| APA_SW_Voltage                         | ADC Physical value | 32       | 32   | @1+  | 1      | 0      | 0   | 25  |
| View_SW_Voltage                        | ADC Physical value | 64       | 32   | @1+  | 1      | 0      | 0   | 25  |
| BPA_SW_Voltage                         | ADC Physical value | 96       | 32   | @1+  | 1      | 0      | 0   | 25  |
| POC_LDO1_RVC_FVC_SENSE_Current_Voltage | ADC Physical value | 128      | 32   | @1+  | 1      | 0      | 0   | 25  |
| POC_LDO2_LH_RH_SENSE_Current_Voltage   | ADC Physical value | 160      | 32   | @1+  | 1      | 0      | 0   | 25  |
| POC_LDO3_FWC_SENSE_Current_Voltage     | ADC Physical value | 192      | 32   | @1+  | 1      | 0      | 0   | 25  |
| SIDERADAR_LR_RR_SENSE_Current_Voltage  | ADC Physical value | 224      | 32   | @1+  | 1      | 0      | 0   | 25  |
| SIDERADAR_RF_LF_SENSE_Current_Voltage  | ADC Physical value | 256      | 32   | @1+  | 1      | 0      | 0   | 25  |
| HEATER_FLR_SENSE_Current_Voltage       | ADC Physical value | 288      | 32   | @1+  | 1      | 0      | 0   | 25  |
| HSD_ULTRASONIC_SENSE_Current_Voltage   | ADC Physical value | 320      | 32   | @1+  | 1      | 0      | 0   | 25  |
| L_BLIS_OUT_Voltage                     | ADC Physical value | 352      | 32   | @1+  | 1      | 0      | 0   | 25  |
| R_BLIS_OUT_Voltage                     | ADC Physical value | 384      | 32   | @1+  | 1      | 0      | 0   | 25  |
| PARK_AID_OUT_Voltage                   | ADC Physical value | 416      | 32   | @1+  | 1      | 0      | 0   | 25  |
| ALGORUNNER_3V3_Voltage                 | ADC Physical value | 448      | 32   | @1+  | 1      | 0      | 0   | 25  |
| ALGORUNNER_1V8_Voltage                 | ADC Physical value | 480      | 32   | @1+  | 1      | 0      | 0   | 25  |
| NTSC_ENCODER_P_Voltage                 | ADC Physical value | 512      | 32   | @1+  | 1      | 0      | 0   | 25  |
| SIDERADAR_LR_PWR_Voltage               | ADC Physical value | 544      | 32   | @1+  | 1      | 0      | 0   | 25  |
| ALGORUNNER_1V0_Voltage                 | ADC Physical value | 576      | 32   | @1+  | 1      | 0      | 0   | 25  |
| ALGORUNNER_1V35_Voltage                | ADC Physical value | 608      | 32   | @1+  | 1      | 0      | 0   | 25  |
| NTSC_ENCODER_N_Voltage                 | ADC Physical value | 640      | 32   | @1+  | 1      | 0      | 0   | 25  |
| VBattU                                 | ADC Physical value | 672      | 32   | @1+  | 1      | 0      | 0   | 25  |
| VISIONCO_3V3_Voltage                   | ADC Physical value | 704      | 32   | @1+  | 1      | 0      | 0   | 25  |
| VISIONCO_1V0_Voltage                   | ADC Physical value | 736      | 32   | @1+  | 1      | 0      | 0   | 25  |
| VISIONCO_1V1_Voltage                   | ADC Physical value | 768      | 32   | @1+  | 1      | 0      | 0   | 25  |
| VISIONCO_1V8_Voltage                   | ADC Physical value | 800      | 32   | @1+  | 1      | 0      | 0   | 25  |
| SIDERADAR_LF_PWR_Voltage               | ADC Physical value | 832      | 32   | @1+  | 1      | 0      | 0   | 25  |
| SIDERADAR_RF_PWR_Voltage               | ADC Physical value | 864      | 32   | @1+  | 1      | 0      | 0   | 25  |
| FLR_PWR_Voltage                        | ADC Physical value | 896      | 32   | @1+  | 1      | 0      | 0   | 25  |
| HEATER_PWR_Voltage                     | ADC Physical value | 928      | 32   | @1+  | 1      | 0      | 0   | 25  |
| SIDERADAR_RR_PWR_Voltage               | ADC Physical value | 960      | 32   | @1+  | 1      | 0      | 0   | 25  |

|                                  |                                                                                                                                      |      |    |     |      |   |     |       |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------|----|-----|------|---|-----|-------|
| AURIX_Rev                        | ADC Physical value                                                                                                                   | 992  | 16 | @1+ | 1    | 0 | 0   | 65535 |
| _HW_Variant                      | ADC Physical value                                                                                                                   | 1008 | 16 | @1+ | 1    | 0 | 0   | 65535 |
| RawNTSC_ENCODER_AURIX_ADC_P      | NA                                                                                                                                   | 1024 | 12 | @1+ | 1    | 0 | 0   | 4095  |
| RawNTSC_ENCODER_AURIX_ADC_N      | NA                                                                                                                                   | 1036 | 12 | @1+ | 1    | 0 | 0   | 4095  |
| VisionCo4OnBdT                   | ADC Physical value                                                                                                                   | 1048 | 8  | @1- | 1    | 0 | -40 | 125   |
| IMAGE2PONBdT                     | ADC Physical value                                                                                                                   | 1056 | 8  | @1- | 1    | 0 | -40 | 125   |
| AlgorunnerOnBdT                  | ADC Physical value                                                                                                                   | 1064 | 8  | @1- | 1    | 0 | -40 | 125   |
| CPYR_agentOnBdT                  | ADC Physical value                                                                                                                   | 1072 | 8  | @1- | 1    | 0 | -40 | 125   |
| RVC_Iout                         | RVC POC Current Sense                                                                                                                | 1080 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| FVC_Iout                         | FVC POC Current Sense                                                                                                                | 1088 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| LH_MIRROR_Iout                   | LH Mirror POC Current Sense                                                                                                          | 1096 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| RH_MIRROR_Iout                   | RH Mirror POC Current Sense                                                                                                          | 1104 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| FWC_MIRROR_Iout                  | FWC POC Current Sense                                                                                                                | 1112 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| SIDERADAR_LR_Iout                | SIDERADAR_LR Current Sense                                                                                                           | 1120 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| SIDERADAR_RR_Iout                | SIDERADAR_RR Current Sense                                                                                                           | 1128 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| SIDERADAR_LF_Iout                | SIDERADAR_LF Current Sense                                                                                                           | 1136 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| SIDERADAR_RF_Iout                | SIDERADAR_RF Current Sense                                                                                                           | 1144 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| MAINRADAR_Iout                   | MAINRADAR Current Sense                                                                                                              | 1152 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| Heater_Iout                      | Heater Current Sense                                                                                                                 | 1160 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| Rear_USSs_Iout                   | Rear Ultrasonics Current Sense                                                                                                       | 1168 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| Front_USSs_Iout                  | Front Ultrasonics Current Sense                                                                                                      | 1176 | 8  | @1+ | 0.05 | 0 | 0   | 12.5  |
| SIDERADAR_LF_PWR_Current_Voltage | SIDERADAR_LF_PWR_Current_Voltage                                                                                                     | 1184 | 8  | @1+ | 0.08 | 0 | 0   | 20.4  |
| SIDERADAR_RF_PWR_Current_Voltage | SIDERADAR_RF_PWR_Current_Voltage                                                                                                     | 1192 | 8  | @1+ | 0.08 | 0 | 0   | 20.4  |
| SIDERADAR_LR_PWR_Current_Voltage | SIDERADAR_LR_PWR_Current_Voltage                                                                                                     | 1200 | 8  | @1+ | 0.08 | 0 | 0   | 20.4  |
| SIDERADAR_RR_PWR_Current_Voltage | SIDERADAR_RR_PWR_Current_Voltage                                                                                                     | 1208 | 8  | @1+ | 0.08 | 0 | 0   | 20.4  |
| FLR_PWR_Current_Voltage          | FLR_PWR_Current_Voltage                                                                                                              | 1216 | 8  | @1+ | 0.08 | 0 | 0   | 20.4  |
| HEATER_PWR_Current_Voltage       | HEATER_PWR_Current_Voltage                                                                                                           | 1224 | 8  | @1+ | 0.08 | 0 | 0   | 20.4  |
| LKA_SW_Status                    | Lane Keeping Switch                                                                                                                  | 1232 | 1  | @1+ | 1    | 0 | 0   | 1     |
| APA_SW_Status                    | Active Park Assist Switch                                                                                                            | 1233 | 1  | @1+ | 1    | 0 | 0   | 1     |
| View_SW_Status                   | Surround View Switch                                                                                                                 | 1234 | 1  | @1+ | 1    | 0 | 0   | 1     |
| BPA_SW_Status                    | Base ParkAid Switch                                                                                                                  | 1235 | 1  | @1+ | 1    | 0 | 0   | 1     |
| L_BLIS_LED_Status                | L_BLIS_LED_Status                                                                                                                    | 1236 | 1  | @1+ | 1    | 0 | 0   | 1     |
| R_BLIS_LED_Status                | R_BLIS_LED_Status                                                                                                                    | 1237 | 1  | @1+ | 1    | 0 | 0   | 1     |
| PARK_AID_LED_Status              | PARK_AID_LED_Status                                                                                                                  | 1238 | 1  | @1+ | 1    | 0 | 0   | 1     |
| AURIX_HSD3_HEATER_EN             | Heater driver enable status<br>0x0 = Turn Off<br>0x1 = Turn On                                                                       | 1239 | 1  | @1+ | 1    | 0 | 0   | 1     |
| HeaterOutputShortCircuit         | Heater short circuit fault<br>0x0 = No fault<br>0x1 = Fault present                                                                  | 1240 | 1  | @1+ | 1    | 0 | 0   | 1     |
| HeaterOutputOpenCircuit          | Heater open circuit fault<br>0x0 = No fault<br>0x1 = Fault present                                                                   | 1241 | 1  | @1+ | 1    | 0 | 0   | 1     |
| HeaterShortAcrossCycles          | Indicates a Short To Ground (STG)<br>condition across a large number of<br>ignition cycles.<br>0x0 = No fault<br>0x1 = Fault present | 1242 | 1  | @1+ | 1    | 0 | 0   | 1     |

|                                     |                                                                                                                                                                    |     |     |          |   |   |   |     |     |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----------|---|---|---|-----|-----|
|                                     | Revision                                                                                                                                                           |     |     |          |   |   |   |     |     |
| RL_SW_Release_Revision              | Rear Left SIDERADAR5 Hardware revision (ex: A3                                                                                                                     | 720 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| RL_SW_Promote_Revision              | Rear Left Updated every stage with Major updates                                                                                                                   | 728 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| RL_SW_Field_Revision                | Rear Left Updated for minor updates                                                                                                                                | 736 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| RL_SW_Field                         | Rear Left Updated for small updates such as Bug fix                                                                                                                | 744 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| VCAN_DBC_Version_Major              | VCAN DBC Version Major                                                                                                                                             | 752 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| VCAN_DBC_Version_Minor              | VCAN DBC Version Minor                                                                                                                                             | 760 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| VCAN_DBC_Version_Revision           | VCAN DBC Version Revision                                                                                                                                          | 768 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| MAINRADAR_PCAN_DBC_Version_Major    | MAINRADAR PCAN DBC Version Major                                                                                                                                   | 776 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| MAINRADAR_PCAN_DBC_Version_Minor    | MAINRADAR PCAN DBC Version Minor                                                                                                                                   | 784 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| MAINRADAR_PCAN_DBC_Version_Revision | MAINRADAR PCAN DBC Version Revision                                                                                                                                | 792 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| SIDERADAR_PCAN_DBC_Version_Major    | SIDERADAR PCAN DBC Version Major                                                                                                                                   | 800 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| SIDERADAR_PCAN_DBC_Version_Minor    | SIDERADAR PCAN DBC Version Minor                                                                                                                                   | 808 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| SIDERADAR_PCAN_DBC_Version_Revision | SIDERADAR PCAN DBC Version Revision                                                                                                                                | 816 | 8   | Unsigned | 1 | 0 | 0 | 255 | N/A |
| _Application_Version                | will provide interface "pi_Lib_SW_Version" to sent SW library version DE_Lib_Version Where DE_Lib_Version is 16 byte ASCII formatted with unused bytes set to NULL | 824 | 128 | Unsigned | 1 | 0 | 0 |     | NA  |

## CFG\_Parameters

CFG\_Parameters(0x103)-STNumber(73)-STVersion(1) **Message CFG\_Parameters (0x103) shall be implemented as follows:**

| gnalName                        | Discription                                                              | StartBit | Size | Sign | Offset | Factor | Min | Max      |
|---------------------------------|--------------------------------------------------------------------------|----------|------|------|--------|--------|-----|----------|
| SensorCfg_FWC_SerNum            | It is an array of 16 bytes<br>[For Stage-F : It is an array of 32 bytes] | 0        | 64   | @1+  | 0      | 1      | 0   | 1.84E+19 |
| VehicleCfg_SelfSteerGradient    | Module Configuration Mode Feature                                        | 64       | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_OverallLength        | D1                                                                       | 80       | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_Wheelbase            | VehicleCfg_OverallLength = D3                                            | 96       | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_FrontAxleToBumper    | VehicleCfg_FrontAxleToBumper = D4                                        | 112      | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_RearAxleToBumper     | VehicleCfg_RearAxleToBumper = D5                                         | 128      | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_Width                | VehicleCfg_Width = D6                                                    | 144      | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_WidthWithMirror      | VehicleCfg_WidthWithMirror = D7                                          | 160      | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_FrntTrckWidthCenter  | VehicleCfg_FrntTrckWidthCenter                                           | 176      | 16   | @1+  | 0      | 1      | 0   | 65536    |
| VehicleCfg_FrntTrckWidthOutside | VehicleCfg_FrntTrckWidthOutside                                          | 192      | 16   | @1+  | 0      | 1      | 0   | 65536    |

|                                 |                                                                                   |     |    |     |   |   |   |       |
|---------------------------------|-----------------------------------------------------------------------------------|-----|----|-----|---|---|---|-------|
| VehicleCfg_RearTrckWidthCenter  | VehicleCfg_RearTrckWidthCenter                                                    | 208 | 16 | @1+ | 0 | 1 | 0 | 65536 |
| VehicleCfg_RearTrckWidthOutside | VehicleCfg_RearTrckWidthOutside                                                   | 224 | 16 | @1+ | 0 | 1 | 0 | 65536 |
| SensorCfg_DPAC_FrontSerNum      | Sensor Configuration DPAC Front serial numbers<br>LVDSSerialNum_Front[] from will | 240 | 16 | @1+ | 0 | 1 | 0 | 65536 |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                              |     |    |     |   |   |   |       |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-----|---|---|---|-------|
|                                    | be stored in this parameter.                                                                                                                                                                                                                                                                                                                                                                                 |     |    |     |   |   |   |       |
| SensorCfg_DPAC_LeftSerNum          | DPAC Left Camera Serial Number                                                                                                                                                                                                                                                                                                                                                                               | 256 | 16 | @1+ | 0 | 1 | 0 | 65536 |
| SensorCfg_DPAC_RightSerNum         | DPAC Right Camera Serial Number                                                                                                                                                                                                                                                                                                                                                                              | 272 | 16 | @1+ | 0 | 1 | 0 | 65536 |
| SensorCfg_DPAC_RearSerNum          | DPAC Rear Camera Serial Number                                                                                                                                                                                                                                                                                                                                                                               | 288 | 16 | @1+ | 0 | 1 | 0 | 65536 |
| SensorCfg_FWC_ZGndToWhlHouse_Right | Z-axis height measurement from ground to the top center of the wheelhouse (including wheel lip molding if applicable)                                                                                                                                                                                                                                                                                        | 304 | 16 | @1+ | 0 | 1 | 0 | 65536 |
| SensorCfg_FWC_ZGndToWhlHouse_Left  | Z-axis height measurement from ground to the top center of the wheelhouse (including wheel lip molding if applicable)                                                                                                                                                                                                                                                                                        | 320 | 16 | @1+ | 0 | 1 | 0 | 65536 |
| FeatureCfg_LINSelLINRVC            | Unsigned integer index into the method 3 LIN Slave Configuration Array for the LIN RVC                                                                                                                                                                                                                                                                                                                       | 336 | 8  | @1+ | 0 | 1 | 0 | 255   |
| VehicleCfg_LINSel_RearCamera2      | VehicleCfg_LINSel_RearCamera2                                                                                                                                                                                                                                                                                                                                                                                | 344 | 8  | @1+ | 0 | 1 | 0 | 255   |
| MarketCfg_Country                  | Vehicle market country configuration parameter.<br>0x0 (Undefined)<br>0x1(Rest of World)<br>0x2 (UK)<br>0x3 (Ireland)<br>0x4 (Canada)<br>0x5 (China)<br>0x6 (Japan)<br>0x7 (USA)<br>0x8 (South_Africa)<br>0x9 (South_Korea)<br>0xA (Australia)<br>0xB (Oman)<br>0xC (Bahrain)<br>0xD (UAE)\<br>0xE (Qatar)<br>0xF- 0xFF (Not_Used)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request. | 352 | 8  | @1+ | 0 | 1 | 0 | 255   |
| VehicleCfg_Engine                  | 0x0 (Undefined)<br>0x1 (I3_NA)<br>0x2 (I3_TC_or_SC)<br>0x3 (I4_NA)<br>0x4 (I4_TC_or_SC)<br>0x5 (I5_NA)<br>0x6 (I5_TC_or_SC)<br>0x7 (V6_NA)<br>0x8 (V6_TC_or_SC)<br>0x9 (V8_NA)<br>0xA (V8_TC_or_SC)<br>0xB (HEV)<br>0xC (PHEV)<br>0xD (BEV)<br>0xE-0xFF(Not Used)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                  | 360 | 8  | @1+ | 0 | 1 | 0 | 255   |
| VehicleCfg_SteeringRatio           | 0.1 – 25.59, w/ 0.01 resolution<br><br>This is obsolete workitem. Please er updated                                                                                                                                                                                                                                                                                                                          | 368 | 16 | @1+ | 0 | 1 | 0 | 255   |
| VehicleCfg_VIN1                    | Vehicle Identification Number 1                                                                                                                                                                                                                                                                                                                                                                              | 384 | 8  | @1+ | 0 | 1 | 0 | 255   |
| VehicleCfg_VIN2                    | Vehicle Identification Number 2                                                                                                                                                                                                                                                                                                                                                                              | 392 | 8  | @1+ | 0 | 1 | 0 | 255   |
| VehicleCfg_VIN3                    | Vehicle Identification Number 3                                                                                                                                                                                                                                                                                                                                                                              | 400 | 8  | @1+ | 0 | 1 | 0 | 255   |
|                                    |                                                                                                                                                                                                                                                                                                                                                                                                              |     |    |     |   |   |   |       |





|                                         |                                                                                                     |     |   |     |   |   |   |     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|-----|
| VehicleCfg_VIN4                         | Vehicle Identification Number 4                                                                     | 408 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN5                         | Vehicle Identification Number 5                                                                     | 416 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN6                         | Vehicle Identification Number 6                                                                     | 424 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN7                         | Vehicle Identification Number 7                                                                     | 432 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN8                         | Vehicle Identification Number 8                                                                     | 440 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN9                         | Vehicle Identification Number 9                                                                     | 448 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN10                        | Vehicle Identification Number 10                                                                    | 456 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN11                        | Vehicle Identification Number 11                                                                    | 464 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN12                        | Vehicle Identification Number 12                                                                    | 472 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN13                        | Vehicle Identification Number 13                                                                    | 480 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN14                        | Vehicle Identification Number 14                                                                    | 488 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN15                        | Vehicle Identification Number 15                                                                    | 496 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN16                        | Vehicle Identification Number 16                                                                    | 504 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VIN17                        | Vehicle Identification Number 17                                                                    | 512 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_FWC_VehicleType              | Used to configure the FWC-relevant Method 3 parameters                                              | 520 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_DPAC_VehicleType             | Used to configure the DPAC-relevant Method 3 parameters<br>0x00=DPAC_VehicleType                    | 528 | 8 | @1+ | 0 | 1 | 0 | 255 |
| ModuleCfg_DiagResetDelay                | 0-12750mS, with 50mS resolution<br><br>Unit is milli-Second                                         | 536 | 8 | @1+ | 0 | 1 | 0 | 255 |
| VehicleCfg_VehicleType                  | Used to select feature-relevant calibration tables<br>0x00=Undefined                                | 544 | 8 | @1+ | 0 | 1 | 0 | 255 |
| SensorCfg_FWC_Pitch                     | it is an array of 4 elements each of 1 byte.                                                        | 552 | 8 | @1+ | 0 | 1 | 0 | 255 |
| SensorCfg_FWC_Yaw                       | it is an array of 4 elements each of 1 byte.                                                        | 560 | 8 | @1+ | 0 | 1 | 0 | 255 |
| SensorCfg_FWC_Roll                      | it is an array of 4 elements each of 1 byte.                                                        | 568 | 8 | @1+ | 0 | 1 | 0 | 255 |
| CustomerSetting_OSW1OffsetIndex_Pers1   | 0x0 (Off) 0x1 (On)                                                                                  | 576 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_OSW1OffsetIndex_Pers2   | 0x0 (Off) 0x1 (On)                                                                                  | 582 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_OSW1OffsetIndex_Pers3   | 0x0 (Off) 0x1 (On)                                                                                  | 588 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_OSW1OffsetIndex_Pers4   | 0x0 (Off) 0x1 (On)                                                                                  | 594 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_OSW1OffsetIndex_Vehicle | 0x0 (Off) 0x1 (On)                                                                                  | 600 | 6 | @1+ | 0 | 1 | 0 | 64  |
| Padding                                 | #N/A                                                                                                | 606 | 2 | @1+ | 0 | 1 | 0 | 1   |
| CustomerSetting_IACCOffset_Pers1        | Index into IACCOffset array                                                                         | 608 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_IACCOffset_Pers2        | Index into IACCOffset array                                                                         | 614 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_IACCOffset_Pers3        | Index into IACCOffset array                                                                         | 620 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_IACCOffset_Pers4        | Index into IACCOffset array                                                                         | 626 | 6 | @1+ | 0 | 1 | 0 | 64  |
| CustomerSetting_IACCOffset_Vehicle      | Index into IACCOffset array                                                                         | 632 | 6 | @1+ | 0 | 1 | 0 | 64  |
| Padding                                 | #N/A                                                                                                | 638 | 2 | @1+ | 0 | 1 | 0 | 1   |
| FeatureCfg_DPAC_TopViewIconColorIndex   | Index into top view color array 0-15                                                                | 640 | 4 | @1+ | 0 | 1 | 0 | 15  |
| MarketCfg_Region                        | 0x0 (Undefined)<br>0x1 (Eu)<br>0x2 (NA)<br>0x3 (SA)<br>0x4 (APA_China)<br>0x5 (APA)<br>0x6 (Africa) | 644 | 4 | @1+ | 0 | 1 | 0 | 15  |

|                                     |                                                                                                                                                                                                         |     |   |     |   |   |   |    |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|----|
|                                     | 0x7 (GCC)<br>0x8 (Australia_NZ)<br>0x9-0xF (NotUsed)<br><br>For "Not Used" values respond with<br>NRC 0x31 for write DID request.                                                                       |     |   |     |   |   |   |    |
| VehicleCfg_ECEVehicleCategory       | 0x0 = Undefined<br>0x1 = M1<br>0x2 = M2<br>0x3 = M3<br>0x4 = N1<br>0x5 = N2<br>0x6 = N3<br>0x7-0xF = Not Used<br><br>For "Not Used" values respond with<br>NRC 0x31 for write DID request.              | 648 | 4 | @1+ | 0 | 1 | 0 | 15 |
| ModuleFeatureCfg_LKS                | Lane Keeping System<br>0x0 (OFF)<br>0x1 (LKAlert)<br>0x2 (LKAlert+LKAid)<br>0x3 (LKAlert+LKAid+LCWA)<br>0x4-0x7 (Not Used)<br><br>For "Not Used" values respond with<br>NRC 0x31 for write DID request. | 652 | 3 | @1+ | 0 | 1 | 0 | 7  |
| ModuleFeatureCfg_TSR                | Traffic Sign Recognition<br>0x0 (Off)<br>0x1 (SLOIF)<br>0x2 (SLIF)<br>0x3-0x8 (Unused)<br><br>For "Not Used" values respond with<br>NRC 0x31 for write DID request.                                     | 655 | 3 | @1+ | 0 | 1 | 0 | 7  |
| FeatureCfg_TSRMode                  | 0x0 (Undefined) 0x1 (OFF) 0x2<br>(CameraOnlyOn) 0x3 (FusionOn)<br>0x4-0x7 (Not Used)                                                                                                                    | 658 | 3 | @1+ | 0 | 1 | 0 | 7  |
| FeatureCfg_DPAC_ParkAssistType      | 0 = None<br>1 = Four_CH<br>2 = Eight_CH<br>3 = Ten_CH<br>4 = Twelve_CHFLK<br>0x5 – 0x7 = Not Used<br><br>For "Not Used" values respond with<br>NRC 0x31 for write DID request.                          | 661 | 3 | @1+ | 0 | 1 | 0 | 7  |
| Padding                             | #N/A                                                                                                                                                                                                    | 664 | 8 | @1+ | 0 | 1 | 0 | 1  |
| CustomerSetting_ACCGapSetting_Pers1 | 0x0 = Not_Used<br>0x1 = Gap1<br>0x2 = Gap2<br>0x3 = Gap3<br>0x4 = Gap4<br>0x5 = Gap5<br>0x6 = Not_Used<br>0x7 = Not_Used<br><br>For "Not Used" values respond with<br>NRC 0x31 for write DID request.   | 672 | 3 | @1+ | 0 | 1 | 0 | 7  |
| CustomerSetting_ACCGapSetting_Pers2 | 0x0 = Not_Used<br>0x1 = Gap1<br>0x2 = Gap2<br>0x3 = Gap3<br>0x4 = Gap4<br>0x5 = Gap5<br>0x6 = Not_Used<br>0x7 = Not_Used                                                                                | 675 | 3 | @1+ | 0 | 1 | 0 | 7  |

|                                       |                                                                                                                                                                                                    |     |   |     |   |   |   |   |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
|                                       | For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                 |     |   |     |   |   |   |   |
| CustomerSetting_ACCGapSetting_Pers3   | 0x0 = Not_Used<br>0x1 = Gap1<br>0x2 = Gap2<br>0x3 = Gap3<br>0x4 = Gap4<br>0x5 = Gap5<br>0x6 = Not_Used<br>0x7 = Not_Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request. | 678 | 3 | @1+ | 0 | 1 | 0 | 7 |
| CustomerSetting_ACCGapSetting_Pers4   | 0x0 = Not_Used<br>0x1 = Gap1<br>0x2 = Gap2<br>0x3 = Gap3<br>0x4 = Gap4<br>0x5 = Gap5<br>0x6 = Not_Used<br>0x7 = Not_Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request. | 681 | 3 | @1+ | 0 | 1 | 0 | 7 |
| CustomerSetting_ACCGapSetting_Vehicle | 0x0 = Not_Used<br>0x1 = Gap1<br>0x2 = Gap2<br>0x3 = Gap3<br>0x4 = Gap4<br>0x5 = Gap5<br>0x6 = Not_Used<br>0x7 = Not_Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request. | 684 | 3 | @1+ | 0 | 1 | 0 | 7 |
| FeatureCfg_PA_ObjectDetectCamViewSrc  | 0x00 - PAM<br>0x01 - CAM<br>0x02 - R_Gear<br>0x03 - PAM_CAM                                                                                                                                        | 687 | 3 | @1+ | 0 | 1 | 0 | 7 |
| VehicleCfg_APASwitchSrcModule         | 0 = None<br>1 = FCDIM<br>2 = SYNC_Gen2<br>3 = SYNC_Gen3<br>4 = ACM                                                                                                                                 | 690 | 3 | @1+ | 0 | 1 | 0 | 7 |
| VehicleCfg_WindshieldType             | 0x0 (Base)<br>0x1 (Acoustic)<br>0x2 (IR_Coated)<br>0x3-0x7 (NotUsed)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                     | 693 | 3 | @1+ | 0 | 1 | 0 | 7 |
| VehicleCfg_LKSActuator                | 0x0 (HapticMotor)<br>0x1 (EPAS)<br>0x2 (AFS)<br>0x3 (Audio)<br>0x4-0x7 (NotUsed)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                         | 696 | 3 | @1+ | 0 | 1 | 0 | 7 |
| VehicleCfg_HeadlampType               | 0x0 (Halogen)<br>0x1 (Xenon)<br>0x2 (LED)<br>0x3-0x7 (NotUsed)<br><br>For "Not Used" values respond with                                                                                           | 699 | 3 | @1+ | 0 | 1 | 0 | 7 |

|                                  |                                                                                                                                                                                                 |     |   |     |   |   |   |   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
|                                  | NRC 0x31 for write DID request.                                                                                                                                                                 |     |   |     |   |   |   |   |
| Padding                          | #N/A                                                                                                                                                                                            | 702 | 2 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_FcwHUD          | Forward Collision Warning Heads-Up<br>0 = Disabled<br>1 = FCW_HUD<br>2 = Advanced_HUD<br>3 = Not Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                 | 704 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_APA             | 0 = None 1 = SAP 2 = FAP 3 = FAP+RePA                                                                                                                                                           | 706 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_TAD             | Trailer Backing Assist<br>0 = Targetless TAD<br>1 = Target TAD<br>2 = Not Applicable<br>3 = Also Not Applicable<br><br>For "Not Applicable" values respond with NRC 0x31 for write DID request. | 708 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_HighBeam        | Auto High Beam<br>0 = Disabled<br>1 = AHBC<br>2 = GFHB_ADB<br>3 = Not Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                            | 710 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_RBA             | 0 = Disabled 1 = RBA 2 = RPAwBrk 3 = CTAwBrk                                                                                                                                                    | 712 | 2 | @1+ | 0 | 1 | 0 | 4 |
| FeatureCfg_LKSMKey               | 0x0 = Undefined<br>0x1 = OFF<br>0x2 = ON<br>0x3 = Not Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                            | 714 | 2 | @1+ | 0 | 1 | 0 | 4 |
| FeatureCfg_LKSStrategy           | Module configuration Strategy<br>0x0 = Undefined<br>0x1 = Non_EuNCAP<br>0x2 = EuNCAP<br>0x3 = Not Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                | 716 | 2 | @1+ | 0 | 1 | 0 | 4 |
| FeatureCfg_DPAC_RearOverlayType  | 0 = Disabled<br>1 = Static<br>2 = NotUsed<br>3 = Full<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                 | 718 | 2 | @1+ | 0 | 1 | 0 | 4 |
| FeatureCfg_DPAC_FrontOverlayType | 0 = Disabled<br>1 = Static<br>2 = NotUsed<br>3 = Full<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                 | 720 | 2 | @1+ | 0 | 1 | 0 | 4 |
| FeatureCfg_DPAC_RearCamType      | [NONE LIN LVDS]<br>0 = None<br>1 = LIN<br>2 = LVDS<br>3 = Not Used                                                                                                                              | 722 | 2 | @1+ | 0 | 1 | 0 | 4 |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |   |     |   |   |   |   |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
|                                      | For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |   |     |   |   |   |   |
| FeatureCfg_PA_Channels               | 0 = CH4<br>1 = CH8<br>2 = CH12<br>3 = Unused<br><br>For "Unused" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 724 | 2 | @1+ | 0 | 1 | 0 | 4 |
| FeatureCfg_SlaveExistsLINRVC         | 0 = Absent<br>1 = Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 726 | 2 | @1+ | 0 | 1 | 0 | 4 |
| FeatureCfg_SlaveExists_RearCamera2   | This is the CHMSL camera<br>0 = Absent<br>1 = Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 728 | 2 | @1+ | 0 | 1 | 0 | 4 |
| MarketCfg_DrivingSide                | For Stage F and later releases, the enumeration in OBJECTSENSE is OPPOSITE of the enumeration in / SPI messages.<br>Convert the enumerations as shown here: OBJECTSENSE Definition: 0x0 = RightSide 0x1 = LeftSide / Definition: 0x0 = LeftSide 0x1 = RightSide<br><br>Default Value = RightSide<br>Previously the definition was:<br>Side of the road on which the CPYR agent travel lane resides (eg. US = RightSide, UK = LeftSide).<br>0x0 (Undefined)<br>0x1 (RightSide)<br>0x2 (LeftSide)<br>0x3 (NotUsed)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request. | 730 | 2 | @1+ | 0 | 1 | 0 | 4 |
| MarketCfg_SpeedLimitSignUnit         | 0x0 (Undefined)<br>0x1 (KPH)<br>0x2 (MPH)<br>0x3 (Not Used)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 732 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_FCWSensitivity_Pers1 | 0x0 = Not_Used<br>0x1 = FCW_Sensitivity_1 (Short Setting)<br>0x2 = FCW_Sensitivity_2 (Normal Setting)<br>0x3 = FCW_Sensitivity_3 (Long Setting)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                  | 734 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_FCWSensitivity_Pers2 | 0x0 = Not_Used<br>0x1 = FCW_Sensitivity_1 (Short Setting)<br>0x2 = FCW_Sensitivity_2 (Normal Setting)<br>0x3 = FCW_Sensitivity_3 (Long Setting)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                  | 736 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_FCWSensitivity_Pers3 | 0x0 = Not_Used<br>0x1 = FCW_Sensitivity_1 (Short Setting)<br>0x2 = FCW_Sensitivity_2 (Normal Setting)<br>0x3 = FCW_Sensitivity_3 (Long Setting)<br><br>For "Not Used" values respond with                                                                                                                                                                                                                                                                                                                                                                                                  | 738 | 2 | @1+ | 0 | 1 | 0 | 4 |

|                                          |                                                                                                                                                                                                                           |     |   |     |   |   |   |   |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
|                                          | NRC 0x31 for write DID request.                                                                                                                                                                                           |     |   |     |   |   |   |   |
| CustomerSetting_FCWSensitivity_Pers4     | 0x0 = Not_Used<br>0x1 = FCW_Sensitivity_1 (Short Setting)<br>0x2 = FCW_Sensitivity_2 (Normal Setting)<br>0x3 = FCW_Sensitivity_3 (Long Setting)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request. | 740 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_FCWSensitivity_Vehicle   | 0x0: Not_Used<br>FCW_Sensitivity_1 (0x1): Short Setting<br>FCW_Sensitivity_2 (0x2): Normal Setting<br>FCW_Sensitivity_3 (0x3): Long Setting                                                                               | 742 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKAlertIntensity_Pers1   | 0x0 = OFF<br>0x1 = Low<br>0x2 = Medium<br>0x3 = High                                                                                                                                                                      | 744 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKAlertIntensity_Pers2   | 0x0 = OFF<br>0x1 = Low<br>0x2 = Medium<br>0x3 = High                                                                                                                                                                      | 746 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKAlertIntensity_Pers3   | 0x0 = OFF<br>0x1 = Low<br>0x2 = Medium<br>0x3 = High                                                                                                                                                                      | 748 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKAlertIntensity_Pers4   | 0x0 = OFF<br>0x1 = Low<br>0x2 = Medium<br>0x3 = High                                                                                                                                                                      | 750 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKAlertIntensity_Vehicle | 0x0 = OFF<br>0x1 = Low<br>0x2 = Medium<br>0x3 = High                                                                                                                                                                      | 752 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSSetting_Pers1         | Lane Keeping System Mode Selection<br>0x0 = Not_Used<br>0x1 = LKAlert<br>0x2 = LKAid<br>0x3 = LKAlert+LKAid<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                     | 754 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSSetting_Pers2         | Lane Keeping System Mode Selection<br>0x0 = Not_Used<br>0x1 = LKAlert<br>0x2 = LKAid<br>0x3 = LKAlert+LKAid<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                     | 756 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSSetting_Pers3         | Lane Keeping System Mode Selection<br>0x0 = Not_Used<br>0x1 = LKAlert<br>0x2 = LKAid<br>0x3 = LKAlert+LKAid<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                     | 758 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSSetting_Pers4         | Lane Keeping System Mode Selection<br>0x0 = Not_Used<br>0x1 = LKAlert<br>0x2 = LKAid<br>0x3 = LKAlert+LKAid                                                                                                               | 760 | 2 | @1+ | 0 | 1 | 0 | 4 |

|                                         |                                                                                                                                                                                       |     |   |     |   |   |   |   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
|                                         | For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                    |     |   |     |   |   |   |   |
| CustomerSetting_LKSSetting_Vehicle      | Lane Keeping System Mode Selection<br>0x0 = Not_Used<br>0x1 = LKAlert<br>0x2 = LKAid<br>0x3 = LKAlert+LKAid<br><br>For "Not Used" values respond with NRC 0x31 for write DID request. | 762 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSLKAMode_Pers1        | 0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)                                                                                                                                       | 764 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSLKAMode_Pers2        | 0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)                                                                                                                                       | 766 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSLKAMode_Pers3        | 0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)                                                                                                                                       | 768 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSLKAMode_Pers4        | 0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)                                                                                                                                       | 770 | 2 | @1+ | 0 | 1 | 0 | 4 |
| CustomerSetting_LKSLKAMode_Vehicle      | 0x0 (OFF) 0x1 (ON) 0x2 (Reduced) 0x3 (Enhanced)                                                                                                                                       | 772 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_PA_LED                       | 0 = Yes<br>1 = No<br>2 = Dimmable<br>3 = Not Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                           | 774 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_RBAAAlertType          | 0 = Off<br>1 = Graphic<br>2 = Text<br>3 = Both                                                                                                                                        | 776 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_FuelType                     | 0 = Gasoline<br>1 = Diesel<br>2 = Hybrid<br>3 = NotUsed<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                     | 778 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_PA_UseWheelTicks       | 0 = Four<br>1 = Two<br>2 = Zero                                                                                                                                                       | 780 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_LksPcaSwitch                 | 0 = Hardwire_LKS<br>1 = Hardwire_PCA<br>2 = NoHardwareSwitchPresent<br>3 = Not Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                         | 782 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_ApaBaSwitch                  | 0 = Hardwire_APA<br>1 = Network_APA<br>2 = NoSwitchPresent<br>3 = Hardwire_BA                                                                                                         | 784 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_CANPA_SwitchCPYR agent | 0 = FCIMA<br>1 = FCIMB<br>2 = NONE<br>3 = ATCM                                                                                                                                        | 786 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_LaneAssist_SwType            | VehicleCfg_LaneAssist_SwType                                                                                                                                                          | 788 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_PrkAidSwitch                 | 0 = HARDWIRE<br>1 = SOFT<br>2 = CAN<br>3 = NONE                                                                                                                                       | 790 | 2 | @1+ | 0 | 1 | 0 | 4 |
|                                         |                                                                                                                                                                                       |     |   |     |   |   |   |   |



|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |   |     |   |   |   |   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
| VehicleCfg_PCAType              | 0x0 = (OFF)<br>0x1 = (RadarFusion)<br>0x2 = (CameraOnly)<br>0x3 = (NotUsed)<br>For "Not Used" values respond with NRC 0x31 for write DID request.<br><br>For Stage-F, Please er below values 0x0<br>= OFF<br>0x1 = RadarFusion<br>0x2 = CameraOnly<br>0x3 = FrontCornerRadarFusion<br><br>For Stage-G, Please er below values 0x0<br>= OFF<br>0x1 = CameraOnly_52deg<br>0x2 = CameraOnly_100deg<br>0x3 = RadarFusion_52deg<br>0x4 = RadarFusion_100deg<br>0x5 = FrontCornerRadarFusion_52deg<br>0x6 = FrontCornerRadarFusion_100deg<br>0x7= NotUsed | 792 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_ACCType              | 0x0 (OFF)<br>0x1 (RadarFusion)<br>0x2-0x3 (Not_Used)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                                                                      | 794 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_GearShiftByWire      | 0 = Not_By_Wire<br>1 = Shift_By_Wire<br>2 = Range_By_Wire<br>3 = Not Used<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                                                 | 796 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_WindshieldHeaterType | 0x0 = (Direct)<br>0x1 = (Network)<br>0x2 = (NoHeaterPresent)<br>0x3 = (NotUsed)<br><br>For "Not Used" values respond with NRC 0x31 for write DID request.                                                                                                                                                                                                                                                                                                                                                                                           | 798 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_DCUIndicator         | 0x0 = None<br>0x1 = Driver Door<br>0x2 = Passenger Door<br>0x3 = Both                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 800 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleCfg_StartStop            | 0 = Not Present<br>1 = Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 802 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleInfo_TowBar              | 0 = Tow_Bar_Not_Present<br>1 = Tow_Bar_Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 804 | 2 | @1+ | 0 | 1 | 0 | 4 |
| VehicleInfo_TrailerTowPresent   | 0 = No<br>1 = Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 806 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_CamSNCheck     | 0 = DISABLED<br>1 = ENABLED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 808 | 2 | @1+ | 0 | 1 | 0 | 4 |
| ModuleFeatureCfg_DistAlert      | Distance Alert<br>0 = Disabled<br>1 = Enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 810 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_DAS            | Driver Alert System<br>0 = Disabled<br>1 = Enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 811 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_IACC           | Intelligent Ade Cruise Control 0 = Disabled<br>1 = Enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 812 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_TJA            | 0x0 (Disabled) 0x1 (Enabled)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 813 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                     |                                                                                                                                                                                            |     |   |     |   |   |   |   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
| ModuleFeatureCfg_WWA                | Wrong-Way Alert<br>0 = Disabled<br>1 = Enabled                                                                                                                                             | 814 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_SBLM               | Sign-Based Light Modes<br>0 = Disabled<br>1 = Enabled                                                                                                                                      | 815 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_SRP                | Suspension Road Preview<br>0 = Disabled<br>1 = Enabled                                                                                                                                     | 816 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_BTT                | 0x0 (Disabled) 0x1 (Enabled)                                                                                                                                                               | 817 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_ATD                | 0x0 (Disabled) 0x1 (Enabled)                                                                                                                                                               | 818 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_ODCV               | 0 = Disabled<br>1 = Enabled                                                                                                                                                                | 819 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_BLIS               | 0x0 (Enabled) 0x1 (Disabled)                                                                                                                                                               | 820 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_CTA                | 0x0 (Enabled) 0x1 (Disabled)                                                                                                                                                               | 821 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_BoundaryAlert      | 0 = Disabled<br>1 = Enabled<br><br>Variable Name<br>For Stage-D =<br>ModuleFeatureCfg_BoundaryAlert<br>For Stage-E = ModuleFeatureCfg_I<br>For Stage-F =<br>ModuleFeatureCfg_BoundaryAlert | 822 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_VOD                | 0x0 (Disabled) 0x1 (Enabled)                                                                                                                                                               | 823 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_AH                 | 0x0 (Disabled) 0x1 (Enabled)                                                                                                                                                               | 824 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_TrailerAssist      | Trailer Assist<br>0 = Disabled<br>1 = Enabled                                                                                                                                              | 825 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_HwyAssist          | 0x0 (Disabled) 0x1 (Enabled)                                                                                                                                                               | 826 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_VideoOverEther     | 0x0 (Disabled) 0x1 (Enabled)                                                                                                                                                               | 827 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_VideoOutput        | 0x0 (Analog) 0x1 (Digital)                                                                                                                                                                 | 828 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_WrdRemoteCamera     | Wired Remote Camera<br>0 = Absent<br>1 = Present                                                                                                                                           | 829 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_OffroadFrontCam     | Off-Road Front Camera Behavior<br>0 = Disabled<br>1 = Enabled                                                                                                                              | 830 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_LowPowerMode        | Low Power Mode for 360° cameras<br>0 = Disabled<br>1 = Enabled                                                                                                                             | 831 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_ParkAssistDisplay   | Visual Park Aid in DPAC<br>0 = Disabled<br>1 = Enabled                                                                                                                                     | 832 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_DoorAjarInd         | 0 = Disabled<br>1 = Enabled                                                                                                                                                                | 833 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_TailgateAjarInd     | 0 = Disabled<br>1 = Enabled                                                                                                                                                                | 834 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_LongTermCalibration | Enable the long-term Adjustment<br>0 = Disabled<br>1 = Enabled                                                                                                                             | 835 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_360Present          | 0 = False<br>1 = True                                                                                                                                                                      | 836 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_FrontCamPresent     | 0 = False<br>1 = True                                                                                                                                                                      | 837 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                        |                                                                                                                         |     |   |     |   |   |   |   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
| FeatureCfg_DPAC_SideCamsPresent        | 0 = False<br>1 = True                                                                                                   | 838 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_RearSplit              | 0 = Off<br>1 = On                                                                                                       | 839 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_TrailerAssistType           | Trailer Backing vs. Reverse guidance<br>0 = TBA<br>1 = TRG Only                                                         | 840 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_DirtyLensDetect        | Enable Dirty Lens Detection<br>0 = Disabled<br>1 = Enabled                                                              | 841 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_WrdRemoteCameraInd     | Control display of Aux Camera<br>Disconnected Icon<br>0 = Disabled<br>1 = Enabled                                       | 842 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_APAF_6pt_Fusion             | 0 = Disabled<br>1 = Enabled                                                                                             | 843 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_LKSignOnStrategy            | Lane keep Ignition On Strategy<br>0 = Off<br>1 = On                                                                     | 844 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BLIS_Left_DCU_Cfg           | 0x0 (Hardwire) 0x1 (Network)                                                                                            | 845 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BLIS_Right_DCU_Cfg          | 0x0 (Hardwire) 0x1 (Network)                                                                                            | 846 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_OffsetView_Enable_Cfg  | 0x0 (Disabled) 0x1 (Enabled)                                                                                            | 847 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_DPAC_CornerView_Enable_Cfg  | 0x0 (Disabled) 0x1 (Enabled)                                                                                            | 848 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_FWC_CameraFoV               | FWC camera type connected with ECU.<br>0x0 = Normal<br>0x1 = Wide<br>Note : is same as this Data-dictionary<br>variable | 849 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BA_Locking                  | 0x0 (Disabled) 0x1 (Enabled)                                                                                            | 850 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BA_Windows                  | 0x0 (Disabled) 0x1 (Enabled)                                                                                            | 851 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BA_UserMode                 | 0x0 (Disabled) 0x1 (Enabled)                                                                                            | 852 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BA_DefaultMode              | 0x0 (Disabled) 0x1 (Enabled)                                                                                            | 853 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BLIS_RearRadar              | 0x0 (Absent) 0x1 (Present)                                                                                              | 854 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_BLIS_FrontRadar             | 0x0 (Absent) 0x1 (Present)                                                                                              | 855 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_CCM_CenterRadar             | 0x0 (Absent) 0x1 (Present)                                                                                              | 856 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_SteWhlSide                  | Side of the CPYR agent vehicle on which<br>the steering wheel resides<br>0 = Lefthand_Drive<br>1 = Righthand_Drive      | 857 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWOnOff_Pers1         | 0x0 = Off<br>0x1 = On                                                                                                   | 858 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWOnOff_Pers2         | 0x0 = Off<br>0x1 = On                                                                                                   | 859 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWOnOff_Pers3         | 0x0 = Off<br>0x1 = On                                                                                                   | 860 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWOnOff_Pers4         | 0x0 = Off<br>0x1 = On                                                                                                   | 861 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWOnOff_Vehicle       | 0x0 = Off<br>0x1 = On                                                                                                   | 862 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWDistanceOnOff_Pers1 | 0x0 = Off<br>0x1 = On                                                                                                   | 863 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWDistanceOnOff_Pers2 | 0x0 = Off<br>0x1 = On                                                                                                   | 864 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWDistanceOnOff_Pers3 | 0x0 = Off                                                                                                               | 865 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                          |                                                    |     |   |     |   |   |   |   |
|------------------------------------------|----------------------------------------------------|-----|---|-----|---|---|---|---|
|                                          | 0x1 = On                                           |     |   |     |   |   |   |   |
| CustomerSetting_FCWDistanceOnOff_Pers4   | 0x0 = Off<br>0x1 = On                              | 866 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_FCWDistanceOnOff_Vehicle | 0x0 = Off<br>0x1 = On                              | 867 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCSelected_Pers1        | Ade Vs Manual Cruise Control 0x0 = No<br>0x1 = Yes | 868 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCSelected_Pers2        | 0x0 = No<br>0x1 = Yes                              | 869 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCSelected_Pers3        | 0x0 = No<br>0x1 = Yes                              | 870 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCSelected_Pers4        | 0x0 = No<br>0x1 = Yes                              | 871 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCSelected_Vehicle      | 0x0 = No<br>0x1 = Yes                              | 872 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSONOff_Pers1           | 0x0 Off<br>0x1 On                                  | 873 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSONOff_Pers2           | 0x0 Off<br>0x1 On                                  | 874 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSONOff_Pers3           | 0x0 Off<br>0x1 On                                  | 875 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSONOff_Pers4           | 0x0 Off<br>0x1 On                                  | 876 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSONOff_Vehicle         | 0x0 Off<br>0x1 On                                  | 877 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSSensitivity_Pers1     | 0x0 Normal<br>0x1 Level2                           | 878 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSSensitivity_Pers2     | 0x0 (Normal) 0x1 (Level2)                          | 879 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSSensitivity_Pers3     | 0x0 Normal<br>0x1 Level2                           | 880 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSSensitivity_Pers4     | 0x0 (Normal) 0x1 (Level2)                          | 881 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LKSSensitivity_Vehicle   | 0x0 Normal<br>0x1 Level2                           | 882 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LCAsSelected_Pers1       | 0x0 (Off) 0x1 (On)                                 | 883 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LCAsSelected_Pers2       | 0x0 (Off) 0x1 (On)                                 | 884 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LCAsSelected_Pers3       | 0x0 (Off) 0x1 (On)                                 | 885 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LCAsSelected_Pers4       | 0x0 (Off) 0x1 (On)                                 | 886 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_LCAsSelected_Vehicle     | 0x0 (Off) 0x1 (On)                                 | 887 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_AHBOOnOff_Pers1          | 0x0 = Off<br>0x1 = On                              | 888 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_AHBOOnOff_Pers2          | 0x0 = Off<br>0x1 = On                              | 889 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_AHBOOnOff_Pers3          | 0x0 = Off<br>0x1 = On                              | 890 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_AHBOOnOff_Pers4          | 0x0 = Off<br>0x1 = On                              | 891 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_AHBOOnOff_Vehicle        | 0x0 = Off<br>0x1 = On                              | 892 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_DASOnOff_Pers1           | 0x0 = Off<br>0x1 = On                              | 893 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_DASOnOff_Pers2           | 0x0 = Off<br>0x1 = On                              | 894 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                          |                                                    |     |   |     |   |   |   |   |

|                                              |                                                            |     |   |     |   |   |   |   |
|----------------------------------------------|------------------------------------------------------------|-----|---|-----|---|---|---|---|
| CustomerSetting_DASOnOff_Pers3               | 0x0 = Off<br>0x1 = On                                      | 895 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_DASOnOff_Pers4               | 0x0 = Off<br>0x1 = On                                      | 896 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_DASOnOff_Vehicle             | 0x0 = Off<br>0x1 = On                                      | 897 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_WWAOnOff_Pers1               | 0x0 = Off<br>0x1 = On                                      | 898 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_WWAOnOff_Pers2               | 0x0 = Off<br>0x1 = On                                      | 899 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_WWAOnOff_Pers3               | 0x0 = Off<br>0x1 = On                                      | 900 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_WWAOnOff_Pers4               | 0x0 = Off<br>0x1 = On                                      | 901 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_WWAOnOff_Vehicle             | 0x0 = Off<br>0x1 = On                                      | 902 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TSRChimeOnOff_Pers1          | 0x0 (Off) 0x1 (On)                                         | 903 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TSRChimeOnOff_Pers2          | 0x0 (Off) 0x1 (On)                                         | 904 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TSRChimeOnOff_Pers3          | 0x0 (Off) 0x1 (On)                                         | 905 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TSRChimeOnOff_Pers4          | 0x0 (Off) 0x1 (On)                                         | 906 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TSRChimeOnOff_Vehicle        | 0x0 (Off) 0x1 (On)                                         | 907 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_OSWOnOff_Pers1               | 0x0 (Off) 0x1 (On)                                         | 908 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_OSWOnOff_Pers2               | 0x0 (Off) 0x1 (On)                                         | 909 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_OSWOnOff_Pers3               | 0x0 (Off) 0x1 (On)                                         | 910 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_OSWOnOff_Pers4               | 0x0 (Off) 0x1 (On)                                         | 911 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_OSWOnOff_Vehicle             | 0x0 (Off) 0x1 (On)                                         | 912 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCAutoSetSpeedOnOff_Pers1   | ACC On/Off Setting<br>0x0 = Off<br>0x1 = On                | 913 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCAutoSetSpeedOnOff_Pers2   | 0x0 = Off<br>0x1 = On                                      | 914 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCAutoSetSpeedOnOff_Pers3   | 0x0 = Off<br>0x1 = On                                      | 915 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCAutoSetSpeedOnOff_Pers4   | 0x0 = Off<br>0x1 = On                                      | 916 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_ACCAutoSetSpeedOnOff_Vehicle | 0x0 Off<br>0x1 On                                          | 917 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TJAOnOff_Pers1               | Traffic Jam Assist On/Off Setting<br>0x0 = Off<br>0x1 = On | 918 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TJAOnOff_Pers2               | Traffic Jam Assist On/Off Setting<br>0x0 = Off<br>0x1 = On | 919 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TJAOnOff_Pers3               | Traffic Jam Assist On/Off Setting<br>0x0 = Off<br>0x1 = On | 920 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TJAOnOff_Pers4               | Traffic Jam Assist On/Off Setting<br>0x0 = Off<br>0x1 = On | 921 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_TJAOnOff_Vehicle             | Traffic Jam Assist On/Off Setting<br>0x0 = Off<br>0x1 = On | 922 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_SoSelected_Pers1             | 0x0 (Off) 0x1 (On)                                         | 923 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_SoSelected_Pers2             | 0x0 (Off) 0x1 (On)                                         | 924 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                             |                                               |     |   |     |   |   |   |   |
|---------------------------------------------|-----------------------------------------------|-----|---|-----|---|---|---|---|
| CustomerSetting_SoSelected_Pers3            | 0x0 (Off) 0x1 (On)                            | 925 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_SoSelected_Pers4            | 0x0 (Off) 0x1 (On)                            | 926 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CustomerSetting_SoSelected_Vehicle          | 0x0 (Off) 0x1 (On)                            | 927 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_LineBasedParking              | 0 = Disabled<br>1 = Enabled                   | 928 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_DrvTube                       | 0 = Disabled<br>1 = Enabled                   | 929 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_VisualDrvTube                 | 0 = Disabled<br>1 = Enabled                   | 930 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_Front                   | Enable Front Park Assist<br>0 = Off<br>1 = On | 931 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_Rear                    | Enable Rear Park Assist<br>0 = Off<br>1 = On  | 932 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_MapTracking                   | 0 = Off<br>1 = On                             | 933 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_MyKey                         | 0 = Disabled<br>1 = Enabled                   | 934 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_SIPa                          | 0 = Off<br>1 = On                             | 935 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_SIPe                          | 0 = Off<br>1 = On                             | 936 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_SnsBlockDetect                | 0 = Disabled<br>1 = Enabled                   | 937 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_TTC_Front_Warn                | 0 = Disabled<br>1 = Enabled                   | 938 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_TTC_Rear_Warn                 | 0 = Disabled<br>1 = Enabled                   | 939 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_TTC_Side_Warn                 | 0 = Disabled<br>1 = Enabled                   | 940 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_SlotFoundSuppression    | 0x0 (Disable)<br>0x1 (Enable)                 | 941 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_FilterActive                  | 0 = Off<br>1 = On                             | 942 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FeatureCfg_PA_WaitForSteering_Display       | 0 = Disabled<br>1 = Enabled                   | 943 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_BidirectionalWhlSensor           | 0 = Not Present<br>1 = Present                | 944 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_DriveType                        | 0 = FWD<br>1 = RWD                            | 945 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_FL_WheelTick_Standstill | 0 = Not Present<br>1 = Present                | 946 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_FR_WheelTick_Standstill | 0 = Not Present<br>1 = Present                | 947 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_RL_WheelTick_Standstill | 0 = Not Present<br>1 = Present                | 948 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_RR_WheelTick_Standstill | 0 = Not Present<br>1 = Present                | 949 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_SOPa                    | 0 = Off<br>1 = On                             | 950 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_SPA                     | 0 = Off<br>1 = On                             | 951 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_VehCondCalc             | 0 = Disabled                                  | 952 | 1 | @1+ | 0 | 1 | 0 | 1 |



|                                      |                                                                              |     |   |     |   |   |   |   |
|--------------------------------------|------------------------------------------------------------------------------|-----|---|-----|---|---|---|---|
|                                      | 1 = Enabled                                                                  |     |   |     |   |   |   |   |
| ModuleFeatureCfg_PA_Lim_BlK_Fun      | 0 = Disabled<br>1 = Enabled                                                  | 953 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_WhlSensorAxle_Cfg         | 0x0 (Front) 0x1 (Rear)                                                       | 954 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_TrllrLghtModulePresent    | 0 = Not Present<br>1 = Present                                               | 955 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_TrllrBrkModulePresent     | 0 = Not Present<br>1 = Present                                               | 956 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_AutoTransStyle            | 0 = Oldstyle<br>1 = Newstyle                                                 | 957 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_ParkBrkType               | 0 = Manual<br>1 = EPB                                                        | 958 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_TransmissionType          | 0 = Automatic<br>1 = Manual                                                  | 959 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_DPAC_FrontCameraSw        | 0 = Network<br>1 = Hardwire                                                  | 960 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_OATSignalCPYR agent | 0 = FCIMA<br>1 = HVAC                                                        | 961 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleFeatureCfg_PA_Hotkey           | 0 = Not Present<br>1 = Present                                               | 962 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_ActiveSuspension          | 0x0 = Not Present<br>0x1 = Present                                           | 963 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_APAAInterface             | 0 = Oldstyle<br>1 = Newstyle                                                 | 964 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleInfo_TrailerLightModule       | 0x0 (Not Present) 0x1 (Present)                                              | 965 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleInfo_TrailerBrakeModule       | 0x0 (Not Present) 0x1 (Present)                                              | 966 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DPAC_EOLAlignmentComplete_Cfg        | 0x0 (False) 0x1 (True)                                                       | 967 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DPAC_SVCAAlignmentComplete_Cfg       | 0x0 (False) 0x1 (True)                                                       | 968 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TADAlignmentComplete_Cfg             | It represents the Alignment completion status of the Trailer Camera i.e. TAD | 969 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehicleCfg_CCM_VehicleType           | Used to select CCM feature-relevant calibration tables                       | 970 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TrailerAngleProtectionBit            | 0x0 (NULL) 0x1 (One_bit)                                                     | 971 | 1 | @1+ | 0 | 1 | 0 | 1 |

## Veh\_Estimation\_Compensation

[Veh\\_Estimation\\_Compensation-STNumber\(12\)-STVersion\(7\)](#)

Message Veh\_Estimation\_Compensation shall be implemented as follows:

| SignalName             | Value Type                | Start Bit | Length |
|------------------------|---------------------------|-----------|--------|
| timestamp_us           | unsigned long<br>long int | 0         | 64     |
| veh_index              | unsigned long<br>int      | 64        | 32     |
| raw_speed_mps          | float                     | 96        | 32     |
| raw_yaw_rate_rps       | float                     | 128       | 32     |
| raw_lat_accel          | float                     | 160       | 32     |
| raw_long_accel         | float                     | 192       | 32     |
| raw_steering_angle_deg | float                     | 224       | 32     |



|                            |                    |     |    |
|----------------------------|--------------------|-----|----|
| comp_steering_angle_deg    | float              | 256 | 32 |
| comp_yaw_rate_unfiltered   | float              | 288 | 32 |
| accel_ped_pos_pct          | float              | 320 | 32 |
| comp_yaw_rate_filtered     | float              | 352 | 32 |
| yaw_rate_bias              | float              | 384 | 32 |
| filt_veh_speed_over_ground | float              | 416 | 32 |
| sensor_sideslip            | float              | 448 | 32 |
| sensor_long_velocity       | float              | 480 | 32 |
| sensor_lat_velocity        | float              | 512 | 32 |
| sideslip_rear_axle         | float              | 544 | 32 |
| vcs_lat_velocity           | float              | 576 | 32 |
| curvature_rear_axle        | float              | 608 | 32 |
| yaw_rate_sa                | float              | 640 | 32 |
| vcs_sideslip               | float              | 672 | 32 |
| vcs_long_velocity          | float              | 704 | 32 |
| raw_speed_qf               | unsigned char      | 736 | 8  |
| raw_yaw_rate_qf            | unsigned char      | 744 | 8  |
| raw_lat_accel_qf           | unsigned char      | 752 | 8  |
| raw_long_accel_qf          | unsigned char      | 760 | 8  |
| wheel_slip                 | unsigned char      | 768 | 8  |
| comp_yaw_rate_qf           | enum               | 776 | 8  |
| comp_steering_angle_qf     | unsigned char      | 784 | 8  |
| yaw_rate_bias_qf           | enum               | 792 | 8  |
| yaw_rate_sa_qf             | unsigned char      | 800 | 8  |
| stationary                 | unsigned char      | 808 | 8  |
| reverse_status             | unsigned char      | 816 | 8  |
| high_beam_status           | unsigned char      | 824 | 8  |
| front_wiper_cmd            | unsigned char      | 832 | 8  |
| brake_pedal_status         | unsigned char      | 840 | 8  |
| turn_signal_status         | unsigned char      | 848 | 8  |
| acc_braking_status         | unsigned char      | 856 | 8  |
| wiper_speed                | unsigned char      | 864 | 8  |
| validity_flags             | unsigned char      | 872 | 8  |
| padding                    | unsigned short int | 880 | 16 |
|                            |                    |     |    |

|                           |               |      |    |
|---------------------------|---------------|------|----|
| StePinComp_An_Est         | float         | 896  | 32 |
| StePinRelInit_An_Sns      | float         | 928  | 32 |
| VehLat2_A_Actl            | float         | 960  | 32 |
| VehLong2_A_Actl           | float         | 992  | 32 |
| VehOverGnd_V_Est          | float         | 1024 | 32 |
| VehYaw_W_Actl             | float         | 1056 | 32 |
| ApedPos_Pc_ActlArb        | float         | 1088 | 32 |
| GearLvrPos_D_Actl         | float         | 1120 | 8  |
| GearRvrse_D_Actl          | float         | 1128 | 8  |
| StePinCompAnEst_D_Qf      | unsigned char | 1136 | 8  |
| VehLatAActl_D_Qf          | unsigned char | 1144 | 8  |
| VehLongAActl_D_Qf         | unsigned char | 1152 | 8  |
| VehYawWActl_D_Qf          | unsigned char | 1160 | 8  |
| ApedPosPcActl_D_Qf        | unsigned char | 1168 | 8  |
| WhlDirRl_D_Actl           | unsigned char | 1176 | 8  |
| WhlDirRr_D_Actl           | unsigned char | 1184 | 8  |
| WhlDirFl_D_Actl           | unsigned char | 1192 | 8  |
| WhlDirFr_D_Actl           | unsigned char | 1200 | 8  |
| VehStop_D_Stat            | unsigned char | 1208 | 8  |
| TracCtlPtActv_B_Actl      | unsigned char | 1216 | 8  |
| AbsActv_B_Actl            | unsigned char | 1224 | 8  |
| AccBrkTotTqMn_B_Actl      | unsigned char | 1232 | 8  |
| HeadLghtHiOn_B_Stat       | unsigned char | 1240 | 8  |
| BpedDrvAppl_D_Actl        | unsigned char | 1248 | 8  |
| TurnLghtSwch_D_Stat       | unsigned char | 1256 | 8  |
| WiprFront_D_Stat          | unsigned char | 1264 | 8  |
| VehicleStationary         | unsigned char | 1272 | 8  |
| periodic_thread_proc      | unsigned char | 1280 | 32 |
| periodic_thread_max       | unsigned int  | 1312 | 32 |
| maxThreadPeriodicity      | unsigned int  | 1344 | 32 |
| Latency_Veh_Velocity_time | unsigned int  | 1376 | 32 |

|                                         |                    |      |    |
|-----------------------------------------|--------------------|------|----|
| Latency_Yaw_Rate_time                   | unsigned int       | 1408 | 32 |
| threadPeriodicityErrorCount             | unsigned int       | 1440 | 16 |
| padding                                 | unsigned short int | 1456 | 16 |
| YawRateBias_Compensation_Major          | unsigned short int | 1472 | 32 |
| YawRateBias_Compensation_Minor          | float              | 1504 | 32 |
| YawRateBias_Compensation_Field          | float              | 1536 | 32 |
| yaw_rate_estimation_major               | float              | 1568 | 32 |
| yaw_rate_estimation_minor               | int                | 1600 | 32 |
| yaw_rate_estimation_field               | int                | 1632 | 32 |
| curvature_and_sideslip_estimation_Major | int                | 1664 | 8  |
| curvature_and_sideslip_estimation_Minor | unsigned char      | 1672 | 8  |
| curvature_and_sideslip_estimation_Field | unsigned char      | 1680 | 8  |

## Active Fault Status

Active\_Fault\_Status-STNumber(75)-STVersion(1) This message is used to report fault status information.

Message Active\_Fault\_Status shall be implemented as fault table defined in [5].

| Signal Name                                                | Description | StartBit | Size | Sign | Offset | Factor | Min | Max |
|------------------------------------------------------------|-------------|----------|------|------|--------|--------|-----|-----|
| VisionCo_3_3V_Supply_Out_Of_Range                          | 0           | 0        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_8V_Supply_Out_Of_Range                          | 0.1         | 1        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_1V_Supply_Out_Of_Range                          | 0.2         | 2        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_0V_Supply_Out_Of_Range                          | 0.3         | 3        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_3_3V_Supply_Out_Of_Range                                | 0.4         | 4        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_1_8V_Supply_Out_Of_Range                                | 0.5         | 5        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_1_35V_Supply_Out_Of_Range                               | 0.6         | 6        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_1_0V_Supply_Out_Of_Range                                | 0.7         | 7        | 1    | @1+  | 0      | 1      | 0   | 1   |
| IMAGE2P_PGOOD_Fail_Fault                                   | 1           | 8        | 1    | @1+  | 0      | 1      | 0   | 1   |
| FVC_CAMPWR_LDO_FailFRONT_CAMERA_FAULT_CKT_ShortToGnd       | 1.1         | 9        | 1    | @1+  | 0      | 1      | 0   | 1   |
| RVC_CAMPWR_LDO_FailREAR_CAMERA_FAULT_CKT_ShortToGnd        | 1.2         | 10       | 1    | @1+  | 0      | 1      | 0   | 1   |
| LH_Mirror_CAMPWR_LDO_FailLEFT_CAMERA_FAULT_CKT_ShortToGnd  | 1.3         | 11       | 1    | @1+  | 0      | 1      | 0   | 1   |
| RH_Mirror_CAMPWR_LDO_FailRIGHT_CAMERA_FAULT_CKT_ShortToGnd | 1.4         | 12       | 1    | @1+  | 0      | 1      | 0   | 1   |
| FWC_CAMPWR_LDO_FailFWC_CAMERA_FAULT_CKT_ShortToGnd         | 1.5         | 13       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 1.6         | 14       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 1.7         | 15       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 2           | 16       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 2.1         | 17       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 2.2         | 18       | 1    | @1+  | 0      | 1      | 0   | 1   |
| MAINRADAR_Output_Open_Circuit                              | 2.3         | 19       | 1    | @1+  | 0      | 1      | 0   | 1   |
| FET_Fault_Across_cycles                                    | 2.4         | 20       | 1    | @1+  | 0      | 1      | 0   | 1   |
| FET_Fail_Across_Cycles                                     | 2.5         | 21       | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_SerialFlash_PowerUp_Fail                                | 2.6         | 22       | 1    | @1+  | 0      | 1      | 0   | 1   |



|                                             |     |    |   |     |   |   |   |   |
|---------------------------------------------|-----|----|---|-----|---|---|---|---|
| System_Over_Temperature                     | 2.7 | 23 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_On_Board_Thermistor_Over_Temp    | 3   | 24 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_On_Board_Thermistor_Over_Temp      | 3.1 | 25 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_On_Board_Thermistor_Over_Temp            | 3.2 | 26 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_On_Board_Thermistor_Over_Temp       | 3.3 | 27 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_MIPS_Over_Temp                     | 3.4 | 28 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_VMP_Over_Temp                      | 3.5 | 29 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Imager_Over_Temp                   | 3.6 | 30 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Boot_OverTemp_Fault                | 3.7 | 31 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_DDR_Over_Temp                      | 4   | 32 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_On_Board_Thermistor_Out_Of_Range | 4.1 | 33 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_On_Board_Thermistor_Out_Of_Range   | 4.2 | 34 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_On_Board_Thermistor_Out_Of_Range    | 4.3 | 35 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_On_Board_Thermistor_Out_Of_Range         | 4.4 | 36 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BatteryLowFault                             | 4.5 | 37 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                    | 4.6 | 38 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                    | 4.7 | 39 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_AURIX_Fault                      | 5   | 40 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_AURIX_Fault                      | 5.1 | 41 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                    | 5.2 | 42 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Short_Across_Cycles                  | 5.3 | 43 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Output_Short_Circuit                 | 5.4 | 44 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Output_Open_Circuit                  | 5.5 | 45 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Short_Fail_Across_Cycles             | 5.6 | 46 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CHMSL_camera_DEC_amplifier_ShortToBatt      | 5.7 | 47 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AUX_camera_DEC_amplifier_ShortToBatt        | 6   | 48 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LKS_Switch_Stuck_On_Fault                   | 6.1 | 49 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APA_Switch_Stuck_On_Fault                   | 6.2 | 50 | 1 | @1+ | 0 | 1 | 0 | 1 |
| _360_Switch_Stuck_On_Fault                  | 6.3 | 51 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BPA_Switch_Stuck_On_Fault                   | 6.4 | 52 | 1 | @1+ | 0 | 1 | 0 | 1 |
| L_BLIS_LED_ShortToBatt                      | 6.5 | 53 | 1 | @1+ | 0 | 1 | 0 | 1 |
| R_BLIS_LED_ShortToBatt                      | 6.6 | 54 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PARK_AID_LED_ShortToBatt                    | 6.7 | 55 | 1 | @1+ | 0 | 1 | 0 | 1 |
| L_BLIS_LED_ShortToGnd                       | 7   | 56 | 1 | @1+ | 0 | 1 | 0 | 1 |
| R_BLIS_LED_ShortToGnd                       | 7.1 | 57 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PARK_AID_LED_ShortToGnd                     | 7.2 | 58 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                    | 7.3 | 59 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AlgoRunner_Error_Fault                      | 7.4 | 60 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                    | 7.5 | 61 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                    | 7.6 | 62 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Pmic_Image2_Reset_Fault                     | 7.7 | 63 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_Message_CRC_Err             | 8   | 64 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_Missing_Message_Err         | 8.1 | 65 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_Missing_Data_Err            | 8.2 | 66 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_MessageReception_Err        | 8.3 | 67 | 1 | @1+ | 0 | 1 | 0 | 1 |

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|                                              |     |     |   |     |   |   |   |   |
|----------------------------------------------|-----|-----|---|-----|---|---|---|---|
| CPYR agent_ENET_VersionMismatch_Err          | 8.4 | 68  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Application_CRC_Fault             | 8.5 | 69  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Callibration_CRC_Fault            | 8.6 | 70  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Cal_Complete_And_Compatible       | 8.7 | 71  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cal_Complete_And_Compatible_SWP3    | 9   | 72  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_MESP_Coding_failure                 | 9.1 | 73  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Application_Complete_And_Compatible | 9.2 | 74  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Application_Checksum                | 9.3 | 75  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Calibration_Checksum                | 9.4 | 76  | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_Appl_Complete_And_Compatible         | 9.5 | 77  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_Frame_Format_Error            | 9.6 | 78  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_Msg_Format_Error              | 9.7 | 79  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_spi_msg_timeout                   | a.0 | 80  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_Tx_Error                      | a.1 | 81  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_CRC_Error                     | a.2 | 82  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spi_version_mismatch                         | a.3 | 83  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spi_Ipc_Comm_Reset                           | a.4 | 84  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Spi_Frame_Format_Error              | a.5 | 85  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_Spi_Msg_Format_Error                | a.6 | 86  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Spi_Msg_Timeout                     | a.7 | 87  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_SPI_CRC_ERROR                       | b.0 | 88  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Serial_Flash_Power_Up_Fail          | b.1 | 89  | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Aurix_FirstTimePowerUpFlashCheck          | b.2 | 90  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Aurix_OTA_Flash_Fail                         | b.3 | 91  | 1 | @1+ | 0 | 1 | 0 | 1 |
| KAM_Design_Read_Fail_Fault                   | b.4 | 92  | 1 | @1+ | 0 | 1 | 0 | 1 |
| KAM_Design_Write_Fail_Fault                  | b.5 | 93  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_DDR_Fault                           | b.6 | 94  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | b.7 | 95  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Boot_Fault                          | c.0 | 96  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Core_Fault                          | c.1 | 97  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Bist_Memory_Fault                   | c.2 | 98  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | c.3 | 99  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Init_Timeout_Fault                  | c.4 | 100 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Core_Dump_Fault                     | c.5 | 101 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parameters_Invalid                           | c.6 | 102 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Unreasonable_Radar_Data                      | c.7 | 103 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_Camera_Wrong_Assembly_Fault              | d.0 | 104 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_shutdown_Invalid_Reason                  | d.1 | 105 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_shutdown_7_POS_DIFF_KAN_VAL              | d.2 | 106 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_9_ROI_DIFF_VC_FCV               | d.3 | 107 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_6_FCV_FCW_TTC                   | d.4 | 108 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_25_CO_FOF_DIFF                  | d.5 | 109 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_30_DT_ERR                       | d.6 | 110 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_18_IMAGER_DATA_ROWS_COLS        | d.7 | 111 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_4_YAW_HORIZON_DEVIATION         | e.0 | 112 | 1 | @1+ | 0 | 1 | 0 | 1 |

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|--------------------------|-----|-----|---|-----|---|---|---|---|
| VISIONCO_GEN_26_CODE_CRC | e.1 | 113 | 1 | @1+ | 0 | 1 | 0 | 1 |
|--------------------------|-----|-----|---|-----|---|---|---|---|



|                                                           |      |     |   |     |   |   |   |   |
|-----------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| VISIONCO_GEN_13_OLD_EGODATA_LAT                           | e.2  | 114 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | e.3  | 115 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | e.4  | 116 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | e.5  | 117 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_31_AEB_PARAM_FFI                             | e.6  | 118 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_29_RFC_ERR                                   | e.7  | 119 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_11_INPUT_CRC_MISMATCH                        | f.0  | 120 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_34_Corrupted_Vehicle_Info_Input_Signals      | f.1  | 121 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_27_Fcv_Sig_Ffi                               | f.2  | 122 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_54_ACC_STL_EXECUTION_FAILED                  | f.3  | 123 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | f.4  | 124 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Missing_Camera_Callibration_Fault                         | f.5  | 125 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_Camera_Type_Fault                                     | f.6  | 126 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | f.7  | 127 | 1 | @1+ | 0 | 1 | 0 | 1 |
| NTSC_Encoder_amplifier_ShortToBatt                        | 10   | 128 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | 10.1 | 129 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | 10.2 | 130 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | 10.3 | 131 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Full_Blockage_Fault                                       | 10.4 | 132 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Partial_Blockage_Fault                                    | 10.5 | 133 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Blurred_Image_Fault                                       | 10.6 | 134 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Splashes_Fault                                            | 10.7 | 135 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Low_Sun_Fault                                             | 11   | 136 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Sun_Rays_Fault                                            | 11.1 | 137 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Foggy_Spots_Fault                                         | 11.2 | 138 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spot_Rays_Fault                                           | 11.3 | 139 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Smearred_Spots_Fault                                      | 11.4 | 140 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SelfGlare_Fault                                           | 11.5 | 141 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Low_Visibility_Fault                                      | 11.6 | 142 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Vision_OutofFocus_Fault                                   | 11.7 | 143 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Vision_OutofCalibration_Fault                             | 12   | 144 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TD_BlockageTest                                           | 12.1 | 145 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_APP_CPS_STL_FAILED                   | 12.2 | 146 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Error_Fault                      | 12.3 | 147 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Fs_Error_Fault                   | 12.4 | 148 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Calibration_Error_Fault          | 12.5 | 149 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Init_Failed_Fault                | 12.6 | 150 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Init_Camera_Init_Fault           | 12.7 | 151 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Video_Grab_Failed_Fault      | 13   | 152 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Camera_Self_Reset_Fault      | 13.1 | 153 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Timeout_Error_Fault          | 13.2 | 154 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Pattern_Test_Fault               | 13.3 | 155 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Cam_Params_Ccft_Crc_Failed_Fault | 13.4 | 156 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_PII_Comparison_Error_Fault           | 13.5 | 157 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_Pv_General_Error_Fault               | 13.6 | 158 | 1 | @1+ | 0 | 1 | 0 | 1 |



|                                                                         |      |     |   |     |   |   |   |   |
|-------------------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| VisionCo_Fatal_Error_Pv_Verificattion_Error_Fault                       | 13.7 | 159 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Error_Fault                                     | 14   | 160 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Error_Fault                                  | 14.1 | 161 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Setting_Fault                | 14.2 | 162 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Registry_Fault               | 14.3 | 163 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Registry_Fault               | 14.4 | 164 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Fault                        | 14.5 | 165 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Fault                     | 14.6 | 166 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Reg_Fault                 | 14.7 | 167 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Open_Blackbox_Fault               | 15   | 168 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Ep_Fault                     | 15.1 | 169 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Ep_Fault                | 15.2 | 170 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Create_Logger_Fault     | 15.3 | 171 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_IL_Fault                | 15.4 | 172 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Check_Reg_Versions_Fault          | 15.5 | 173 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_C_R_Error                                                  | 15.6 | 174 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Init_Vdi_Failure                                                        | 15.7 | 175 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Init_I2C_Failure                                                        | 16   | 176 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_No_Free_Header_Failure                                             | 16.1 | 177 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Issue_Failure                                                      | 16.2 | 178 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Acquire_Failure                                                    | 16.3 | 179 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Release_Failure                                                    | 16.4 | 180 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Mal_Func                                                   | 16.5 | 181 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Ctrl_Flt                                                   | 16.6 | 182 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Vdo_Flt                                                    | 16.7 | 183 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Video_Error_Fault                  | 17   | 184 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_I2C_Not_On_Time_Video_Error_Fault      | 17.1 | 185 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Logical_Video_Error_Fault              | 17.2 | 186 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Inconsistent_Frames_Video_Error_Fault  | 17.3 | 187 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Verification_Failure_Video_Error_Fault | 17.4 | 188 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Invalid_Video_Header_Fault             | 17.5 | 189 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Error_Reg_Flag_Fault                   | 17.6 | 190 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Reg_Write_Hil_Failure_Error_Fault      | 17.7 | 191 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Zero_Histogram_Error_Fault             | 18   | 192 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Sum_Not_Match_Histogram_Error_Fault    | 18.1 | 193 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Internal_Error_Fault               | 18.2 | 194 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_No_Buffer_Fault                | 18.3 | 195 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Buffer_Invalid_Format_Fault    | 18.4 | 196 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Timeout_Fault                  | 18.5 | 197 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Overflow_Fault            | 18.6 | 198 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Underflow_Fault           | 18.7 | 199 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Fifo_Fault              | 19   | 200 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Weights_Fault           | 19.1 | 201 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Gamma_Fault             | 19.2 | 202 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Hist_Fault              | 19.3 | 203 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                                                  |      |     |   |     |   |   |   |   |
|------------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Start_After_Start_Fault | 19.4 | 204 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Core_Id_Fault           | 19.5 | 205 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Frame_Count_Read_Fail_Fault     | 19.6 | 206 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Histograms_Mismatch_Fault       | 19.7 | 207 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Ccft_Decisions_Mismatch_Fault   | 1a.0 | 208 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Header_Verification_Error_Fault | 1a.1 | 209 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Shutdown_Fault          | 1a.2 | 210 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Shot_Out_Of_Sync_Fault          | 1a.3 | 211 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_temperature_Fault                               | 1a.4 | 212 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Error_Out_Signal_Internal_Fault                         | 1a.5 | 213 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Error_Out_Temperature_Fault                             | 1a.6 | 214 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCM_LostCom                                                      | 1a.7 | 215 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TCM_LostCom                                                      | 1b.0 | 216 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GSM_LostCom                                                      | 1b.1 | 217 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_LostCom                                                      | 1b.2 | 218 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SECM_LostCom                                                     | 1b.3 | 219 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PSCM_LostCom                                                     | 1b.4 | 220 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BCM_LostCom                                                      | 1b.5 | 221 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_LostCom                                                      | 1b.6 | 222 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_LostCom                                                      | 1b.7 | 223 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HUD_LostCom                                                      | 1c.0 | 224 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SCCM_LostCom                                                     | 1c.1 | 225 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACM_LostCom                                                      | 1c.2 | 226 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DDM_LostCom                                                      | 1c.3 | 227 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PDM_LostCom                                                      | 1c.4 | 228 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APIM_LostCom                                                     | 1c.5 | 229 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                         | 1c.6 | 230 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCMH_LostCom                                                     | 1c.7 | 231 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VDM_LostCom                                                      | 1d.0 | 232 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCM_InvalidData                                                  | 1d.1 | 233 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TCM_InvalidData                                                  | 1d.2 | 234 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GSM_InvalidData                                                  | 1d.3 | 235 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidData                                                  | 1d.4 | 236 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SECM_InvalidData                                                 | 1d.5 | 237 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PSCM_InvalidData                                                 | 1d.6 | 238 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BCM_InvalidData                                                  | 1d.7 | 239 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_InvaildData                                                  | 1e.0 | 240 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SCCM_InvalidData                                                 | 1e.1 | 241 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidData                                                  | 1e.2 | 242 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DDM_InvalidData                                                  | 1e.3 | 243 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PDM_InvalidData                                                  | 1e.4 | 244 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HVAC_InvalidData                                                 | 1e.5 | 245 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APIM_InvalidData                                                 | 1e.6 | 246 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                         | 1e.7 | 247 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                         | 1f.0 | 248 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCMH_InvalidData                                                 | 1f.1 | 249 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                                             |      |     |   |     |   |   |   |   |
|-------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| VDM_InvalidData                                             | 1f.2 | 250 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_InvalidRollCnt                                          | 1f.3 | 251 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidRollCnt                                          | 1f.4 | 252 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidChkSum                                           | 1f.5 | 253 | 1 | @1+ | 0 | 1 | 0 | 1 |
| reserved                                                    | 1f.6 | 254 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VCAN_Bus_Off_Fault                                          | 1f.7 | 255 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FLRCAN_Bus_Off_Fault                                        | 20   | 256 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RF_LF_CAN_Bus_Off_Fault                           | 20.1 | 257 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_LR_CAN_Bus_Off_Fault                           | 20.2 | 258 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_PCAN_Missing_Message_Fault                        | 20.3 | 259 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_PCAN_Range_Check_Fault                            | 20.4 | 260 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 20.5 | 261 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FR_PCAN_Range_Check_Fault                         | 20.6 | 262 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FL_PCAN_Range_Check_Fault                         | 20.7 | 263 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RL_PCAN_Range_Check_Fault                         | 21   | 264 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_PCAN_Range_Check_Fault                         | 21.1 | 265 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CarCfg_Implausible_Denied_Param_23                          | 21.2 | 266 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CarCfg_Implausible_Denied_Param_96                          | 21.3 | 267 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_Software_Compatibility_Fault                      | 21.4 | 268 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_Software_Compatibility_Fault                      | 21.5 | 269 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GCCConfigurationNotconfigured_CarCfg_Undefined_Denied       | 21.6 | 270 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_Invalid_Hardware_configuration                    | 21.7 | 271 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidRollCnt                                          | 22   | 272 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidChkSum                                           | 22.1 | 273 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Brake_Temperature_Too_High                                  | 22.2 | 274 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Camera_Module_Alignment_or_Adjustment_Incorrect_RearCamera_ | 22.3 | 275 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DPAC_Vision_System_Camera_Alignment_or_Adjustment_Incorrect | 22.4 | 276 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Internal_Control_Module_Software_Incompatibility            | 22.5 | 277 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Initial_Configuration_Not_Complete                          | 22.6 | 278 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Control_Module_Configuration_Incompatible                   | 22.7 | 279 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Bus_Failure                                 | 23   | 280 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Not_Configured                              | 23.1 | 281 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Fault                                       | 23.2 | 282 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Bus_Failure                                    | 23.3 | 283 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Not_Configured                                 | 23.4 | 284 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Fault                                          | 23.5 | 285 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Front_USS_HSD_FailureFrontUSS_Ckt_ShortToGnd                | 23.6 | 286 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_USS_HSD_FailureRearUSS_Ckt_ShortToGnd                  | 23.7 | 287 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Front_USS_HSD_ShortToBattFrontUSS_Ckt_ShortToBatt           | 24   | 288 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_USS_HSD_ShortToBattRearUSS_Ckt_ShortToBatt             | 24.1 | 289 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Ignition_Status_Unknown_Fault                               | 24.2 | 290 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Ignition_Status_Missing_Fault                               | 24.3 | 291 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Veh_V_ActlEng_Missing_Fault                                 | 24.4 | 292 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_Missing_Fault                              | 24.5 | 293 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_Fault                                      | 24.6 | 294 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                               |      |     |   |     |   |   |   |   |
|-----------------------------------------------|------|-----|---|-----|---|---|---|---|
| VehVActlEng_D_Qf_No_data_exists_Fault         | 24.7 | 295 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_MES_49_Data_Path_Corruption          | 25   | 296 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_MES_50_VMP_SANITY_FAILED             | 25.1 | 297 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 25.2 | 298 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_40_PED_SANITY_FAULT                  | 25.3 | 299 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_39_PED_OVERLAP_FAULT                 | 25.4 | 300 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 25.5 | 301 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 25.6 | 302 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 25.7 | 303 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 26   | 304 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 26.1 | 305 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_48_ALGO_PITCH_MISMATCH               | 26.2 | 306 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_45_ROAD_SFCONF_VALIDATION            | 26.3 | 307 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_42_ROAD_PREDICTION_VALIDATION        | 26.4 | 308 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_33_ROAD_LANE_DISQUALIFICATINO_FAILED | 26.5 | 309 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TrlrLampCnnct_B_Actl_Missing_Fault            | 26.6 | 310 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_MissingMsg_Fault        | 26.7 | 311 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_MissingMsg_Fault          | 27   | 312 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_DCUFault                | 27.1 | 313 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_DCUFault                  | 27.2 | 314 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_ON_OFF_Mismatch_Fault   | 27.3 | 315 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_ON_OFF_Mismatch_Fault     | 27.4 | 316 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IgnKeyType_D_Actl_Unknown_Fault               | 27.5 | 317 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IgnKeyType_D_ActlMissing_Fault                | 27.6 | 318 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parklamp_Status_Missing_Fault                 | 27.7 | 319 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Application_CRC_Fault                      | 28   | 320 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 28.1 | 321 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_DDR_RAM_CRC_Fault                          | 28.2 | 322 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 28.3 | 323 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 28.4 | 324 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 28.5 | 325 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 28.6 | 326 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 28.7 | 327 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29   | 328 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29.1 | 329 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29.2 | 330 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29.3 | 331 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29.4 | 332 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29.5 | 333 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29.6 | 334 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 29.7 | 335 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_INT_OSC_TEST_FAIL                          | 2a.0 | 336 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_DPAC_Process_Fault                       | 2a.1 | 337 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_FWC_Process_Fault                        | 2a.2 | 338 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2a.3 | 339 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                               |      |     |   |     |   |   |   |   |
|-----------------------------------------------|------|-----|---|-----|---|---|---|---|
| TR_Stack_Overflow_Fault                       | 2a.4 | 340 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2a.5 | 341 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2a.6 | 342 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2a.7 | 343 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2b.0 | 344 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_DDR_Fault                                  | 2b.1 | 345 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Boot_Fault                                 | 2b.2 | 346 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2b.3 | 347 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Appl_Complete_And_Compatible               | 2b.4 | 348 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Core_Exception_Fault                       | 2b.5 | 349 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Control_ConfigReg_Fault                    | 2b.6 | 350 | 1 | @1+ | 0 | 1 | 0 | 1 |
| reserved                                      | 2b.7 | 351 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Message_CRC_Err                       | 2c.0 | 352 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Missing_Message_Err                   | 2c.1 | 353 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Missing_Data_Err                      | 2c.2 | 354 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_MessageReception_Err                  | 2c.3 | 355 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_VersionMismatch_Err                   | 2c.4 | 356 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Communication_Drv_Err                 | 2c.5 | 357 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2c.6 | 358 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2c.7 | 359 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.0 | 360 | 1 | @1+ | 0 | 1 | 0 | 1 |
| QSPI_Flash_Illegal_Instruction_Err            | 2d.1 | 361 | 1 | @1+ | 0 | 1 | 0 | 1 |
| QSPI_Communication_Drv_Err                    | 2d.2 | 362 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.3 | 363 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.4 | 364 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.5 | 365 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_PERFPROC_IntMemFail                | 2d.6 | 366 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_PERFPROC_IntElectronicFail         | 2d.7 | 367 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LEFT_CAMERA_FAULT_CKT_ShortToBatt             | 2e.0 | 368 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RIGHT_CAMERA_FAULT_CKT_ShortToBatt            | 2e.1 | 369 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BA_Switch_Stuck_On_Fault                      | 2e.2 | 370 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.3 | 371 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.4 | 372 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.5 | 373 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.6 | 374 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.7 | 375 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.0 | 376 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.1 | 377 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.2 | 378 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.3 | 379 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.4 | 380 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.5 | 381 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.6 | 382 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.7 | 383 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_21_CAMERA_HEIGHT_LIMIT_EXCEEDED  | 30   | 384 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_44_ROAD_SANITY_CORRECTION_VALIDATION | 30.1 | 385 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                            |      |     |   |     |   |   |   |   |
|--------------------------------------------|------|-----|---|-----|---|---|---|---|
| VISIONCO_53_DECELERATION_DBS_OVERLAP_FAULT | 30.2 | 386 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_24_CCFT_DIFF_RESULTS          | 30.3 | 387 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GearRvrse_D_Actl_Missing_Fault             | 30.4 | 388 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_TCM_LostComm                           | 30.5 | 389 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WheelData_Missing_Fault                    | 30.6 | 390 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_BCM_LostComm                           | 30.7 | 391 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_RCM_LostCom                            | 31   | 392 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_ABS_InvalidData                        | 31.1 | 393 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_TCM_InvalidData                        | 31.2 | 394 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GearRvrse_D_Actl_invalid_Fault             | 31.3 | 395 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_IPC_LostCom                            | 31.4 | 396 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Inhibited_Fault_Flag                   | 31.5 | 397 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Validity_Fault_Flag                    | 31.6 | 398 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LaneFusion_Validity_Fault_Flag             | 31.7 | 399 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                   | 32   | 400 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                   | 32.1 | 401 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                   | 32.2 | 402 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_CAMERA_FAULT_CKT_ShortToBatt           | 32.3 | 403 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftRearSOD_Ckt_ShortToBatt                | 32.4 | 404 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftRearSOD_Ckt_ShortToGnd                 | 32.5 | 405 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightRearSOD_Ckt_ShortToBatt               | 32.6 | 406 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightRearSOD_Ckt_ShortToGnd                | 32.7 | 407 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftFrontSOD_Ckt_ShortToBatt               | 33   | 408 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftFrontSOD_Ckt_ShortToGnd                | 33.1 | 409 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightFrontSOD_Ckt_ShortToBatt              | 33.2 | 410 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightFrontSOD_Ckt_ShortToGnd               | 33.3 | 411 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FRONT_CAMERA_FAULT_CKT_ShortToBatt         | 33.4 | 412 | 1 | @1+ | 0 | 1 | 0 | 1 |
| REAR_CAMERA_FAULT_CKT_ShortToBatt          | 33.5 | 413 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                   | 33.6 | 414 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_915E31_PerfProcessor                   | 33.7 | 415 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4008_FAULT_CONDITION                 | 34   | 416 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4096_FAULT_CONDITION                 | 34.1 | 417 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4098_FAULT_CONDITION                 | 34.2 | 418 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4208_FAULT_CONDITION                 | 34.3 | 419 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4296_FAULT_CONDITION                 | 34.4 | 420 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4298_FAULT_CONDITION                 | 34.5 | 421 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C08_FAULT_CONDITION                 | 34.6 | 422 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C96_FAULT_CONDITION                 | 34.7 | 423 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C98_FAULT_CONDITION                 | 35   | 424 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4808_FAULT_CONDITION                 | 35.1 | 425 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4896_FAULT_CONDITION                 | 35.2 | 426 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4898_FAULT_CONDITION                 | 35.3 | 427 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5008_FAULT_CONDITION                 | 35.4 | 428 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5096_FAULT_CONDITION                 | 35.5 | 429 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5098_FAULT_CONDITION                 | 35.6 | 430 | 1 | @1+ | 0 | 1 | 0 | 1 |



|                                         |      |     |   |     |   |   |   |   |
|-----------------------------------------|------|-----|---|-----|---|---|---|---|
| DTC_93F308_FAULT_CONDITION              | 35.7 | 431 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F396_FAULT_CONDITION              | 36   | 432 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F398_FAULT_CONDITION              | 36.1 | 433 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3608_FAULT_CONDITION              | 36.2 | 434 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3696_FAULT_CONDITION              | 36.3 | 435 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3698_FAULT_CONDITION              | 36.4 | 436 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3808_FAULT_CONDITION              | 36.5 | 437 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3896_FAULT_CONDITION              | 36.6 | 438 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3898_FAULT_CONDITION              | 36.7 | 439 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D08_FAULT_CONDITION              | 37   | 440 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D96_FAULT_CONDITION              | 37.1 | 441 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D98_FAULT_CONDITION              | 37.2 | 442 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4408_FAULT_CONDITION              | 37.3 | 443 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4496_FAULT_CONDITION              | 37.4 | 444 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4498_FAULT_CONDITION              | 37.5 | 445 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4608_FAULT_CONDITION              | 37.6 | 446 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4696_FAULT_CONDITION              | 37.7 | 447 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4698_FAULT_CONDITION              | 38   | 448 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F408_FAULT_CONDITION              | 38.1 | 449 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F496_FAULT_CONDITION              | 38.2 | 450 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F498_FAULT_CONDITION              | 38.3 | 451 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93039E_FAULT_CONDITION              | 38.4 | 452 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_915E31_FAULT_CONDITION              | 38.5 | 453 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_948E31_FAULT_CONDITION              | 38.6 | 454 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_92BE31_FAULT_CONDITION              | 38.7 | 455 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_92BF31_FAULT_CONDITION              | 39   | 456 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_IMAGE2                       | 39.1 | 457 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00089_IMAGE2                       | 39.2 | 458 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FL_PCAN_Missing_Message_Fault | 39.3 | 459 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FR_PCAN_Missing_Message_Fault | 39.4 | 460 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RL_PCAN_Missing_Message_Fault | 39.5 | 461 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_PCAN_Missing_Message_Fault | 39.6 | 462 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidData                        | 39.7 | 463 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidRollCnt                     | 3a.0 | 464 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidChkSum                      | 3a.1 | 465 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TRM_InvalidData                         | 3a.2 | 466 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_LostComm                           | 3a.3 | 467 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HVAC_LostComm                           | 3a.4 | 468 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TRM_LostComm                            | 3a.5 | 469 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ATCM_LostComm                           | 3a.6 | 470 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HCM_LostComm                            | 3a.7 | 471 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parklamp_Status_Unknown_Fault           | 3b.0 | 472 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Litval_Unknown_Fault                    | 3b.1 | 473 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Litval_Missing_Fault                    | 3b.2 | 474 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TurnLghtSwitch_D_Stat_Missing_Fault     | 3b.3 | 475 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                         |      |     |   |     |   |   |   |   |

|                                 |      |     |   |     |   |   |   |   |
|---------------------------------|------|-----|---|-----|---|---|---|---|
| Btt_L_Actl_MissingMsg_Fault     | 3b.4 | 476 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ATD_Mismatch_Fault              | 3b.5 | 477 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TLM_Mismatch_Fault              | 3b.6 | 478 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TBM_Mismatch_Fault              | 3b.7 | 479 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                        | 3c.0 | 480 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRr_D_Actl_Missing         | 3c.1 | 481 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRl_D_Actl_Missing         | 3c.2 | 482 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFl_D_Actl_Missing         | 3c.3 | 483 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFr_D_Actl_Missing         | 3c.4 | 484 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRr_D_Actl_Failed_Fault    | 3c.5 | 485 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRl_D_Actl_Failed_Fault    | 3c.6 | 486 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFr_D_Actl_Failed_Fault    | 3c.7 | 487 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFl_D_Actl_Failed_Fault    | 3d.0 | 488 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaBrk_D_Stat_Denied_Fault      | 3d.1 | 489 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaBrk_D_Stat_Missing_Fault     | 3d.2 | 490 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Isig_brake_timeout_fault        | 3d.3 | 491 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                        | 3d.4 | 492 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaYBrkEnbl_B_Rq_Mismatch_Fault | 3d.5 | 493 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DVR_SELECT_STAT_Fault           | 3d.6 | 494 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_IMAGE2               | 3d.7 | 495 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBDeniedByABS                 | 3e.0 | 496 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBDeniedByPCM                 | 3e.1 | 497 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBImplausByABS                | 3e.2 | 498 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBImplausByPCM                | 3e.3 | 499 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AGENT_InvalidData               | 3e.4 | 500 | 1 | @1+ | 0 | 1 | 0 | 1 |
| unused                          | 3e.5 | 501 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplausByABS                 | 3e.6 | 502 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplausByPCM                 | 3e.7 | 503 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplausAccelByABS            | 3f.0 | 504 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCFailToCancelByPCM            | 3f.1 | 505 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCDeniedbyABS                  | 3f.2 | 506 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTC_ACCDeniedByDIM           | 3f.3 | 507 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleCMbB_ASILB     | 3f.4 | 508 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TJAimplausByPSCM                | 3f.5 | 509 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleHA_FSM         | 3f.6 | 510 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleFAPA_FSM       | 3f.7 | 511 | 1 | @1+ | 0 | 1 | 0 | 1 |

## Latch\_Fault\_Status

Latch\_Fault\_Status-STNumber(76)-STVersion(1) This message is used to report fault status information.

Message Latch\_Fault\_Status shall be implemented as fault table defined in [5].

| Signal Name                       | Description | StartBit | Size | Sign | Offset | Factor | Min | Max |
|-----------------------------------|-------------|----------|------|------|--------|--------|-----|-----|
| VisionCo_3_3V_Supply_Out_Of_Range | 0           | 0        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_8V_Supply_Out_Of_Range | 0.1         | 1        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_1V_Supply_Out_Of_Range | 0.2         | 2        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_0V_Supply_Out_Of_Range | 0.3         | 3        | 1    | @1+  | 0      | 1      | 0   | 1   |
|                                   |             |          |      |      |        |        |     |     |

|                                                            |     |    |   |     |   |   |   |   |
|------------------------------------------------------------|-----|----|---|-----|---|---|---|---|
| TR_3_3V_Supply_Out_Of_Range                                | 0.4 | 4  | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_1_8V_Supply_Out_Of_Range                                | 0.5 | 5  | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_1_35V_Supply_Out_Of_Range                               | 0.6 | 6  | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_1_0V_Supply_Out_Of_Range                                | 0.7 | 7  | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_PGOOD_Fail_Fault                                   | 1   | 8  | 1 | @1+ | 0 | 1 | 0 | 1 |
| FVC_CAMPWR_LDO_FailFRONT_CAMERA_FAULT_CKT_ShortToGnd       | 1.1 | 9  | 1 | @1+ | 0 | 1 | 0 | 1 |
| RVC_CAMPWR_LDO_FailREAR_CAMERA_FAULT_CKT_ShortToGnd        | 1.2 | 10 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LH_Mirror_CAMPWR_LDO_FailLEFT_CAMERA_FAULT_CKT_ShortToGnd  | 1.3 | 11 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RH_Mirror_CAMPWR_LDO_FailRIGHT_CAMERA_FAULT_CKT_ShortToGnd | 1.4 | 12 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_CAMPWR_LDO_FailFWC_CAMERA_FAULT_CKT_ShortToGnd         | 1.5 | 13 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 1.6 | 14 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 1.7 | 15 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 2   | 16 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 2.1 | 17 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 2.2 | 18 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_Output_Open_Circuit                              | 2.3 | 19 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FET_Fault_Across_cycles                                    | 2.4 | 20 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FET_Fail_Across_Cycles                                     | 2.5 | 21 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_SerialFlash_PowerUp_Fail                                | 2.6 | 22 | 1 | @1+ | 0 | 1 | 0 | 1 |
| System_Over_Temperature                                    | 2.7 | 23 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_On_Board_Thermistor_Over_Temp                   | 3   | 24 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_On_Board_Thermistor_Over_Temp                     | 3.1 | 25 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_On_Board_Thermistor_Over_Temp                           | 3.2 | 26 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_On_Board_Thermistor_Over_Temp                      | 3.3 | 27 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_MIPS_Over_Temp                                    | 3.4 | 28 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_VMP_Over_Temp                                     | 3.5 | 29 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Imager_Over_Temp                                  | 3.6 | 30 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Boot_OverTemp_Fault                               | 3.7 | 31 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_DDR_Over_Temp                                     | 4   | 32 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_On_Board_Thermistor_Out_Of_Range                | 4.1 | 33 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_On_Board_Thermistor_Out_Of_Range                  | 4.2 | 34 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_On_Board_Thermistor_Out_Of_Range                   | 4.3 | 35 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_On_Board_Thermistor_Out_Of_Range                        | 4.4 | 36 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BatteryLowFault                                            | 4.5 | 37 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 4.6 | 38 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 4.7 | 39 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_AURIX_Fault                                     | 5   | 40 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_AURIX_Fault                                     | 5.1 | 41 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                   | 5.2 | 42 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Short_Across_Cycles                                 | 5.3 | 43 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Output_Short_Circuit                                | 5.4 | 44 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Output_Open_Circuit                                 | 5.5 | 45 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Short_Fail_Across_Cycles                            | 5.6 | 46 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CHMSL_camera_DEC_amplifier_ShortToBatt                     | 5.7 | 47 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AUX_camera_DEC_amplifier_ShortToBatt                       | 6   | 48 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LKS_Switch_Stuck_On_Fault                                  | 6.1 | 49 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                              |     |    |   |     |   |   |   |   |
|----------------------------------------------|-----|----|---|-----|---|---|---|---|
| APA_Switch_Stuck_On_Fault                    | 6.2 | 50 | 1 | @1+ | 0 | 1 | 0 | 1 |
| _360_Switch_Stuck_On_Fault                   | 6.3 | 51 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BPA_Switch_Stuck_On_Fault                    | 6.4 | 52 | 1 | @1+ | 0 | 1 | 0 | 1 |
| L_BLIS_LED_ShortToBatt                       | 6.5 | 53 | 1 | @1+ | 0 | 1 | 0 | 1 |
| R_BLIS_LED_ShortToBatt                       | 6.6 | 54 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PARK_AID_LED_ShortToBatt                     | 6.7 | 55 | 1 | @1+ | 0 | 1 | 0 | 1 |
| L_BLIS_LED_ShortToGnd                        | 7   | 56 | 1 | @1+ | 0 | 1 | 0 | 1 |
| R_BLIS_LED_ShortToGnd                        | 7.1 | 57 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PARK_AID_LED_ShortToGnd                      | 7.2 | 58 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 7.3 | 59 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AlgoRunner_Error_Fault                       | 7.4 | 60 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 7.5 | 61 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 7.6 | 62 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Pmic_Image2_Reset_Fault                      | 7.7 | 63 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_Message_CRC_Err              | 8   | 64 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_Missing_Message_Err          | 8.1 | 65 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_Missing_Data_Err             | 8.2 | 66 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_MessageReception_Err         | 8.3 | 67 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_ENET_VersionMismatch_Err          | 8.4 | 68 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_Application_CRC_Fault             | 8.5 | 69 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_Callibration_CRC_Fault            | 8.6 | 70 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_Cal_Complete_And_Compatible       | 8.7 | 71 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cal_Complete_And_Compatible_SWP3    | 9   | 72 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_MESP_Coding_failure                 | 9.1 | 73 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Application_Complete_And_Compatible | 9.2 | 74 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Application_Checksum                | 9.3 | 75 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Calibration_Checksum                | 9.4 | 76 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_Appl_Complete_And_Compatible         | 9.5 | 77 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_Spi_Frame_Format_Error            | 9.6 | 78 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_Spi_Msg_Format_Error              | 9.7 | 79 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_spi_msg_timeout                   | a.0 | 80 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_Spi_Tx_Error                      | a.1 | 81 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR_agent_Spi_CRC_Error                     | a.2 | 82 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spi_version_mismatch                         | a.3 | 83 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spi_Ipc_Comm_Reset                           | a.4 | 84 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Spi_Frame_Format_Error              | a.5 | 85 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_Spi_Msg_Format_Error                | a.6 | 86 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Spi_Msg_Timeout                     | a.7 | 87 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_SPI_CRC_ERROR                       | b.0 | 88 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Serial_Flash_Power_Up_Fail          | b.1 | 89 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Aurix_FirstTimePowerUpFlashCheck          | b.2 | 90 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Aurix_OTA_Flash_Fail                         | b.3 | 91 | 1 | @1+ | 0 | 1 | 0 | 1 |
| KAM_Design_Read_Fail_Fault                   | b.4 | 92 | 1 | @1+ | 0 | 1 | 0 | 1 |
| KAM_Design_Write_Fail_Fault                  | b.5 | 93 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_DDR_Fault                           | b.6 | 94 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                                      |      |     |   |     |   |   |   |   |
|------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| Reserved                                             | b.7  | 95  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Boot_Fault                                  | c.0  | 96  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Core_Fault                                  | c.1  | 97  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Bist_Memory_Fault                           | c.2  | 98  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | c.3  | 99  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Init_Timeout_Fault                          | c.4  | 100 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Core_Dump_Fault                             | c.5  | 101 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parameters_Invalid                                   | c.6  | 102 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Unreasonable_Radar_Data                              | c.7  | 103 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_Camera_Wrong_Assembly_Fault                      | d.0  | 104 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_shutdown_Invalid_Reason                          | d.1  | 105 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_shutdown_7_POS_DIFF_KAN_VAL                      | d.2  | 106 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_9_ROI_DIFF_VC_FCV                       | d.3  | 107 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_6_FCV_FCW_TTC                           | d.4  | 108 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_25_CO_FOF_DIFF                          | d.5  | 109 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_30_DT_ERR                               | d.6  | 110 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_18_IMAGER_DATA_ROWS_COLS                | d.7  | 111 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_4_YAW_HORIZON_DEVIATION                 | e.0  | 112 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_26_CODE_CRC                             | e.1  | 113 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_13_OLD_EGODATA_LAT                      | e.2  | 114 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | e.3  | 115 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | e.4  | 116 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | e.5  | 117 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_31_AEB_PARAM_FFI                        | e.6  | 118 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_29_RFC_ERR                              | e.7  | 119 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_11_INPUT_CRC_MISMATCH                   | f.0  | 120 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_34_Corrupted_Vehicle_Info_Input_Signals | f.1  | 121 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_27_Fcv_Sig_Ffi                          | f.2  | 122 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_54_ACC_STL_EXECUTION_FAILED             | f.3  | 123 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | f.4  | 124 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Missing_Camera_Callibration_Fault                    | f.5  | 125 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_Camera_Type_Fault                                | f.6  | 126 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | f.7  | 127 | 1 | @1+ | 0 | 1 | 0 | 1 |
| NTSC_Encoder_amplifier_ShortToBatt                   | 10   | 128 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | 10.1 | 129 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | 10.2 | 130 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | 10.3 | 131 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Full_Blockage_Fault                                  | 10.4 | 132 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Partial_Blockage_Fault                               | 10.5 | 133 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Blurred_Image_Fault                                  | 10.6 | 134 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Splashes_Fault                                       | 10.7 | 135 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Low_Sun_Fault                                        | 11   | 136 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Sun_Rays_Fault                                       | 11.1 | 137 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Foggy_Spots_Fault                                    | 11.2 | 138 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spot_Rays_Fault                                      | 11.3 | 139 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                                      |      |     |   |     |   |   |   |   |

|                                                                     |      |     |   |     |   |   |   |   |
|---------------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| Smeared_Spots_Fault                                                 | 11.4 | 140 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SelfGlare_Fault                                                     | 11.5 | 141 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Low_Visibility_Fault                                                | 11.6 | 142 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Vision_OutofFocus_Fault                                             | 11.7 | 143 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Vision_OutofCalibration_Fault                                       | 12   | 144 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TD_BlockageTest                                                     | 12.1 | 145 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_APP_CPS_STL_FAILED                             | 12.2 | 146 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Error_Fault                                | 12.3 | 147 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Fs_Error_Fault                             | 12.4 | 148 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Calibration_Error_Fault                    | 12.5 | 149 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Init_Failed_Fault                          | 12.6 | 150 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Init_Camera_Init_Fault                     | 12.7 | 151 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Video_Grab_Failed_Fault                | 13   | 152 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Camera_Self_Reset_Fault                | 13.1 | 153 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Timeout_Error_Fault                    | 13.2 | 154 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Pattern_Test_Fault                         | 13.3 | 155 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Cam_Params_Ccft_Crc_Failed_Fault           | 13.4 | 156 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_PII_Comparison_Error_Fault                     | 13.5 | 157 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_Pv_General_Error_Fault                         | 13.6 | 158 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_Pv_Verificattion_Error_Fault                   | 13.7 | 159 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Error_Fault                                 | 14   | 160 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Error_Fault                              | 14.1 | 161 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Setting_Fault            | 14.2 | 162 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Registry_Fault           | 14.3 | 163 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Registry_Fault           | 14.4 | 164 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Fault                    | 14.5 | 165 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Fault                 | 14.6 | 166 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Reg_Fault             | 14.7 | 167 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Open_Blackbox_Fault           | 15   | 168 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Ep_Fault                 | 15.1 | 169 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Ep_Fault            | 15.2 | 170 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Create_Logger_Fault | 15.3 | 171 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_IL_Fault            | 15.4 | 172 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Check_Reg_Versions_Fault      | 15.5 | 173 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_C_R_Error                                              | 15.6 | 174 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Init_Vdi_Failure                                                    | 15.7 | 175 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Init_I2C_Failure                                                    | 16   | 176 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_No_Free_Header_Failure                                         | 16.1 | 177 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Issue_Failure                                                  | 16.2 | 178 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Acquire_Failure                                                | 16.3 | 179 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Release_Failure                                                | 16.4 | 180 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Mal_Func                                               | 16.5 | 181 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Ctrl_Flt                                               | 16.6 | 182 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Vdo_Flt                                                | 16.7 | 183 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Video_Error_Fault              | 17   | 184 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_I2C_Not_On_Time_Video_Error_Fault  | 17.1 | 185 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                                                         |      |     |   |     |   |   |   |   |
|-------------------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| VisionCo_Camera1_VideoErrorFlags_Logical_Video_Error_Fault              | 17.2 | 186 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Inconsistent_Frames_Video_Error_Fault  | 17.3 | 187 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Verification_Failure_Video_Error_Fault | 17.4 | 188 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Invalid_Video_Header_Fault             | 17.5 | 189 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Error_Reg_Flag_Fault                   | 17.6 | 190 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Reg_Write_Hil_Failure_Error_Fault      | 17.7 | 191 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Zero_Histogram_Error_Fault             | 18   | 192 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Sum_Not_Match_Histogram_Error_Fault    | 18.1 | 193 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Internal_Error_Fault               | 18.2 | 194 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_No_Buffer_Fault                | 18.3 | 195 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Buffer_Invalid_Format_Fault    | 18.4 | 196 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Timeout_Fault                  | 18.5 | 197 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Overflow_Fault            | 18.6 | 198 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Underflow_Fault           | 18.7 | 199 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Fifo_Fault              | 19   | 200 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Weights_Fault           | 19.1 | 201 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Gamma_Fault             | 19.2 | 202 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Hist_Fault              | 19.3 | 203 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Start_After_Start_Fault        | 19.4 | 204 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Core_Id_Fault                  | 19.5 | 205 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Frame_Count_Read_Fail_Fault            | 19.6 | 206 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Histograms_Mismatch_Fault              | 19.7 | 207 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Ccft_Decisions_Mismatch_Fault          | 1a.0 | 208 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Header_Verification_Error_Fault        | 1a.1 | 209 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Shutdown_Fault                 | 1a.2 | 210 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Shot_Out_Of_Sync_Fault                 | 1a.3 | 211 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_temperature_Fault                                      | 1a.4 | 212 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Error_Out_Signal_Internal_Fault                                | 1a.5 | 213 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Error_Out_Temperature_Fault                                    | 1a.6 | 214 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCM_LostCom                                                             | 1a.7 | 215 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TCM_LostCom                                                             | 1b.0 | 216 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GSM_LostCom                                                             | 1b.1 | 217 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_LostCom                                                             | 1b.2 | 218 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SECM_LostCom                                                            | 1b.3 | 219 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PSCM_LostCom                                                            | 1b.4 | 220 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BCM_LostCom                                                             | 1b.5 | 221 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_LostCom                                                             | 1b.6 | 222 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_LostCom                                                             | 1b.7 | 223 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HUD_LostCom                                                             | 1c.0 | 224 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SCCM_LostCom                                                            | 1c.1 | 225 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACM_LostCom                                                             | 1c.2 | 226 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DDM_LostCom                                                             | 1c.3 | 227 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PDM_LostCom                                                             | 1c.4 | 228 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APIM_LostCom                                                            | 1c.5 | 229 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                                | 1c.6 | 230 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                                             |      |     |   |     |   |   |   |   |
|-------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| PCMH_LostCom                                                | 1c.7 | 231 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VDM_LostCom                                                 | 1d.0 | 232 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCM_InvalidData                                             | 1d.1 | 233 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TCM_InvalidData                                             | 1d.2 | 234 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GSM_InvalidData                                             | 1d.3 | 235 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidData                                             | 1d.4 | 236 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SECM_InvalidData                                            | 1d.5 | 237 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PSCM_InvalidData                                            | 1d.6 | 238 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BCM_InvalidData                                             | 1d.7 | 239 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_InvaiddData                                             | 1e.0 | 240 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SCCM_InvalidData                                            | 1e.1 | 241 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidData                                             | 1e.2 | 242 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DDM_InvalidData                                             | 1e.3 | 243 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PDM_InvalidData                                             | 1e.4 | 244 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HVAC_InvalidData                                            | 1e.5 | 245 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APIM_InvalidData                                            | 1e.6 | 246 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 1e.7 | 247 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 1f.0 | 248 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCMH_InvalidData                                            | 1f.1 | 249 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VDM_InvalidData                                             | 1f.2 | 250 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_InvaiddRollCnt                                          | 1f.3 | 251 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidRollCnt                                          | 1f.4 | 252 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidChkSum                                           | 1f.5 | 253 | 1 | @1+ | 0 | 1 | 0 | 1 |
| reserved                                                    | 1f.6 | 254 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VCAN_Bus_Off_Fault                                          | 1f.7 | 255 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FLRCAN_Bus_Off_Fault                                        | 20   | 256 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RF_LF_CAN_Bus_Off_Fault                           | 20.1 | 257 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_LR_CAN_Bus_Off_Fault                           | 20.2 | 258 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_PCAN_Missing_Message_Fault                        | 20.3 | 259 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_PCAN_Range_Check_Fault                            | 20.4 | 260 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 20.5 | 261 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FR_PCAN_Range_Check_Fault                         | 20.6 | 262 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FL_PCAN_Range_Check_Fault                         | 20.7 | 263 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RL_PCAN_Range_Check_Fault                         | 21   | 264 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_PCAN_Range_Check_Fault                         | 21.1 | 265 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CarCfg_Implausible_Denied_Param_23                          | 21.2 | 266 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CarCfg_Implausible_Denied_Param_96                          | 21.3 | 267 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_Software_Compatibility_Fault                      | 21.4 | 268 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_Software_Compatibility_Fault                      | 21.5 | 269 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GCCCConfigurationNotconfigured_CarCfg_Undefined_Denied      | 21.6 | 270 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_Invalid_Hardware_configuration                    | 21.7 | 271 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidRollCnt                                          | 22   | 272 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidChkSum                                           | 22.1 | 273 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Brake_Temperature_Too_High                                  | 22.2 | 274 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Camera_Module_Alignment_or_Adjustment_Incorrect_RearCamera_ | 22.3 | 275 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                                             |      |     |   |     |   |   |   |   |



|                                                             |      |     |   |     |   |   |   |   |
|-------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| DPAC_Vision_System_Camera_Alignment_or_Adjustment_Incorrect | 22.4 | 276 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Internal_Control_Module_Software_Incompatibility            | 22.5 | 277 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Initial_Configuration_Not_Complete                          | 22.6 | 278 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Control_Module_Configuration_Incompatible                   | 22.7 | 279 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Bus_Failure                                 | 23   | 280 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Not_Configured                              | 23.1 | 281 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Fault                                       | 23.2 | 282 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Bus_Failure                                    | 23.3 | 283 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Not_Configured                                 | 23.4 | 284 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Fault                                          | 23.5 | 285 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Front_USS_HSD_FailureontUSS_Ckt_ShortToGnd                  | 23.6 | 286 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_USS_HSD_FailureRearUSS_Ckt_ShortToGnd                  | 23.7 | 287 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Front_USS_HSD_ShortToBattFrontUSS_Ckt_ShortToBatt           | 24   | 288 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_USS_HSD_ShortToBattRearUSS_Ckt_ShortToBatt             | 24.1 | 289 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Ignition_Status_Unknown_Fault                               | 24.2 | 290 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Ignition_Status_Missing_Fault                               | 24.3 | 291 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Veh_V_ActlEng_Missing_Fault                                 | 24.4 | 292 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_Missing_Fault                              | 24.5 | 293 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_Fault                                      | 24.6 | 294 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_No_data_exists_Fault                       | 24.7 | 295 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_MES_49_Data_Path_Corruption                        | 25   | 296 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_MES_50_VMP_SANITY_FAILED                           | 25.1 | 297 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 25.2 | 298 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_40_PED_SANITY_FAULT                                | 25.3 | 299 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_39_PED_OVERLAP_FAULT                               | 25.4 | 300 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 25.5 | 301 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 25.6 | 302 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 25.7 | 303 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 26   | 304 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 26.1 | 305 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_48_ALGO_PITCH_MISMATCH                             | 26.2 | 306 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_45_ROAD_SFCONF_VALIDATION                          | 26.3 | 307 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_42_ROAD_PREDICTION_VALIDATION                      | 26.4 | 308 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_33_ROAD_LANE_DISQUALIFICATIO_FAILED                | 26.5 | 309 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TrlrLampCnct_B_Actl_Missing_Fault                           | 26.6 | 310 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_MissingMsg_Fault                      | 26.7 | 311 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_MissingMsg_Fault                        | 27   | 312 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_DCUFault                              | 27.1 | 313 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_DCUFault                                | 27.2 | 314 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_ON_OFF_Mismatch_Fault                 | 27.3 | 315 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_ON_OFF_Mismatch_Fault                   | 27.4 | 316 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IgnKeyType_D_Actl_Unknown_Fault                             | 27.5 | 317 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IgnKeyType_D_ActlMissing_Fault                              | 27.6 | 318 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parklamp_Status_Missing_Fault                               | 27.7 | 319 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Application_CRC_Fault                                    | 28   | 320 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 28.1 | 321 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                    |      |     |   |     |   |   |   |   |
|------------------------------------|------|-----|---|-----|---|---|---|---|
| TR_DDR_RAM_CRC_Fault               | 28.2 | 322 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 28.3 | 323 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 28.4 | 324 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 28.5 | 325 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 28.6 | 326 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 28.7 | 327 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29   | 328 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29.1 | 329 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29.2 | 330 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29.3 | 331 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29.4 | 332 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29.5 | 333 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29.6 | 334 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 29.7 | 335 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_INT_OSC_TEST_FAIL               | 2a.0 | 336 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_DPAC_Process_Fault            | 2a.1 | 337 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_FWC_Process_Fault             | 2a.2 | 338 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2a.3 | 339 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Stack_Overflow_Fault            | 2a.4 | 340 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2a.5 | 341 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2a.6 | 342 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2a.7 | 343 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2b.0 | 344 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_DDR_Fault                       | 2b.1 | 345 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Boot_Fault                      | 2b.2 | 346 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2b.3 | 347 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Appl_Complete_And_Compatible    | 2b.4 | 348 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Core_Exception_Fault            | 2b.5 | 349 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Control_ConfigReg_Fault         | 2b.6 | 350 | 1 | @1+ | 0 | 1 | 0 | 1 |
| reserved                           | 2b.7 | 351 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Message_CRC_Err            | 2c.0 | 352 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Missing_Message_Err        | 2c.1 | 353 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Missing_Data_Err           | 2c.2 | 354 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_MessageReception_Err       | 2c.3 | 355 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_VersionMismatch_Err        | 2c.4 | 356 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Communication_Drv_Err      | 2c.5 | 357 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2c.6 | 358 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2c.7 | 359 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2d.0 | 360 | 1 | @1+ | 0 | 1 | 0 | 1 |
| QSPI_Flash_Illegal_Instruction_Err | 2d.1 | 361 | 1 | @1+ | 0 | 1 | 0 | 1 |
| QSPI_Communication_Drv_Err         | 2d.2 | 362 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2d.3 | 363 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2d.4 | 364 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 2d.5 | 365 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_PERFPROC_IntMemFail     | 2d.6 | 366 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                               |      |     |   |     |   |   |   |   |
|-----------------------------------------------|------|-----|---|-----|---|---|---|---|
| DTC_F00049_PERFPROC_IntElectronicFail         | 2d.7 | 367 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LEFT_CAMERA_FAULT_CKT_ShortToBatt             | 2e.0 | 368 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RIGHT_CAMERA_FAULT_CKT_ShortToBatt            | 2e.1 | 369 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BA_Switch_Stuck_On_Fault                      | 2e.2 | 370 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.3 | 371 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.4 | 372 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.5 | 373 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.6 | 374 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.7 | 375 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.0 | 376 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.1 | 377 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.2 | 378 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.3 | 379 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.4 | 380 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.5 | 381 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.6 | 382 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.7 | 383 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_21_CAMERA_HEIGHT_LIMIT_EXCEEDED  | 30   | 384 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_44_ROAD_SANITY_CORRECTION_VALIDATION | 30.1 | 385 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_53_DECELERATION_DBS_OVERLAP_FAULT    | 30.2 | 386 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_24_CCFT_DIFF_RESULTS             | 30.3 | 387 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GearRvrse_D_Actl_Missing_Fault                | 30.4 | 388 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_TCM_LostComm                              | 30.5 | 389 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WheelData_Missing_Fault                       | 30.6 | 390 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_BCM_LostComm                              | 30.7 | 391 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_RCM_LostCom                               | 31   | 392 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_ABS_InvalidData                           | 31.1 | 393 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_TCM_InvalidData                           | 31.2 | 394 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GearRvrse_D_Actl_invalid_Fault                | 31.3 | 395 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_IPC_LostCom                               | 31.4 | 396 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Inhibited_Fault_Flag                      | 31.5 | 397 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Validity_Fault_Flag                       | 31.6 | 398 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LaneFusion_Validity_Fault_Flag                | 31.7 | 399 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 32   | 400 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 32.1 | 401 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 32.2 | 402 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_CAMERA_FAULT_CKT_ShortToBatt              | 32.3 | 403 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftRearSOD_Ckt_ShortToBatt                   | 32.4 | 404 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftRearSOD_Ckt_ShortToGnd                    | 32.5 | 405 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightRearSOD_Ckt_ShortToBatt                  | 32.6 | 406 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightRearSOD_Ckt_ShortToGnd                   | 32.7 | 407 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftFrontSOD_Ckt_ShortToBatt                  | 33   | 408 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftFrontSOD_Ckt_ShortToGnd                   | 33.1 | 409 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightFrontSOD_Ckt_ShortToBatt                 | 33.2 | 410 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightFrontSOD_Ckt_ShortToGnd                  | 33.3 | 411 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                               |      |     |   |     |   |   |   |   |

|                                    |      |     |   |     |   |   |   |   |
|------------------------------------|------|-----|---|-----|---|---|---|---|
| FRONT_CAMERA_FAULT_CKT_ShortToBatt | 33.4 | 412 | 1 | @1+ | 0 | 1 | 0 | 1 |
| REAR_CAMERA_FAULT_CKT_ShortToBatt  | 33.5 | 413 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 33.6 | 414 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_915E31_PerfProcessor           | 33.7 | 415 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4008_FAULT_CONDITION         | 34   | 416 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4096_FAULT_CONDITION         | 34.1 | 417 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4098_FAULT_CONDITION         | 34.2 | 418 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4208_FAULT_CONDITION         | 34.3 | 419 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4296_FAULT_CONDITION         | 34.4 | 420 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4298_FAULT_CONDITION         | 34.5 | 421 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C08_FAULT_CONDITION         | 34.6 | 422 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C96_FAULT_CONDITION         | 34.7 | 423 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C98_FAULT_CONDITION         | 35   | 424 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4808_FAULT_CONDITION         | 35.1 | 425 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4896_FAULT_CONDITION         | 35.2 | 426 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4898_FAULT_CONDITION         | 35.3 | 427 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5008_FAULT_CONDITION         | 35.4 | 428 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5096_FAULT_CONDITION         | 35.5 | 429 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5098_FAULT_CONDITION         | 35.6 | 430 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F308_FAULT_CONDITION         | 35.7 | 431 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F396_FAULT_CONDITION         | 36   | 432 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F398_FAULT_CONDITION         | 36.1 | 433 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3608_FAULT_CONDITION         | 36.2 | 434 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3696_FAULT_CONDITION         | 36.3 | 435 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3698_FAULT_CONDITION         | 36.4 | 436 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3808_FAULT_CONDITION         | 36.5 | 437 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3896_FAULT_CONDITION         | 36.6 | 438 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3898_FAULT_CONDITION         | 36.7 | 439 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D08_FAULT_CONDITION         | 37   | 440 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D96_FAULT_CONDITION         | 37.1 | 441 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D98_FAULT_CONDITION         | 37.2 | 442 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4408_FAULT_CONDITION         | 37.3 | 443 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4496_FAULT_CONDITION         | 37.4 | 444 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4498_FAULT_CONDITION         | 37.5 | 445 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4608_FAULT_CONDITION         | 37.6 | 446 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4696_FAULT_CONDITION         | 37.7 | 447 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4698_FAULT_CONDITION         | 38   | 448 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F408_FAULT_CONDITION         | 38.1 | 449 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F496_FAULT_CONDITION         | 38.2 | 450 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F498_FAULT_CONDITION         | 38.3 | 451 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93039E_FAULT_CONDITION         | 38.4 | 452 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_915E31_FAULT_CONDITION         | 38.5 | 453 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_948E31_FAULT_CONDITION         | 38.6 | 454 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_92BE31_FAULT_CONDITION         | 38.7 | 455 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_92BF31_FAULT_CONDITION         | 39   | 456 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_IMAGE2                  | 39.1 | 457 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                         |      |     |   |     |   |   |   |   |
|-----------------------------------------|------|-----|---|-----|---|---|---|---|
| DTC_F00089_IMAGE2                       | 39.2 | 458 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FL_PCAN_Missing_Message_Fault | 39.3 | 459 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FR_PCAN_Missing_Message_Fault | 39.4 | 460 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RL_PCAN_Missing_Message_Fault | 39.5 | 461 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_PCAN_Missing_Message_Fault | 39.6 | 462 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidData                        | 39.7 | 463 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidRollCnt                     | 3a.0 | 464 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidChkSum                      | 3a.1 | 465 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TRM_InvalidData                         | 3a.2 | 466 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_LostComm                           | 3a.3 | 467 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HVAC_LostComm                           | 3a.4 | 468 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TRM_LostComm                            | 3a.5 | 469 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ATCM_LostComm                           | 3a.6 | 470 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HCM_LostComm                            | 3a.7 | 471 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parklamp_Status_Unknown_Fault           | 3b.0 | 472 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Litval_Unknown_Fault                    | 3b.1 | 473 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Litval_Missing_Fault                    | 3b.2 | 474 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TurnLghtSwitch_D_Stat_Missing_Fault     | 3b.3 | 475 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Btt_L_Actl_MissingMsg_Fault             | 3b.4 | 476 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ATD_Mismatch_Fault                      | 3b.5 | 477 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TLM_Mismatch_Fault                      | 3b.6 | 478 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TBM_Mismatch_Fault                      | 3b.7 | 479 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                | 3c.0 | 480 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRr_D_Actl_Missing                 | 3c.1 | 481 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRl_D_Actl_Missing                 | 3c.2 | 482 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFl_D_Actl_Missing                 | 3c.3 | 483 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFr_D_Actl_Missing                 | 3c.4 | 484 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRr_D_Actl_Failed_Fault            | 3c.5 | 485 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRl_D_Actl_Failed_Fault            | 3c.6 | 486 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFr_D_Actl_Failed_Fault            | 3c.7 | 487 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFl_D_Actl_Failed_Fault            | 3d.0 | 488 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaBrk_D_Stat_Denied_Fault              | 3d.1 | 489 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaBrk_D_Stat_Missing_Fault             | 3d.2 | 490 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Isig_brake_timeout_fault                | 3d.3 | 491 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                | 3d.4 | 492 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaYBrkEnbl_B_Rq_Mismatch_Fault         | 3d.5 | 493 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DVR_SELECT_STAT_Fault                   | 3d.6 | 494 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_IMAGE2                       | 3d.7 | 495 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBDeniedByABS                         | 3e.0 | 496 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBDeniedByPCM                         | 3e.1 | 497 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBImplusByABS                         | 3e.2 | 498 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBImplusByPCM                         | 3e.3 | 499 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AGENT_InvalidData                       | 3e.4 | 500 | 1 | @1+ | 0 | 1 | 0 | 1 |
| unused                                  | 3e.5 | 501 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplusByABS                          | 3e.6 | 502 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                             |      |     |   |     |   |   |   |   |
|-----------------------------|------|-----|---|-----|---|---|---|---|
| ACCImplausByPCM             | 3e.7 | 503 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplausAccelByABS        | 3f.0 | 504 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCFailToCancelByPCM        | 3f.1 | 505 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCDeniedbyABS              | 3f.2 | 506 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTC_ACCDeniedByDIM       | 3f.3 | 507 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleCMBB_ASILB | 3f.4 | 508 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TJAImplausByPSCM            | 3f.5 | 509 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleHA_FSM     | 3f.6 | 510 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleFAPA_FSM   | 3f.7 | 511 | 1 | @1+ | 0 | 1 | 0 | 1 |

## History\_Fault\_Status

History\_Fault\_Status-STNumber(77)-STVersion(1) This message is used to report fault status information.

Message History\_Fault\_Status shall be implemented as fault table defined in [5].

| Signal Name                                                | Description | StartBit | Size | Sign | Offset | Factor | Min | Max |
|------------------------------------------------------------|-------------|----------|------|------|--------|--------|-----|-----|
| VisionCo_3_3V_Supply_Out_Of_Range                          | 0           | 0        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_8V_Supply_Out_Of_Range                          | 0.1         | 1        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_1V_Supply_Out_Of_Range                          | 0.2         | 2        | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_1_0V_Supply_Out_Of_Range                          | 0.3         | 3        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_3_3V_Supply_Out_Of_Range                                | 0.4         | 4        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_1_8V_Supply_Out_Of_Range                                | 0.5         | 5        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_1_35V_Supply_Out_Of_Range                               | 0.6         | 6        | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_1_0V_Supply_Out_Of_Range                                | 0.7         | 7        | 1    | @1+  | 0      | 1      | 0   | 1   |
| IMAGE2P_PGOOD_Fail_Fault                                   | 1           | 8        | 1    | @1+  | 0      | 1      | 0   | 1   |
| FVC_CAMPWR_LDO_FailFRONT_CAMERA_FAULT_CKT_ShortToGnd       | 1.1         | 9        | 1    | @1+  | 0      | 1      | 0   | 1   |
| RVC_CAMPWR_LDO_FailREAR_CAMERA_FAULT_CKT_ShortToGnd        | 1.2         | 10       | 1    | @1+  | 0      | 1      | 0   | 1   |
| LH_Mirror_CAMPWR_LDO_FailLEFT_CAMERA_FAULT_CKT_ShortToGnd  | 1.3         | 11       | 1    | @1+  | 0      | 1      | 0   | 1   |
| RH_Mirror_CAMPWR_LDO_FailRIGHT_CAMERA_FAULT_CKT_ShortToGnd | 1.4         | 12       | 1    | @1+  | 0      | 1      | 0   | 1   |
| FWC_CAMPWR_LDO_FailFWC_CAMERA_FAULT_CKT_ShortToGnd         | 1.5         | 13       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 1.6         | 14       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 1.7         | 15       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 2           | 16       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 2.1         | 17       | 1    | @1+  | 0      | 1      | 0   | 1   |
| Reserved                                                   | 2.2         | 18       | 1    | @1+  | 0      | 1      | 0   | 1   |
| MAINRADAR_Output_Open_Circuit                              | 2.3         | 19       | 1    | @1+  | 0      | 1      | 0   | 1   |
| FET_Fault_Across_cycles                                    | 2.4         | 20       | 1    | @1+  | 0      | 1      | 0   | 1   |
| FET_Fail_Across_Cycles                                     | 2.5         | 21       | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_SerialFlash_PowerUp_Fail                                | 2.6         | 22       | 1    | @1+  | 0      | 1      | 0   | 1   |
| System_Over_Temperature                                    | 2.7         | 23       | 1    | @1+  | 0      | 1      | 0   | 1   |
| CPYR_agent_On_Board_Thermistor_Over_Temp                   | 3           | 24       | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_On_Board_Thermistor_Over_Temp                     | 3.1         | 25       | 1    | @1+  | 0      | 1      | 0   | 1   |
| TR_On_Board_Thermistor_Over_Temp                           | 3.2         | 26       | 1    | @1+  | 0      | 1      | 0   | 1   |
| IMAGE2P_On_Board_Thermistor_Over_Temp                      | 3.3         | 27       | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_MIPS_Over_Temp                                    | 3.4         | 28       | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_VMP_Over_Temp                                     | 3.5         | 29       | 1    | @1+  | 0      | 1      | 0   | 1   |
| VisionCo_Imager_Over_Temp                                  | 3.6         | 30       | 1    | @1+  | 0      | 1      | 0   | 1   |
|                                                            |             |          |      |      |        |        |     |     |

|                                              |     |    |   |     |   |   |   |   |
|----------------------------------------------|-----|----|---|-----|---|---|---|---|
| VisionCo_Boot_OverTemp_Fault                 | 3.7 | 31 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_DDR_Over_Temp                       | 4   | 32 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_On_Board_Thermistor_Out_Of_Range  | 4.1 | 33 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_On_Board_Thermistor_Out_Of_Range    | 4.2 | 34 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IMAGE2P_On_Board_Thermistor_Out_Of_Range     | 4.3 | 35 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_On_Board_Thermistor_Out_Of_Range          | 4.4 | 36 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BatteryLowFault                              | 4.5 | 37 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 4.6 | 38 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 4.7 | 39 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_AURIX_Fault                       | 5   | 40 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_AURIX_Fault                       | 5.1 | 41 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 5.2 | 42 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Short_Across_Cycles                   | 5.3 | 43 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Output_Short_Circuit                  | 5.4 | 44 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Output_Open_Circuit                   | 5.5 | 45 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Heater_Short_Fail_Across_Cycles              | 5.6 | 46 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CHMSL_camera_DEC_amplifier_ShortToBatt       | 5.7 | 47 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AUX_camera_DEC_amplifier_ShortToBatt         | 6   | 48 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LKS_Switch_Stuck_On_Fault                    | 6.1 | 49 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APA_Switch_Stuck_On_Fault                    | 6.2 | 50 | 1 | @1+ | 0 | 1 | 0 | 1 |
| _360_Switch_Stuck_On_Fault                   | 6.3 | 51 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BPA_Switch_Stuck_On_Fault                    | 6.4 | 52 | 1 | @1+ | 0 | 1 | 0 | 1 |
| L_BLIS_LED_ShortToBatt                       | 6.5 | 53 | 1 | @1+ | 0 | 1 | 0 | 1 |
| R_BLIS_LED_ShortToBatt                       | 6.6 | 54 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PARK_AID_LED_ShortToBatt                     | 6.7 | 55 | 1 | @1+ | 0 | 1 | 0 | 1 |
| L_BLIS_LED_ShortToGnd                        | 7   | 56 | 1 | @1+ | 0 | 1 | 0 | 1 |
| R_BLIS_LED_ShortToGnd                        | 7.1 | 57 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PARK_AID_LED_ShortToGnd                      | 7.2 | 58 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 7.3 | 59 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AlgoRunner_Error_Fault                       | 7.4 | 60 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 7.5 | 61 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 7.6 | 62 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Pmic_Image2_Reset_Fault                      | 7.7 | 63 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_ENET_Message_CRC_Err              | 8   | 64 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_ENET_Missing_Message_Err          | 8.1 | 65 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_ENET_Missing_Data_Err             | 8.2 | 66 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_ENET_MessageReception_Err         | 8.3 | 67 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_ENET_VersionMismatch_Err          | 8.4 | 68 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Application_CRC_Fault             | 8.5 | 69 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Callibration_CRC_Fault            | 8.6 | 70 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Cal_Complete_And_Compatible       | 8.7 | 71 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cal_Complete_And_Compatible_SWP3    | 9   | 72 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_MESP_Coding_failure                 | 9.1 | 73 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Application_Complete_And_Compatible | 9.2 | 74 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Application_Checksum                | 9.3 | 75 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Calibration_Checksum                | 9.4 | 76 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                                      |     |     |   |     |   |   |   |   |
|------------------------------------------------------|-----|-----|---|-----|---|---|---|---|
| IMAGE2P_Appl_Complete_And_Compatible                 | 9.5 | 77  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_Frame_Format_Error                    | 9.6 | 78  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_Msg_Format_Error                      | 9.7 | 79  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_spi_msg_timeout                           | a.0 | 80  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_Tx_Error                              | a.1 | 81  | 1 | @1+ | 0 | 1 | 0 | 1 |
| CPYR agent_Spi_CRC_Error                             | a.2 | 82  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spi_version_mismatch                                 | a.3 | 83  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spi_Ipc_Comm_Reset                                   | a.4 | 84  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Spi_Frame_Format_Error                      | a.5 | 85  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_Spi_Msg_Format_Error                        | a.6 | 86  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Spi_Msg_Timeout                             | a.7 | 87  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_SPI_CRC_ERROR                               | b.0 | 88  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Serial_Flash_Power_Up_Fail                  | b.1 | 89  | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Aurix_FirstTimePowerUpFlashCheck                  | b.2 | 90  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Aurix_OTA_Flash_Fail                                 | b.3 | 91  | 1 | @1+ | 0 | 1 | 0 | 1 |
| KAM_Design_Read_Fail_Fault                           | b.4 | 92  | 1 | @1+ | 0 | 1 | 0 | 1 |
| KAM_Design_Write_Fail_Fault                          | b.5 | 93  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_DDR_Fault                                   | b.6 | 94  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | b.7 | 95  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Boot_Fault                                  | c.0 | 96  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Core_Fault                                  | c.1 | 97  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Bist_Memory_Fault                           | c.2 | 98  | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | c.3 | 99  | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Init_Timeout_Fault                          | c.4 | 100 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Core_Dump_Fault                             | c.5 | 101 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parameters_Invalid                                   | c.6 | 102 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Unreasonable_Radar_Data                              | c.7 | 103 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_Camera_Wrong_Assembly_Fault                      | d.0 | 104 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_shutdown_Invalid_Reason                          | d.1 | 105 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_shutdown_7_POS_DIFF_KAN_VAL                      | d.2 | 106 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_9_ROI_DIFF_VC_FCV                       | d.3 | 107 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_6_FCV_FCW_TTC                           | d.4 | 108 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Shutdown_25_CO_FOF_DIFF                          | d.5 | 109 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_30_DT_ERR                               | d.6 | 110 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_18_IMAGER_DATA_ROWS_COLS                | d.7 | 111 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_4_YAW_HORIZON_DEVIATION                 | e.0 | 112 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_26_CODE_CRC                             | e.1 | 113 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_13_OLD_EGODATA_LAT                      | e.2 | 114 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | e.3 | 115 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | e.4 | 116 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                             | e.5 | 117 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_31_AEB_PARAM_FFI                        | e.6 | 118 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_29_RFC_ERR                              | e.7 | 119 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_11_INPUT_CRC_MISMATCH                   | f.0 | 120 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_34_Corrupted_Vehicle_Info_Input_Signals | f.1 | 121 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                                      |     |     |   |     |   |   |   |   |



|                                                           |      |     |   |     |   |   |   |   |
|-----------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| AEB_Shutdown_27_Fcv_Sig_Ffi                               | f.2  | 122 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_54_ACC_STL_EXECUTION_FAILED                  | f.3  | 123 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | f.4  | 124 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Missing_Camera_Callibration_Fault                         | f.5  | 125 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_Camera_Type_Fault                                     | f.6  | 126 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | f.7  | 127 | 1 | @1+ | 0 | 1 | 0 | 1 |
| NTSC_Encoder_amplifier_ShortToBatt                        | 10   | 128 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | 10.1 | 129 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | 10.2 | 130 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                  | 10.3 | 131 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Full_Blockage_Fault                                       | 10.4 | 132 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Partial_Blockage_Fault                                    | 10.5 | 133 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Blurred_Image_Fault                                       | 10.6 | 134 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Splashes_Fault                                            | 10.7 | 135 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Low_Sun_Fault                                             | 11   | 136 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Sun_Rays_Fault                                            | 11.1 | 137 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Foggy_Spots_Fault                                         | 11.2 | 138 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Spot_Rays_Fault                                           | 11.3 | 139 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Smearred_Spots_Fault                                      | 11.4 | 140 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SelfGlare_Fault                                           | 11.5 | 141 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Low_Visibility_Fault                                      | 11.6 | 142 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Vision_OutofFocus_Fault                                   | 11.7 | 143 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Vision_OutofCalibration_Fault                             | 12   | 144 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TD_BlockageTest                                           | 12.1 | 145 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_APP_CPS_STL_FAILED                   | 12.2 | 146 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Error_Fault                      | 12.3 | 147 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Fs_Error_Fault                   | 12.4 | 148 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Calibration_Error_Fault          | 12.5 | 149 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Init_Failed_Fault                | 12.6 | 150 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Init_Camera_Init_Fault           | 12.7 | 151 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Video_Grab_Failed_Fault      | 13   | 152 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Camera_Self_Reset_Fault      | 13.1 | 153 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_I2C_Timeout_Error_Fault          | 13.2 | 154 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Pattern_Test_Fault               | 13.3 | 155 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_App_Cam_Params_Ccft_Crc_Failed_Fault | 13.4 | 156 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_PII_Comparison_Error_Fault           | 13.5 | 157 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_Pv_General_Error_Fault               | 13.6 | 158 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Fatal_Error_Pv_Verificattion_Error_Fault         | 13.7 | 159 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Error_Fault                       | 14   | 160 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Error_Fault                    | 14.1 | 161 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Setting_Fault  | 14.2 | 162 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Load_Registry_Fault | 14.3 | 163 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Registry_Fault | 14.4 | 164 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Fault          | 14.5 | 165 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Fault       | 14.6 | 166 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                                           |      |     |   |     |   |   |   |   |

|                                                                         |      |     |   |     |   |   |   |   |
|-------------------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Bb_Reg_Fault                 | 14.7 | 167 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Open_Blackbox_Fault               | 15   | 168 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Init_Ep_Fault                     | 15.1 | 169 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Ep_Fault                | 15.2 | 170 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_Create_Logger_Fault     | 15.3 | 171 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Post_Init_IL_Fault                | 15.4 | 172 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Minor_Error_Bm_Em_Err_Failed_Check_Reg_Versions_Fault          | 15.5 | 173 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_C_R_Error                                                  | 15.6 | 174 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Init_Vdi_Failure                                                        | 15.7 | 175 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Init_I2C_Failure                                                        | 16   | 176 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_No_Free_Header_Failure                                             | 16.1 | 177 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Issue_Failure                                                      | 16.2 | 178 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Acquire_Failure                                                    | 16.3 | 179 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Grab_Release_Failure                                                    | 16.4 | 180 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Mal_Func                                                   | 16.5 | 181 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Ctrl_Flt                                                   | 16.6 | 182 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Cam_Vdo_Flt                                                    | 16.7 | 183 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Video_Error_Fault                  | 17   | 184 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_I2C_Not_On_Time_Video_Error_Fault      | 17.1 | 185 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Logical_Video_Error_Fault              | 17.2 | 186 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Inconsistent_Frames_Video_Error_Fault  | 17.3 | 187 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Verification_Failure_Video_Error_Fault | 17.4 | 188 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Invalid_Video_Header_Fault             | 17.5 | 189 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Error_Reg_Flag_Fault                   | 17.6 | 190 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Reg_Write_Hil_Failure_Error_Fault      | 17.7 | 191 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Zero_Histogram_Error_Fault             | 18   | 192 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Sum_Not_Match_Histogram_Error_Fault    | 18.1 | 193 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Internal_Error_Fault               | 18.2 | 194 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_No_Buffer_Fault                | 18.3 | 195 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Buffer_Invalid_Format_Fault    | 18.4 | 196 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Timeout_Fault                  | 18.5 | 197 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Overflow_Fault            | 18.6 | 198 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Fifo_Underflow_Fault           | 18.7 | 199 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Fifo_Fault              | 19   | 200 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Weights_Fault           | 19.1 | 201 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Gamma_Fault             | 19.2 | 202 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Parity_Hist_Fault              | 19.3 | 203 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Start_After_Start_Fault        | 19.4 | 204 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Core_Id_Fault                  | 19.5 | 205 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Frame_Count_Read_Fail_Fault            | 19.6 | 206 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Histograms_Mismatch_Fault              | 19.7 | 207 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Ccft_Decisions_Mismatch_Fault          | 1a.0 | 208 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Header_Verification_Error_Fault        | 1a.1 | 209 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Vdi_Err_Shutdown_Fault                 | 1a.2 | 210 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_VideoErrorFlags_Shot_Out_Of_Sync_Fault                 | 1a.3 | 211 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Camera1_temperature_Fault                                      | 1a.4 | 212 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                          |      |     |   |     |   |   |   |   |
|------------------------------------------|------|-----|---|-----|---|---|---|---|
| VisionCo_Error_Out_Signal_Internal_Fault | 1a.5 | 213 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionCo_Error_Out_Temperature_Fault     | 1a.6 | 214 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCM_LostCom                              | 1a.7 | 215 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TCM_LostCom                              | 1b.0 | 216 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GSM_LostCom                              | 1b.1 | 217 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_LostCom                              | 1b.2 | 218 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SECM_LostCom                             | 1b.3 | 219 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PSCM_LostCom                             | 1b.4 | 220 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BCM_LostCom                              | 1b.5 | 221 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_LostCom                              | 1b.6 | 222 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_LostCom                              | 1b.7 | 223 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HUD_LostCom                              | 1c.0 | 224 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SCCM_LostCom                             | 1c.1 | 225 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACM_LostCom                              | 1c.2 | 226 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DDM_LostCom                              | 1c.3 | 227 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PDM_LostCom                              | 1c.4 | 228 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APIM_LostCom                             | 1c.5 | 229 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                 | 1c.6 | 230 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCMH_LostCom                             | 1c.7 | 231 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VDM_LostCom                              | 1d.0 | 232 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCM_InvalidData                          | 1d.1 | 233 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TCM_InvalidData                          | 1d.2 | 234 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GSM_InvalidData                          | 1d.3 | 235 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidData                          | 1d.4 | 236 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SECM_InvalidData                         | 1d.5 | 237 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PSCM_InvalidData                         | 1d.6 | 238 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BCM_InvalidData                          | 1d.7 | 239 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_InvaildData                          | 1e.0 | 240 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SCCM_InvalidData                         | 1e.1 | 241 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidData                          | 1e.2 | 242 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DDM_InvalidData                          | 1e.3 | 243 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PDM_InvalidData                          | 1e.4 | 244 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HVAC_InvalidData                         | 1e.5 | 245 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APIM_InvalidData                         | 1e.6 | 246 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                 | 1e.7 | 247 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                 | 1f.0 | 248 | 1 | @1+ | 0 | 1 | 0 | 1 |
| PCMH_InvalidData                         | 1f.1 | 249 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VDM_InvalidData                          | 1f.2 | 250 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IPC_InvaildRollCnt                       | 1f.3 | 251 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidRollCnt                       | 1f.4 | 252 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RCM_InvalidChkSum                        | 1f.5 | 253 | 1 | @1+ | 0 | 1 | 0 | 1 |
| reserved                                 | 1f.6 | 254 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VCAN_Bus_Off_Fault                       | 1f.7 | 255 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FLRCAN_Bus_Off_Fault                     | 20   | 256 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RF_LF_CAN_Bus_Off_Fault        | 20.1 | 257 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                                             |      |     |   |     |   |   |   |   |
|-------------------------------------------------------------|------|-----|---|-----|---|---|---|---|
| SIDERADAR_RR_LR_CAN_Bus_Off_Fault                           | 20.2 | 258 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_PCAN_Missing_Message_Fault                        | 20.3 | 259 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_PCAN_Range_Check_Fault                            | 20.4 | 260 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 20.5 | 261 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FR_PCAN_Range_Check_Fault                         | 20.6 | 262 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FL_PCAN_Range_Check_Fault                         | 20.7 | 263 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RL_PCAN_Range_Check_Fault                         | 21   | 264 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_PCAN_Range_Check_Fault                         | 21.1 | 265 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CarCfg_Implausible_Denied_Param_23                          | 21.2 | 266 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CarCfg_Implausible_Denied_Param_96                          | 21.3 | 267 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_Software_Compatibility_Fault                      | 21.4 | 268 | 1 | @1+ | 0 | 1 | 0 | 1 |
| MAINRADAR_Software_Compatibility_Fault                      | 21.5 | 269 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GCCConfigurationNotconfigured_CarCfg_Undefined_Denied       | 21.6 | 270 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_Invalid_Hardware_configuration                    | 21.7 | 271 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidRollCnt                                          | 22   | 272 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ABS_InvalidChkSum                                           | 22.1 | 273 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Brake_Temperature_Too_High                                  | 22.2 | 274 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Camera_Module_Alignment_or_Adjustment_Incorrect_RearCamera_ | 22.3 | 275 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DPAC_Vision_System_Camera_Alignment_or_Adjustment_Incorrect | 22.4 | 276 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Internal_Control_Module_Software_Incompatibility            | 22.5 | 277 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Initial_Configuration_Not_Complete                          | 22.6 | 278 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Control_Module_Configuration_Incompatible                   | 22.7 | 279 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Bus_Failure                                 | 23   | 280 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Not_Configured                              | 23.1 | 281 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LIN_Rear_Camera_Fault                                       | 23.2 | 282 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Bus_Failure                                    | 23.3 | 283 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Not_Configured                                 | 23.4 | 284 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_Camera2_Fault                                          | 23.5 | 285 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Front_USS_HSD_FailureFrontUSS_Ckt_ShortToGnd                | 23.6 | 286 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_USS_HSD_FailureRearUSS_Ckt_ShortToGnd                  | 23.7 | 287 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Front_USS_HSD_ShortToBattFrontUSS_Ckt_ShortToBatt           | 24   | 288 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Rear_USS_HSD_ShortToBattRearUSS_Ckt_ShortToBatt             | 24.1 | 289 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Ignition_Status_Unknown_Fault                               | 24.2 | 290 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Ignition_Status_Missing_Fault                               | 24.3 | 291 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Veh_V_ActlEng_Missing_Fault                                 | 24.4 | 292 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_Missing_Fault                              | 24.5 | 293 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_Fault                                      | 24.6 | 294 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehVActlEng_D_Qf_No_data_exists_Fault                       | 24.7 | 295 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_MES_49_Data_Path_Corruption                        | 25   | 296 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_MES_50_VMP_SANITY_FAILED                           | 25.1 | 297 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 25.2 | 298 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_40_PED_SANITY_FAULT                                | 25.3 | 299 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_39_PED_OVERLAP_FAULT                               | 25.4 | 300 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 25.5 | 301 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                                    | 25.6 | 302 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                                             |      |     |   |     |   |   |   |   |

|                                              |      |     |   |     |   |   |   |   |
|----------------------------------------------|------|-----|---|-----|---|---|---|---|
| Reserved                                     | 25.7 | 303 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 26   | 304 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 26.1 | 305 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_48_ALGO_PITCH_MISMATCH              | 26.2 | 306 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_45_ROAD_SFCONF_VALIDATION           | 26.3 | 307 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_42_ROAD_PREDICTION_VALIDATION       | 26.4 | 308 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_33_ROAD_LANE_DISQUALIFICATIO_FAILED | 26.5 | 309 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TrlRlmpCnct_B_Actl_Missing_Fault             | 26.6 | 310 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_MissingMsg_Fault       | 26.7 | 311 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_MissingMsg_Fault         | 27   | 312 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_DCUFault               | 27.1 | 313 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_DCUFault                 | 27.2 | 314 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatDriverSide_ON_OFF_Mismatch_Fault  | 27.3 | 315 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BLISLEDStatPassSide_ON_OFF_Mismatch_Fault    | 27.4 | 316 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IgnKeyType_D_Actl_Unknown_Fault              | 27.5 | 317 | 1 | @1+ | 0 | 1 | 0 | 1 |
| IgnKeyType_D_ActlMissing_Fault               | 27.6 | 318 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parklamp_Status_Missing_Fault                | 27.7 | 319 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Application_CRC_Fault                     | 28   | 320 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 28.1 | 321 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_DDR_RAM_CRC_Fault                         | 28.2 | 322 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 28.3 | 323 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 28.4 | 324 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 28.5 | 325 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 28.6 | 326 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 28.7 | 327 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29   | 328 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29.1 | 329 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29.2 | 330 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29.3 | 331 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29.4 | 332 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29.5 | 333 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29.6 | 334 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 29.7 | 335 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_INT_OSC_TEST_FAIL                         | 2a.0 | 336 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_DPAC_Process_Fault                      | 2a.1 | 337 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_FWC_Process_Fault                       | 2a.2 | 338 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 2a.3 | 339 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Stack_Overflow_Fault                      | 2a.4 | 340 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 2a.5 | 341 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 2a.6 | 342 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 2a.7 | 343 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 2b.0 | 344 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_DDR_Fault                                 | 2b.1 | 345 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Boot_Fault                                | 2b.2 | 346 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                     | 2b.3 | 347 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Appl_Complete_And_Compatible              | 2b.4 | 348 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                               |      |     |   |     |   |   |   |   |
|-----------------------------------------------|------|-----|---|-----|---|---|---|---|
| TR_Core_Exception_Fault                       | 2b.5 | 349 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_Control_ConfigReg_Fault                    | 2b.6 | 350 | 1 | @1+ | 0 | 1 | 0 | 1 |
| reserved                                      | 2b.7 | 351 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Message_CRC_Err                       | 2c.0 | 352 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Missing_Message_Err                   | 2c.1 | 353 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Missing_Data_Err                      | 2c.2 | 354 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_MessageReception_Err                  | 2c.3 | 355 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_VersionMismatch_Err                   | 2c.4 | 356 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TR_ENET_Communication_Drv_Err                 | 2c.5 | 357 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2c.6 | 358 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2c.7 | 359 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.0 | 360 | 1 | @1+ | 0 | 1 | 0 | 1 |
| QSPI_Flash_Illegal_Instruction_Err            | 2d.1 | 361 | 1 | @1+ | 0 | 1 | 0 | 1 |
| QSPI_Communication_Drv_Err                    | 2d.2 | 362 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.3 | 363 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.4 | 364 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2d.5 | 365 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_PERFPROC_IntMemFail                | 2d.6 | 366 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_PERFPROC_IntElectronicFail         | 2d.7 | 367 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LEFT_CAMERA_FAULT_CKT_ShortToBatt             | 2e.0 | 368 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RIGHT_CAMERA_FAULT_CKT_ShortToBatt            | 2e.1 | 369 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BA_Switch_Stuck_On_Fault                      | 2e.2 | 370 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.3 | 371 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.4 | 372 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.5 | 373 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.6 | 374 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2e.7 | 375 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.0 | 376 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.1 | 377 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.2 | 378 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.3 | 379 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.4 | 380 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.5 | 381 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.6 | 382 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                      | 2f.7 | 383 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_21_CAMERA_HEIGHT_LIMIT_EXCEEDED  | 30   | 384 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_44_ROAD_SANITY_CORRECTION_VALIDATION | 30.1 | 385 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_53_DECELERATION_DBS_OVERLAP_FAULT    | 30.2 | 386 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VISIONCO_GEN_24_CCFT_DIFF_RESULTS             | 30.3 | 387 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GearRvrse_D_Actl_Missing_Fault                | 30.4 | 388 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_TCM_LostComm                              | 30.5 | 389 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WheelData_Missing_Fault                       | 30.6 | 390 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_BCM_LostComm                              | 30.7 | 391 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_RCM_LostCom                               | 31   | 392 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_ABS_InvalidData                           | 31.1 | 393 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                    |      |     |   |     |   |   |   |   |
|------------------------------------|------|-----|---|-----|---|---|---|---|
| SOD_TCM_InvalidData                | 31.2 | 394 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GearRvrse_D_Actl_invalid_Fault     | 31.3 | 395 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOD_IPC_LostCom                    | 31.4 | 396 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Inhibited_Fault_Flag           | 31.5 | 397 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AEB_Validity_Fault_Flag            | 31.6 | 398 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LaneFusion_Validity_Fault_Flag     | 31.7 | 399 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 32   | 400 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 32.1 | 401 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 32.2 | 402 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FWC_CAMERA_FAULT_CKT_ShortToBatt   | 32.3 | 403 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftRearSOD_Ckt_ShortToBatt        | 32.4 | 404 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftRearSOD_Ckt_ShortToGnd         | 32.5 | 405 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightRearSOD_Ckt_ShortToBatt       | 32.6 | 406 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightRearSOD_Ckt_ShortToGnd        | 32.7 | 407 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftFrontSOD_Ckt_ShortToBatt       | 33   | 408 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LeftFrontSOD_Ckt_ShortToGnd        | 33.1 | 409 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightFrontSOD_Ckt_ShortToBatt      | 33.2 | 410 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RightFrontSOD_Ckt_ShortToGnd       | 33.3 | 411 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FRONT_CAMERA_FAULT_CKT_ShortToBatt | 33.4 | 412 | 1 | @1+ | 0 | 1 | 0 | 1 |
| REAR_CAMERA_FAULT_CKT_ShortToBatt  | 33.5 | 413 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                           | 33.6 | 414 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_915E31_PerfProcessor           | 33.7 | 415 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4008_FAULT_CONDITION         | 34   | 416 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4096_FAULT_CONDITION         | 34.1 | 417 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4098_FAULT_CONDITION         | 34.2 | 418 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4208_FAULT_CONDITION         | 34.3 | 419 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4296_FAULT_CONDITION         | 34.4 | 420 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4298_FAULT_CONDITION         | 34.5 | 421 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C08_FAULT_CONDITION         | 34.6 | 422 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C96_FAULT_CONDITION         | 34.7 | 423 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929C98_FAULT_CONDITION         | 35   | 424 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4808_FAULT_CONDITION         | 35.1 | 425 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4896_FAULT_CONDITION         | 35.2 | 426 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4898_FAULT_CONDITION         | 35.3 | 427 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5008_FAULT_CONDITION         | 35.4 | 428 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5096_FAULT_CONDITION         | 35.5 | 429 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B5098_FAULT_CONDITION         | 35.6 | 430 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F308_FAULT_CONDITION         | 35.7 | 431 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F396_FAULT_CONDITION         | 36   | 432 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F398_FAULT_CONDITION         | 36.1 | 433 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3608_FAULT_CONDITION         | 36.2 | 434 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3696_FAULT_CONDITION         | 36.3 | 435 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3698_FAULT_CONDITION         | 36.4 | 436 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3808_FAULT_CONDITION         | 36.5 | 437 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B3896_FAULT_CONDITION         | 36.6 | 438 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                    |      |     |   |     |   |   |   |   |

|                                         |      |     |   |     |   |   |   |   |
|-----------------------------------------|------|-----|---|-----|---|---|---|---|
| DTC_9B3898_FAULT_CONDITION              | 36.7 | 439 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D08_FAULT_CONDITION              | 37   | 440 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D96_FAULT_CONDITION              | 37.1 | 441 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_929D98_FAULT_CONDITION              | 37.2 | 442 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4408_FAULT_CONDITION              | 37.3 | 443 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4496_FAULT_CONDITION              | 37.4 | 444 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4498_FAULT_CONDITION              | 37.5 | 445 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4608_FAULT_CONDITION              | 37.6 | 446 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4696_FAULT_CONDITION              | 37.7 | 447 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_9B4698_FAULT_CONDITION              | 38   | 448 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F408_FAULT_CONDITION              | 38.1 | 449 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F496_FAULT_CONDITION              | 38.2 | 450 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93F498_FAULT_CONDITION              | 38.3 | 451 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_93039E_FAULT_CONDITION              | 38.4 | 452 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_915E31_FAULT_CONDITION              | 38.5 | 453 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_948E31_FAULT_CONDITION              | 38.6 | 454 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_92BE31_FAULT_CONDITION              | 38.7 | 455 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_92BF31_FAULT_CONDITION              | 39   | 456 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00042_IMAGE2                       | 39.1 | 457 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00089_IMAGE2                       | 39.2 | 458 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FL_PCAN_Missing_Message_Fault | 39.3 | 459 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_FR_PCAN_Missing_Message_Fault | 39.4 | 460 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RL_PCAN_Missing_Message_Fault | 39.5 | 461 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SIDERADAR_RR_PCAN_Missing_Message_Fault | 39.6 | 462 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidData                        | 39.7 | 463 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidRollCnt                     | 3a.0 | 464 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_InvalidChkSum                      | 3a.1 | 465 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TRM_InvalidData                         | 3a.2 | 466 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DSCM_LostComm                           | 3a.3 | 467 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HVAC_LostComm                           | 3a.4 | 468 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TRM_LostComm                            | 3a.5 | 469 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ATCM_LostComm                           | 3a.6 | 470 | 1 | @1+ | 0 | 1 | 0 | 1 |
| HCM_LostComm                            | 3a.7 | 471 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Parklamp_Status_Unknown_Fault           | 3b.0 | 472 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Litval_Unknown_Fault                    | 3b.1 | 473 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Litval_Missing_Fault                    | 3b.2 | 474 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TurnLghtSwitch_D_Stat_Missing_Fault     | 3b.3 | 475 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Btt_L_Actl_MissingMsg_Fault             | 3b.4 | 476 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ATD_Mismatch_Fault                      | 3b.5 | 477 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TLM_Mismatch_Fault                      | 3b.6 | 478 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TBM_Mismatch_Fault                      | 3b.7 | 479 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                                | 3c.0 | 480 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRr_D_Actl_Missing                 | 3c.1 | 481 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRl_D_Actl_Missing                 | 3c.2 | 482 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFl_D_Actl_Missing                 | 3c.3 | 483 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFr_D_Actl_Missing                 | 3c.4 | 484 | 1 | @1+ | 0 | 1 | 0 | 1 |



|                                 |      |     |   |     |   |   |   |   |
|---------------------------------|------|-----|---|-----|---|---|---|---|
| WhlDirRr_D_Actl_Failed_Fault    | 3c.5 | 485 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirRl_D_Actl_Failed_Fault    | 3c.6 | 486 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFr_D_Actl_Failed_Fault    | 3c.7 | 487 | 1 | @1+ | 0 | 1 | 0 | 1 |
| WhlDirFl_D_Actl_Failed_Fault    | 3d.0 | 488 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaBrk_D_Stat_Denied_Fault      | 3d.1 | 489 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaBrk_D_Stat_Missing_Fault     | 3d.2 | 490 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Isig_brake_timeout_fault        | 3d.3 | 491 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Reserved                        | 3d.4 | 492 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CtaYBrkEnbl_B_Rq_Mismatch_Fault | 3d.5 | 493 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DVR_SELECT_STAT_Fault           | 3d.6 | 494 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DTC_F00049_IMAGE2               | 3d.7 | 495 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBDeniedByABS                 | 3e.0 | 496 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBDeniedByPCM                 | 3e.1 | 497 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBImplausByABS                | 3e.2 | 498 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CMbBImplausByPCM                | 3e.3 | 499 | 1 | @1+ | 0 | 1 | 0 | 1 |
| AGENT_InvalidData               | 3e.4 | 500 | 1 | @1+ | 0 | 1 | 0 | 1 |
| unused                          | 3e.5 | 501 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplausByABS                 | 3e.6 | 502 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplausByPCM                 | 3e.7 | 503 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCImplausAccelByABS            | 3f.0 | 504 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCFailToCancelByPCM            | 3f.1 | 505 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ACCDeniedbyABS                  | 3f.2 | 506 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTC_ACCDeniedByDIM           | 3f.3 | 507 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleCMbB_ASILB     | 3f.4 | 508 | 1 | @1+ | 0 | 1 | 0 | 1 |
| TJAImplausByPSCM                | 3f.5 | 509 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleHA_FSM         | 3f.6 | 510 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SetDTCImplausibleFAPA_FSM       | 3f.7 | 511 | 1 | @1+ | 0 | 1 | 0 | 1 |

## CPYR AGENT\_SPI\_Debug

CPYR AGENT\_SPI\_Debug(0x203)-STNumber(78)-STVersion(1) Message CPYR AGENT\_SPI\_Debug (0X203) shall be implemented as follows:

| Signal Name                     | Description                                                                   | Startbit | Length [Bit] | Value Type | Factor | Offset | Minimum | Maximum | Unit |
|---------------------------------|-------------------------------------------------------------------------------|----------|--------------|------------|--------|--------|---------|---------|------|
| CPYR agentSpiDllRxBadHdrCtr     | CPYR agent receives frame with "D" nibble not equal to 8, 9, or A.            | 0        | 8            | Unsigned   | 1      | 0      | 0       | 255     | NA   |
| CPYR agentSpiDllRxLongFrameCtr  | SPI driver indicates incorrect frame (not 128 bytes)                          | 8        | 8            | Unsigned   | 1      | 0      | 0       | 255     | NA   |
| CPYR agentSpiDllRxShortFrameCtr | SPI driver indicates incorrect frame (not 128 bytes)                          | 16       | 8            | Unsigned   | 1      | 0      | 0       | 255     | NA   |
| CPYR agentSpiTpRxBadHdrCtr      | The transport layer on the CPYR agent detects an unexpected header or length. | 24       | 8            | Unsigned   | 1      | 0      | 0       | 255     | NA   |
| CPYR agentSpiTpRxBadSeqCtr      | The transport layer on the CPYR agent receives an unexpected frame.           | 32       | 8            | Unsigned   | 1      | 0      | 0       | 255     | NA   |

## Temperature

Temperature (0x206)-STNumber(79)-STVersion(1) Message \_Temperature (0x206) shall be implemented as follows:

| Signal Name         | Description                                                                                                               | Startbit | Length [Bit] | Value Type | Factor | Offset | Minimum | Maximum | Unit |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|----------|--------------|------------|--------|--------|---------|---------|------|
| CPYR agentOnBdT     | CPYR agent Temperature                                                                                                    | 0        | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| VisionCo4OnBdT      | VisionCo4 Temperature                                                                                                     | 8        | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| IMAGE2POnBdT        | IMAGE2P Temperature                                                                                                       | 16       | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| AlgorunnerOnBdT     | Algorunner Temperature                                                                                                    | 24       | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| imagerTemperature   | Imager temperature<br>0x80 = Unknown                                                                                      | 32       | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| mipsTemperature     | MIPS temperature<br>0x80 = Unknown                                                                                        | 40       | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| vmpTemperature      | VMP temperature<br>0x80 = Unknown                                                                                         | 48       | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| ddrTemperature      | DDR Temperature<br>0 = Range OK<br>1 = Below Spec<br>2 = Above Spec<br>3 = Above 85C<br>255 = Error (if the read failed.) | 56       | 8            | Signed     | 1      | 0      | -40     | 125     | C    |
| CPYR agent_ThermSts | 00 = NORMAL<br>01 = OVERTEMP<br>10 = INVALID                                                                              | 64       | 2            | Signed     | 1      | 0      | -2      | 1       | NA   |
| VisionCo_ThermSts   | 00 = NORMAL<br>01 = OVERTEMP<br>10 = INVALID                                                                              | 66       | 2            | Signed     | 1      | 0      | -2      | 1       | NA   |
| Algorunner_ThermSts | 00 = NORMAL<br>01 = OVERTEMP<br>10 = INVALID                                                                              | 68       | 2            | Signed     | 1      | 0      | -2      | 1       | NA   |
| IMAGE2P_ThermSts    | 00 = NORMAL<br>01 = OVERTEMP<br>10 = INVALID                                                                              | 70       | 2            | Signed     | 1      | 0      | -2      | 1       | NA   |

### OBJECTSENSE\_Signals

Message OBJECTSENSE\_Signals shall be implemented as

**ST Number (80)**

| SIGNALS                                | Description                                                     | StartBit | Length [Bit] | Value Type | Offset | Factor | Minimum | Maximum |
|----------------------------------------|-----------------------------------------------------------------|----------|--------------|------------|--------|--------|---------|---------|
| OBJECTSENSEYawRate                     | Yaw Rate                                                        | 0        | 32           | @1-        | 0      | 0.0002 | -2      | 2       |
| OBJECTSENSEYawRate_Est                 | Wheel-Speed based YawRate                                       | 32       | 32           | @1-        | 0      | 0.0002 | -2      | 2       |
| OBJECTSENSE_FiltVehicleSpeedOverGround | Vehicle Speed over ground                                       | 64       | 32           | @1-        | 0      | 0.01   | 0       | 191.13  |
| OBJECTSENSE_V_CPYR agent_Long_front    | CPYR agent Vehicle Longitudinal Velocity relative to the ground | 96       | 32           | @1-        | 0      | 1      | -20     | 100     |
| OBJECTSENSE_V_CPYR agent_Lat_front     | CPYR agent Vehicle                                              | 128      | 32           | @1-        | 0      | 1      | -20     | 20      |

|                                               |                                                                                                              |     |    |     |   |      |      |          |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----|----|-----|---|------|------|----------|
|                                               | Lateral Velocity relative to the ground                                                                      |     |    |     |   |      |      |          |
| StePinComp_An_Est                             | Steering Wheel Angle                                                                                         | 160 | 32 | @1- | 0 | 0.1  | 0.1  | 25.59    |
| VehicleCfg_SteeringRatio                      | Steering Gear Ratio                                                                                          | 192 | 32 | @1- | 0 | 0.1  | 0.1  | 25.59    |
| VehicleCfg_Wheelbase                          | Wheelbase                                                                                                    | 224 | 32 | @1- | 0 | 1    | 0    | 65535    |
| VISION_TIMESTAMP                              | Timestamp of FWC vision data to determine delay / to compensate latency (synchronized with CPYR agent timer) | 256 | 32 | @1+ | 0 | 1    | 0    | 4.29E+09 |
| CPYR_AGENT_CURRENT_TIME                       | Current timer value of CPYR agent $\mu$ Controller to determine delay / to compensate latency                | 288 | 32 | @1+ | 0 | 1    | 0    | 4.29E+09 |
| OBJECTSENSE_FiltVehicleSpeedOverGround_signed | Signed Vehicle Speed                                                                                         | 320 | 32 | @1- | 0 | 0.01 | -100 | 100      |
| OBJECTSENSE_FUS_LookIndex                     | OBJECTSENSE Scan Index                                                                                       | 352 | 16 | @1+ | 0 | 1    | 0    | 65535    |
| OBJECTSENSE_VIS_GrabIndex                     | Vision Frame Index                                                                                           | 368 | 16 | @1+ | 0 | 1    | 0    | 65535    |
| OBJECTSENSE_DrivingDirection                  | Driving Direction<br>UNKNOWN (0x0)<br>BACKWARD (0x1)<br>FORWARD (0x2)                                        | 384 | 2  | @1+ | 0 | 1    | 0    | 2        |
| OBJECTSENSE_DrivingReverse                    | Not_Reverse (0x0) Reverse (0x1)                                                                              | 386 | 1  | @1+ | 0 | 1    | 0    | 1        |
| RadarSystemNotOperational_FrCtr               | OBJECTSENSE Status                                                                                           | 387 | 1  | @1+ | 0 | 1    | 0    | 1        |
| RadarSystemNotOperational_FrLft               | OBJECTSENSE Status                                                                                           | 388 | 1  | @1+ | 0 | 1    | 0    | 1        |
| RadarSystemNotOperational_FrRt                | OBJECTSENSE Status                                                                                           | 389 | 1  | @1+ | 0 | 1    | 0    | 1        |
| RadarSystemNotOperational_RrLft               | OBJECTSENSE Status                                                                                           | 390 | 1  | @1+ | 0 | 1    | 0    | 1        |
| RadarSystemNotOperational_RrRt                | OBJECTSENSE Status                                                                                           | 391 | 1  | @1+ | 0 | 1    | 0    | 1        |
| VisionSystemNotOperational_FWC                | OBJECTSENSE Status                                                                                           | 392 | 1  | @1+ | 0 | 1    | 0    | 1        |
| PerfSOCNotOperational                         | OBJECTSENSE Status                                                                                           | 393 | 1  | @1+ | 0 | 1    | 0    | 1        |
| VisSystemNotOperational_RDPAC                 | OBJECTSENSE Status                                                                                           | 394 | 1  | @1+ | 0 | 1    | 0    | 1        |

|                                 |                       |     |   |     |   |   |   |   |
|---------------------------------|-----------------------|-----|---|-----|---|---|---|---|
| VisSystemNotOperational_360     | OBJECTSENSE<br>Status | 395 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_DPAC_ProcessNotOperational | OBJECTSENSE<br>Status | 396 | 1 | @1+ | 0 | 1 | 0 | 1 |
| Perf_FWC_ProcessNotOperational  | OBJECTSENSE<br>Status | 397 | 1 | @1+ | 0 | 1 | 0 | 1 |
| USLSystemNotOperational         | OBJECTSENSE<br>Status | 398 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ParkAidLEDNotOperational        | OBJECTSENSE<br>Status | 399 | 1 | @1+ | 0 | 1 | 0 | 1 |

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|                                |                       |     |   |     |   |   |   |   |
|--------------------------------|-----------------------|-----|---|-----|---|---|---|---|
| BLISLEDNotOperational          | OBJECTSENSE<br>Status | 400 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APA_SwitchNotOperational       | OBJECTSENSE<br>Status | 401 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BA_SwitchNotOperational        | OBJECTSENSE<br>Status | 402 | 1 | @1+ | 0 | 1 | 0 | 1 |
| BPA_SwitchNotOperational       | OBJECTSENSE<br>Status | 403 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LKS_SwitchNotOperational       | OBJECTSENSE<br>Status | 404 | 1 | @1+ | 0 | 1 | 0 | 1 |
| View_SwitchNotOperational      | OBJECTSENSE<br>Status | 405 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisSystemNotOperational_LINRVC | OBJECTSENSE<br>Status | 406 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisSystemNotOperational_CHMSL  | OBJECTSENSE<br>Status | 407 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentOutOfRange_FrCtr | OBJECTSENSE<br>Status | 408 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentOutOfRange_FrLft | OBJECTSENSE<br>Status | 409 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentOutOfRange_FrRt  | OBJECTSENSE<br>Status | 410 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentOutOfRange_RrLft | OBJECTSENSE<br>Status | 411 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentOutOfRange_RrRt  | OBJECTSENSE<br>Status | 412 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionAlignmentOutOfRange_FWC  | OBJECTSENSE<br>Status | 413 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarBlockageDetected_FrCtr    | OBJECTSENSE<br>Status | 414 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarBlockageDetected_FrLft    | OBJECTSENSE<br>Status | 415 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarBlockageDetected_FrRt     | OBJECTSENSE<br>Status | 416 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarBlockageDetected_RrLft    | OBJECTSENSE<br>Status | 417 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarBlockageDetected_RrRt     | OBJECTSENSE<br>Status | 418 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionBlockageDetected_FWC     | OBJECTSENSE<br>Status | 419 | 1 | @1+ | 0 | 1 | 0 | 1 |
| USLBlockageDetected_Front      | OBJECTSENSE<br>Status | 420 | 1 | @1+ | 0 | 1 | 0 | 1 |
| USLBlockageDetected_Rear       | OBJECTSENSE<br>Status | 421 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarExtConditionsNotOK_FrCtr  | OBJECTSENSE<br>Status | 422 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarExtConditionsNotOK_FrLft  | OBJECTSENSE<br>Status | 423 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarExtConditionsNotOK_FrRt   | OBJECTSENSE<br>Status | 424 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarExtConditionsNotOK_RrLft  | OBJECTSENSE<br>Status | 425 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarExtConditionsNotOK_RrRt   | OBJECTSENSE<br>Status | 426 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionExtConditionsNotOK_FWC   | OBJECTSENSE<br>Status | 427 | 1 | @1+ | 0 | 1 | 0 | 1 |
| SOCExtConditionsNotOK_Perf     | OBJECTSENSE<br>Status | 428 | 1 | @1+ | 0 | 1 | 0 | 1 |

|                                |                       |     |   |     |   |   |   |   |
|--------------------------------|-----------------------|-----|---|-----|---|---|---|---|
| SOCExtConditionsNotOK_DPAC     | OBJECTSENSE<br>Status | 429 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentIncomplete_FrCtr | OBJECTSENSE<br>Status | 430 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentIncomplete_FrLft | OBJECTSENSE<br>Status | 431 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentIncomplete_FrRt  | OBJECTSENSE<br>Status | 432 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentIncomplete_RrLft | OBJECTSENSE<br>Status | 433 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentIncomplete_RrRt  | OBJECTSENSE<br>Status | 434 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionAlignmentIncomplete_FWC  | OBJECTSENSE<br>Status | 435 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionAlignmentIncomplete_360  | OBJECTSENSE<br>Status | 436 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionAlignmentIncomplete_DRVC | OBJECTSENSE<br>Status | 437 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionAlignmentNotStarted_FWC  | OBJECTSENSE<br>Status | 438 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentNotStarted_FrCtr | OBJECTSENSE<br>Status | 439 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentNotStarted_FrLft | OBJECTSENSE<br>Status | 440 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentNotStarted_FrRt  | OBJECTSENSE<br>Status | 441 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentNotStarted_RrLft | OBJECTSENSE<br>Status | 442 | 1 | @1+ | 0 | 1 | 0 | 1 |
| RadarAlignmentNotStarted_RrRt  | OBJECTSENSE<br>Status | 443 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VisionSensorDataUpdating_FWC   | OBJECTSENSE<br>Status | 444 | 1 | @1+ | 0 | 1 | 0 | 1 |
|                                |                       |     |   |     |   |   |   |   |

|                                |                    |     |   |     |   |   |   |   |
|--------------------------------|--------------------|-----|---|-----|---|---|---|---|
| PerfSOCDataUpdating            | OBJECTSENSE Status | 445 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CentralConfigError             | OBJECTSENSE Status | 446 | 1 | @1+ | 0 | 1 | 0 | 1 |
| ModuleConfigError              | OBJECTSENSE Status | 447 | 1 | @1+ | 0 | 1 | 0 | 1 |
| APAImplausibleError            | OBJECTSENSE Status | 448 | 1 | @1+ | 0 | 1 | 0 | 1 |
| CmbbImplausibleError           | OBJECTSENSE Status | 449 | 1 | @1+ | 0 | 1 | 0 | 1 |
| LaneFusionImplausibleError     | OBJECTSENSE Status | 450 | 1 | @1+ | 0 | 1 | 0 | 1 |
| GearPosImplausibleError        | OBJECTSENSE Status | 451 | 1 | @1+ | 0 | 1 | 0 | 1 |
| YawRateImplausibleError        | OBJECTSENSE Status | 452 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehAccelImplausibleError       | OBJECTSENSE Status | 453 | 1 | @1+ | 0 | 1 | 0 | 1 |
| FirstRowBckDrvImplausibleError | OBJECTSENSE Status | 454 | 1 | @1+ | 0 | 1 | 0 | 1 |
| VehSpdImplausibleError         | OBJECTSENSE Status | 455 | 1 | @1+ | 0 | 1 | 0 | 1 |
| DrvEngageLevelImplausibleError | OBJECTSENSE Status | 456 | 1 | @1+ | 0 | 1 | 0 | 1 |

## SOD\_Signals

SOD\_Signals(0x400)-STNumber(81)-STVersion(1) **This message provides**

Message SOD\_Signals (0x400) shall be implemented as follows:

| Signal Name       | Description                                                                                                                                                                                                                                                                            | Startbit | Length [Bit] | Value Type | Factor | Offset | Minimum | Maximum | Unit  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|------------|--------|--------|---------|---------|-------|
| GearLvrPos_D_Actl | init value 14;<br>0x0: Park<br>0x1: Reverse<br>0x2: Neutral<br>0x3: Drive<br>0x4: Sport_DriveSport<br>0x5:low<br>0x6: 1st<br>0x7: 2nd<br>0x8:3rd<br>0x9: 4th<br>0xA: 5th<br>0xB: 6th<br>0xC:undefined_treat_as_fault<br>0xD: undefined_treat_as_fault<br>0xE:unknown pos<br>0xF: Fault | 0        | 4            | NA         | 1      | 0      | 0       | 0xF     | N/A   |
| GearRvrse_D_Actl  | init value 7;<br>0x0 inactive not confirmed<br>0x1: inactive confirmed<br>0x2: Active not confirmed<br>0x3: Active confirmed<br>0x4 - 0x6 not used<br>0x7: Fault                                                                                                                       | 4        | 3            | NA         | 1      | 0      | 0       | 0x7     | N/A   |
| raw_long_accel    |                                                                                                                                                                                                                                                                                        | 7        | 32           | float      | 0.01   | -40    | -40     | 41.89   | m/s^2 |
| raw_lat_accel     |                                                                                                                                                                                                                                                                                        | 39       | 32           | float      | 0.01   | -40    | -40     | 41.89   | m/s^2 |
| SodSnsX_D_Stat    | 0 - Clear<br>1 - Blocked<br>2 - System Failure (fault)<br>3 - Second Warning Audio                                                                                                                                                                                                     | 71       | 2            | NA         | 1      | 0      | 0       | 3       | NA    |
| SodX_D_Stat       | 0x0: Off 0x1: Trailer_Tow_Off 0x2: On 0x3: Disable 0x4: invalid 0x5-0x7: not used                                                                                                                                                                                                      | 73       | 2            | NA         | 1      | 0      | 0       | 4       | NA    |
| BttX_D_Stat       | 0x0: NotDetermined 0x1: Connected 0x2: Pending 0x3: NotConnected 0x4: OffTemp 0x5: Off 0x6: Disable 0x7: not used                                                                                                                                                                      | 75       | 3            | NA         | 1      | 0      | 0       | 7       | NA    |
| BttX_D_RqDvr      | 0x0: Null 0x1: NoRequest 0x2: Request 0x3: not used 0                                                                                                                                                                                                                                  | 78       | 3            | NA         | 1      | 0      | 0       | 3       | NA    |
|                   | hx0A-64 is valid length. hx7E (12.6) is no                                                                                                                                                                                                                                             |          |              |            |        |        |         |         |       |

|                         |                                                                                                                                                                                                                                                                        |     |    |       |   |   |          |        |        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-------|---|---|----------|--------|--------|
|                         | trailer                                                                                                                                                                                                                                                                |     |    |       |   |   |          |        |        |
| TrlrLampCnnect_B_Actl   | Trailer connect signal from the Trailer Lighting Module (TLM): Connected / not connected                                                                                                                                                                               | 113 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| BTT_ENABLE_DISABLE      | Enables and disables BLIS with Trailer Tow feature. Configured per program. Default is 0. Configured via DID 0xDE00                                                                                                                                                    | 114 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| ATD_ENABLE_DISABLE      | Enables and disables the ATD function. Default is 0.                                                                                                                                                                                                                   | 115 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| BLIS_ENABLE_DISABLE     | BLIS enable/disable state shall be set by the global parameter.<br>0x0: Enable 0x1: Disable                                                                                                                                                                            | 116 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| Sod_D_Rq                | 0 = OFF, 1 = BLIS On Secondary Warning OFF, 2 = BLIS ON Secondary Warning ON, 3 = unknown                                                                                                                                                                              | 117 | 2  | NA    | 1 | 0 | 0        | 3      | NA     |
| SODX_ATD_Clutter        | 00 = PENDING 01 = CLUTTER DETECTED<br>02 = NO CLUTTER 03 = SEARCH TERMINATED                                                                                                                                                                                           | 119 | 2  | NA    | 1 | 0 | 0        | 3      | NA     |
| SodAlrtX_D_Stat         | 0 – Lamp Off 1 – Lamp On 2 – Flash 3 – Bulb Prove Out                                                                                                                                                                                                                  | 121 | 2  | NA    | 1 | 0 | 0        | 3      | NA     |
| TurnLghtSwch_D_Stat     | 2 0 – Off 1 – Left 2 – Right 3 – Unused                                                                                                                                                                                                                                | 123 | 2  | NA    | 1 | 0 | 0        | 3      | NA     |
| raw_steering_angle_deg  |                                                                                                                                                                                                                                                                        | 125 | 32 | float | 1 | 0 | -1600.00 | 1676.7 | degree |
| BalrX_D_Stat            | BAX_D_Stat denotes Boundary Alert feature On/Off status. Output of ON/OFF STD                                                                                                                                                                                          | 157 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| BalrSnsX_D_Falt         | BASnsLeft_D_Stat is used to denote the Boundary Alert Sensor Fault Status. States of signal are Clear(0x0), Blocked(0x1), SystemFailure(0x2) or Invalid(0x3)                                                                                                           | 158 | 2  | NA    | 1 | 0 | 0        | 3      | NA     |
| RstrnlmpactEvtntStatus  | This input signal sends the restraints impact event status to the SOD from the RCM. States of Signal are Normal(0x00), Not_Used_1(0x01), Not_Used_2(0x02), Threshold_1_Exceeded(0x03), Not_Used_3(0x04), Threshold_2_Exceeded(0x05), Not_Used_4(0x06) or Invalid(0x07) | 160 | 3  | NA    | 1 | 0 | 0        | 7      | NA     |
| BalrSwch_D_Stat         | Boundary Alert switch status for user. States of signal are Null(0x0), NotPressed(0x1) or Pressed(0x2)                                                                                                                                                                 | 163 | 2  | NA    | 1 | 0 | 0        | 1      | NA     |
| DrStatDrv_B_Actl        | Driver door open status. (Closed or Ajar)                                                                                                                                                                                                                              | 165 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| DrStatPsngr_B_Actl      | Passenger door open status. (Closed or Ajar)                                                                                                                                                                                                                           | 166 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| DrStatRl_B_Actl         | ear Left door open status. (Closed or Ajar)                                                                                                                                                                                                                            | 167 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| DrStatRr_B_Actl         | Rear Right door open status. (Closed or Ajar)                                                                                                                                                                                                                          | 168 | 1  | NA    | 1 | 0 | 0        | 1      | NA     |
| WndwUpDrvFront_D_Falt   | Driver Front Door Fault status. States of window fault signals are Null(0x0), No fault(0x1) or Fault(0x2)                                                                                                                                                              | 169 | 2  | NA    | 1 | 0 | 0        | 2      | NA     |
| WndwUpDrvRear_D_Falt    | Driver Rear Door Fault status. States of window fault signals are Null(0x0), No fault(0x1) or Fault(0x2)                                                                                                                                                               | 171 | 2  | NA    | 1 | 0 | 0        | 2      | NA     |
| WndwUpPsngrFront_D_Falt | Passenger Rear Door Fault status. States of window fault signals are Null(0x0), No fault(0x1) or Fault(0x2)                                                                                                                                                            | 173 | 2  | NA    | 1 | 0 | 0        | 2      | NA     |
| BalrMde_D_Rq            | Request from driver input to change BA mode. States of signal Null(0x00), Low(0x1), Med(0x2) or High(0x3)                                                                                                                                                              | 175 | 2  | NA    | 1 | 0 | 0        | 3      | NA     |
| CtalnnrX_D_Stat         | 0x0 = CTA Initializing 0x1 = CTA Standby State 0x2 = CTA Not Reporting State 0x3 = CTA Reporting State                                                                                                                                                                 | 177 | 2  | NA    | 1 | 0 | 0        | 3      | NA     |



# VisionCo SPI Messages

## VisionCo DASP Messages

The following table lists all Defined Serial Protocol Messages that shall be transmitted from the VisionCo and received by the CPYR agent. Implementation shall be according to the specifications given in the following subsections:

| Data ID | Message Name                       | Modes           | Periodic Rate (Hz) | Rate Tolerance (ms) | Description                                                   |
|---------|------------------------------------|-----------------|--------------------|---------------------|---------------------------------------------------------------|
| 0xA0    | VISIONCO_BASE_DIAG_STATUS_MSG      | APP / SPC / TAC | 18                 | +/- 10              | VisionCo4 base diagnostic status message.                     |
| 0xA1    | VISIONCO_APP_DIAG_STATUS_MSG       | APP/ SPC        | 18                 | +/- 10              | VisionCo4 application diagnostic status message.              |
| 0xA2    | VISIONCO_DYNAMIC_CAL_MSG           | APP/ SPC        | 18                 | +/- 10              | VisionCo4 Service Point Calibration (SPC) and AutoFix output. |
| 0xA3    | VISIONCO_TAC_CAL_MSG               | TAC             | 18                 | +/- 10              | VisionCo4 Target Auto-Calibration (TAC) output.               |
| 0xE0    | VISIONCO_VIS_ROAD_DATA_MSG         | APP             | 18                 | +/- 10              | VisionCo4 road detection data message.                        |
| 0xE1    | VISIONCO_VIS_LIGHT_SENSOR_DATA_MSG | APP             | 18                 | +/- 10              | VisionCo4 AHBC function data message.                         |
| 0xE2    | VISIONCO_VIS_TRAFFIC_SIGNS_MSG     | APP             | 18                 | +/- 10              | Traffic signs as detected by vision subsystem.                |
| 0xE3    | VISIONCO_VIS_AEB_INFO_MSG          | APP             | 18                 | +/- 10              | AEB decisions ( for braking or warning) based on vision.      |
| 0xE4    | VISIONCO_VIS_OBSTACLES_MSG         | APP             | 18                 | +/- 10              | Vision objects as detected by vision subsystem.               |
| 0xE5    | VISIONCO_VIS_FAILSAFE_MSG          | APP             | 18                 | +/- 10              | Vision failsafes                                              |
| 0xE6    | VISIONCO_VIS_SRP_MSG               | APP             | 18                 | +/- 10              | Suspension Road Preview                                       |
| 0xE7    | VISIONCO_VIS_SAFETY_MSG            | APP             | 18                 | +/- 10              | SCFM and Challenge and Response                               |

Note: Do not assign a Data ID to 0x5E.

All messages shall be transmitted periodically at the rate specified in attribute "Periodic Rate" with a tolerance as specified in "Rate Tolerance".

## VISIONCO\_APP\_DIAG\_STATUS\_MSG (0xA1)

VISIONCO\_APP\_DIAG\_STATUS\_MSG(0xA1)-STNumber(41)-STVersion(1) The DASP Data in the VisionCo App Diag Status Message shall be as shown below.

| Member Name          | Type                              | Array Size | Description                                                                          | Range           | Units |
|----------------------|-----------------------------------|------------|--------------------------------------------------------------------------------------|-----------------|-------|
| timestamp_ms         | uint32_t                          | n/a        | Timestamp to determine if message has been updated.                                  | 0 to $2^{32}-1$ | ms    |
| visionDiagnosticInfo | VisionCo_vision_diagnostic_info_t | n/a        | This contains information about vision AGC control settings for day/night detection. | n/a             | n/a   |

## VISIONCO\_DYNAMIC\_CAL\_MSG (0xA2)

103400,  -VISIONCO\_DYNAMIC\_CAL\_MSG(0xA2)-STNumber(42)-STVersion(1) The VisionCo SPC and AutoFix Outputs Message shall be as shown below.

| Member Name      | Type                                  | Array Size | Description                                          | Range           | Units |
|------------------|---------------------------------------|------------|------------------------------------------------------|-----------------|-------|
| timestamp_ms     | uint32_t                              | n/a        | Timestamp to determine if message has been updated.  | 0 to $2^{32}-1$ | ms    |
| dynamicCalParams | VisionCo_vision_calibration_dynamic_t | n/a        | The SPC and AutoFix results with a 64 bit timestamp. | n/a             | n/a   |

## VISIONCO\_TAC\_CAL\_MSG (0xA3)

VISIONCO\_TAC\_CAL\_MSG(0xA3)-STNumber(43)-STVersion(1) The VisionCo TAC Outputs Message shall be as shown below.

| Member Name  | Type                                 | Array Size | Description                                         | Range           | Units |
|--------------|--------------------------------------|------------|-----------------------------------------------------|-----------------|-------|
| timestamp_ms | uint32_t                             | n/a        | Timestamp to determine if message has been updated. | 0 to $2^{32}-1$ | ms    |
| tacCalParams | VisionCo_vision_calibration_status_t | n/a        | The TAC output results with a 64 bit timestamp.     | n/a             | n/a   |

## VISIONCO\_VIS\_ROAD\_DATA\_MSG (0xE0)

VISIONCO\_VIS\_ROAD\_DATA\_MSG(0xE0)-STNumber(44)-STVersion(1) The DASP Data in the VisionCo Road Data Message shall be as shown below.

| Member Name  | Type                 | Array Size | Description                                                                                | Range           | Units |
|--------------|----------------------|------------|--------------------------------------------------------------------------------------------|-----------------|-------|
| timestamp_ms | uint32_t             | n/a        | Timestamp to determine if message has been updated.                                        | 0 to $2^{32}-1$ | ms    |
| roadInfo     | VisionCo_road_info_t | n/a        | Information about the road including lane markers and road center with a 64 bit timestamp. | n/a             | n/a   |

## VISIONCO\_VIS\_LIGHT\_SENSOR\_DATA\_MSG (0xE1)

VISIONCO\_VIS\_LIGHT\_SENSOR\_DATA\_MSG (0xE1) - ST Number (45) - ST Version (1)

The DASP Data in the VisionCo Light Sensor Data Message shall be as shown below.

| Member Name           | Type                               | Array Size | Description                                                  | Range           | Units |
|-----------------------|------------------------------------|------------|--------------------------------------------------------------|-----------------|-------|
| timestamp_ms          | uint32_t                           | n/a        | Timestamp to determine if message has been updated.          | 0 to $2^{32}-1$ | ms    |
| activeLightSensorInfo | visionq_active_light_sensor_info_t | n/a        | The active light sensor information with a 64 bit timestamp. | n/a             | n/a   |

## VISIONCO\_VIS\_TRAFFIC\_SIGNS\_MSG (0xE2)

10VISIONCO\_VIS\_TRAFFIC\_SIGNS\_MSG (0xE2) - ST Number (46) - ST Version (1)

The DASP Data in the VisionCo Traffic Signs Message shall be as shown below.

| Member Name  | Type                         | Array Size | Description                                                                                | Range           | Units |
|--------------|------------------------------|------------|--------------------------------------------------------------------------------------------|-----------------|-------|
| timestamp_ms | uint32_t                     | n/a        | Timestamp to determine if message has been updated.                                        | 0 to $2^{32}-1$ | ms    |
| tsrInfo      | VisionCo_traffic_sign_info_t | n/a        | Current traffic sign detections as received from vision subsystem with a 64 bit timestamp. | n/a             | n/a   |

## VISIONCO\_VIS\_AEB\_INFO\_MSG (0xE3)

VISIONCO\_VIS\_AEB\_INFO\_MSG(0xE3)-STNumber(47)-STVersion(1) The DASP Data in the VisionCo Vision AEB Message shall be as shown below.

| Member Name  | Type                       | Array Size | Description                                                                  | Range                   | Units |
|--------------|----------------------------|------------|------------------------------------------------------------------------------|-------------------------|-------|
| timestamp_ms | uint32_t                   | n/a        | Timestamp to determine if message has been updated.                          | 0 to 2 <sup>32</sup> -1 | ms    |
| visAEB       | VisionCo_vision_AEB_info_t | n/a        | Vision only AEB decisions ( for warning or braking) with a 64 bit timestamp. | n/a                     | n/a   |

## VISIONCO\_VIS\_OBSTACLES\_MSG (0xE4)

VISIONCO\_VIS\_OBSTACLES\_MSG(0xE4)-STNumber(48)-STVersion(1) The DASP Data in the VisionCo Vision Obstacles Message shall be as shown below.

| Member Name  | Type                             | Array Size | Description                                                                                 | Range                   | Units |
|--------------|----------------------------------|------------|---------------------------------------------------------------------------------------------|-------------------------|-------|
| timestamp_ms | uint32_t                         | n/a        | Timestamp to determine if message has been updated.                                         | 0 to 2 <sup>32</sup> -1 | ms    |
| visObjects   | VisionCo_vision_obstacles_info_t | 1          | Vision only obstacles as received from Vision Processing subsystem with a 64 bit timestamp. | n/a                     | n/a   |

## VISIONCO\_VIS\_FAILSAFE\_MSG (0xE5)

VISIONCO\_VIS\_FAILSAFE\_MSG(0xE5)-STNumber(49)-STVersion(1) The DASP Data in the VisionCo Vision Failsafe Message shall be as shown below.

| Member Name  | Type                        | Array Size | Description                                                | Range                   | Units |
|--------------|-----------------------------|------------|------------------------------------------------------------|-------------------------|-------|
| timestamp_ms | uint32_t                    | n/a        | Timestamp to determine if message has been updated.        | 0 to 2 <sup>32</sup> -1 | ms    |
| visFailsafes | VisionCo_vision_failsafes_t | 1          | Vision Failsafes as defined in VisionCo_vision_failsafes_t | n/a                     | n/a   |

## VISIONCO\_VIS\_SRP\_MSG (0xE6)

VISIONCO\_VIS\_SRP\_MSG(0xE6)-STNumber(50)-STVersion(1) The DASP Data in the VisionCo Vision SRP Message shall be as shown below.

| Member Name  | Type     | Array Size | Description                                         | Range                   | Units |
|--------------|----------|------------|-----------------------------------------------------|-------------------------|-------|
| timestamp_ms | uint32_t | n/a        | Timestamp to determine if message has been updated. | 0 to 2 <sup>32</sup> -1 | ms    |

## VISIONCO\_DIAGNOSTIC\_MSG (0x41)

VISIONCO\_DIAGNOSTIC\_MSG (0x41)- ST Number (51) - ST Version (1)

Message 0x41 Fault Information from “Application Diagnostics Message.xlsx” document (previously named “BPP diagnostic protocol” ).

| Name                        | Bit Offset | Byte Offset | Length | Type | Value                                                                                                                                        | Resolution | Offset | Range From | Range To | Unit        | Description                                                                       |
|-----------------------------|------------|-------------|--------|------|----------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|------------|----------|-------------|-----------------------------------------------------------------------------------|
| Reserved_1                  | 0          | 0           | 8      | uint |                                                                                                                                              | 1          | 0      | 0          | 0        | NA          |                                                                                   |
| Application_Message_Version | 8          | 1           | 8      | uint | 1                                                                                                                                            | 1          | 0      | 1          | 1        | NA          |                                                                                   |
| Main_State                  | 16         | 2           | 8      | uint | 0x0 = UNKNOWN<br>0x1 = PENDING<br>0x2 = RUNNING_VISION<br>0x3 = BOOT<br>0x92 = PENDING_VISION<br>0x85 = PENDING_INIT_DV<br>0xAA = RUNNING_DV | 1          | 0      | 0          | 255      | Enum        | Application main state.                                                           |
| Sub_State                   | 24         | 3           | 8      | uint |                                                                                                                                              | 1          | 0      | 0          | 255      | NA          | Application sub-state.                                                            |
| VisionQ_Process_Index       | 32         | 4           | 32     | uint |                                                                                                                                              | 1          | 0      | 0          | 4.29E+09 | NA          | Frame number                                                                      |
| VisionQ_Timestamp           | 64         | 8           | 32     | uint |                                                                                                                                              | 1          | 0      | 0          | 4.29E+09 | millisecond | The time when the image was grabbed.<br>Used for synchronization with algorithmic |

|                                |     |    |    |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |     |          |             |                                                                                                                                                                                                                                                                                                              |
|--------------------------------|-----|----|----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VisionCo_Current_Timestamp     | 96  | 12 | 32 | uint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | 0   | 4.29E+09 | millisecond | The actual time read from the VisionCo. Used for synchronization with algorithmic output.                                                                                                                                                                                                                    |
| Application_Diagnostics_part_1 | 128 | 16 | 16 | uint | See description below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 | 0 | 0   | 65535    | bitwise     | Application Diagnostics (first short)                                                                                                                                                                                                                                                                        |
| Application_Diagnostics_part_2 | 144 | 18 | 16 | uint | See description below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 | 0 | 0   | 65535    | bitwise     | Application Diagnostics (second short)                                                                                                                                                                                                                                                                       |
| Fatal_Error                    | 160 | 20 | 8  | uint | 0=APP_OK<br>1=APP_ERROR<br>10=APP_FS_ERROR<br>11=APP_CALIBRATION_ERROR<br>20=APP_INIT_FAILED<br>21=APP_INIT_CAMERA_INIT<br>50=APP_I2C_VIDEO_GRAB_FAILED<br>51=APP_I2C_CAMERA_SELF_RESET<br>52=APP_I2C_TIMEOUT_ERROR<br>70=APP_PATTERN_TEST<br>80=APP_CAM_PARAMS_CCFT_CRC_FAILED<br>81=PLL_COMPARISON_ERROR<br>90=PV_GENERAL_ERROR<br>91=PV_VERIFICATION_ERROR                                                                                                                                            | 1 | 0 | 0   | 91       | Enum        | Error that requires SW reset. VisionCo cannot continue to function. If happens continuously, needs to go to Service Point. Note - as long as this Enum is 0, there is no fatal error.<br><br>In the event of any fatal errors, only message 0x41 will continue to be sent and no DASP messages will be sent. |
| Reserved_2                     | 168 | 21 | 8  | uint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | 0   | 0        | NA          |                                                                                                                                                                                                                                                                                                              |
| Minor_Error                    | 176 | 22 | 16 | uint | 0=BM_OK<br>1=BM_ERROR<br>5001=BM_EM_ERROR<br>5002=BM_EM_ERR_FAILED_LOAD_SETTING<br>5003=BM_EM_ERR_FAILED_LOAD_REGISTRY<br>5004=BM_EM_ERR_FAILED_INIT_REGISTRY<br>5005=BM_EM_ERR_FAILED_INIT<br>5006=BM_EM_ERR_FAILED_INIT_BB<br>5007=BM_EM_ERR_FAILED_INIT_BB_REG<br>5008=BM_EM_ERR_FAILED_OPEN_BLACKBOX<br>5009=BM_EM_ERR_FAILED_INIT_EP<br>5010=BM_EM_ERR_FAILED_POST_INIT_EP<br>5011=BM_EM_ERR_FAILED_INIT_CREATE_LOGGER<br>5012=BM_EM_ERR_FAILED_INIT_IL<br>5013=BM_EM_ERR_FAILED_CHECK_REG_VERSIONS | 1 | 0 | 0   | 5013     | Enum        | BPP Minor error. problem exists in one of the elements (e.g - EDR problem). In this case, VisionCo main processing loop continue running as usual.<br><br>NOTE: For now only mapped to EDR errors.                                                                                                           |
| Temperature_MIPS               | 192 | 24 | 16 | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | MIPS temperature                                                                                                                                                                                                                                                                                             |
| Temperature_VMP                | 208 | 26 | 16 | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | VMP temperature                                                                                                                                                                                                                                                                                              |
| Temperature_DDR                | 224 | 28 | 8  | uint | 0 – temperature is 85°C; default resh.<br>1 – SDRAM low temperature operating limit exceeded.<br>2 – SDRAM high temperature operating limit exceeded.<br>3 – temperature is much greater than 85°C; 0.25x resh, with derating.<br>4 – temperature is much less than 85°C; default resh.<br>5 – temperature is less than 85°C; default resh. 6 – temperature is greater than 85°C; 0.5x resh. 7 – temperature is greater than 85°C; 0.25x resh, no derating                                               | 1 | 0 | 0   | 3        | Enum        | DDR Temperature                                                                                                                                                                                                                                                                                              |
| Reserved_3                     | 232 | 29 | 8  | uint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | 0   | 0        | NA          |                                                                                                                                                                                                                                                                                                              |
| CFG_status                     | 240 | 30 | 16 | uint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | 0   | 65535    | bitwise     | TBD                                                                                                                                                                                                                                                                                                          |
| SPI_status                     | 256 | 32 | 16 | uint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | 0   | 65535    | bitwise     | bit 0 - CRC SPI error<br>bit 1 - CRC SPI continuous error (more than 3 times in a row)<br>bit 2 - sequence SPI error<br>bit 3 - sequence SPI continuous error (more than 3 times in a row)<br>bits 4,5 - reserved<br>bits 6-13 - SPI health, not functional yet                                              |
| Reserved_4                     | 272 | 34 | 8  | uint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | 0   | 0        | NA          |                                                                                                                                                                                                                                                                                                              |
|                                | 280 | 35 | 8  | uint | bit 0 - camera1 information valid<br>bit 1 - camera2 information valid<br>bit 2 - camera3 information valid<br>bit 3 - camera4 information valid<br>bit 4 - camera5 information valid<br>bit 5 - camera6 information valid<br>bit 6 - camera7 information valid<br>bit 7 - camera8 information valid                                                                                                                                                                                                     | 1 | 0 | 0   | 7        | bitwise     | For systems with multiple cameras, the bits indicate which cameras are valid.                                                                                                                                                                                                                                |
| Camera1_temperature            | 288 | 36 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera1 temperature                                                                                                                                                                                                                                                                                          |
| Camera2_temperature            | 296 | 37 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera2 temperature                                                                                                                                                                                                                                                                                          |
| Camera3_temperature            | 304 | 38 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera3 temperature                                                                                                                                                                                                                                                                                          |
| Camera4_temperature            | 312 | 39 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera4 temperature                                                                                                                                                                                                                                                                                          |
| Camera5_temperature            | 320 | 40 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera5 temperature                                                                                                                                                                                                                                                                                          |
| Camera6_temperature            | 328 | 41 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera6 temperature                                                                                                                                                                                                                                                                                          |
| Camera7_temperature            | 336 | 42 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera7 temperature                                                                                                                                                                                                                                                                                          |
| Camera8_temperature            | 344 | 43 | 8  | sint |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1 | 0 | -40 | 125      | deg         | Camera8 temperature                                                                                                                                                                                                                                                                                          |

|                         |     |     |    |      |  |   |   |   |          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|-----|-----|----|------|--|---|---|---|----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Camera1_VideoErrorRange | 352 | 44  | 8  | uint |  | 1 | 0 | 0 | 255      | bitwise | VideoErrorRange – This is <b>not implemented</b> yet! 4 MSB bits are how many video errors were in this frame. 4 LSB bits are when was the first error, it will be the last 4 bits of the frame index and not all the number.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Camera2_VideoErrorRange | 360 | 45  | 8  | uint |  | 1 | 0 | 0 | 255      | bitwise | Not functional yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Camera3_VideoErrorRange | 368 | 46  | 8  | uint |  | 1 | 0 | 0 | 255      | bitwise | Not functional yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Camera4_VideoErrorRange | 376 | 47  | 8  | uint |  | 1 | 0 | 0 | 255      | bitwise | Not functional yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Camera5_VideoErrorRange | 384 | 48  | 8  | uint |  | 1 | 0 | 0 | 255      | bitwise | Not functional yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Camera6_VideoErrorRange | 392 | 49  | 8  | uint |  | 1 | 0 | 0 | 255      | bitwise | Not functional yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Camera8_VideoErrorRange | 408 | 51  | 8  | uint |  | 1 | 0 | 0 | 255      | bitwise | Not functional yet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Camera1_VideoErrorFlags | 416 | 52  | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | NO_VIDEO_ERROR = 0x0,<br>VDI_VIDEO_ERROR = 0x1,<br>I2C_NOT_ON_TIME_VIDEO_ERROR = 0x2,<br>LOGICAL_VIDEO_ERROR = 0x4,<br>INCONSISTENT_FRAMES_VIDEO_ERROR = 0x8,<br>VERIFICATION_FAILURE_VIDEO_ERROR = 0x10,<br>INVALID_VIDEO_HEADER = 0x20,<br>ERROR_REG_FLAG = 0x40,<br>REG_WRITE_HIL_FAILURE_ERROR = 0x80,<br>ZERO_HISTOGRAM_ERROR = 0x100,<br>SUM_NOT_MATCH_HISTOGRAM_ERROR = 0x200,<br>VDI_INTERNAL_ERROR = 0x400,<br>VDI_ERR_NO_BUFFERS = 0x800,<br>VDI_ERR_BUFFER_INVALID_FORMAT = 0x1000,<br>VDI_ERR_TIMEOUT = 0x2000,<br>VDI_ERR_FIFO_OVERFLOW = 0x4000,<br>VDI_ERR_FIFO_UNDERFLOW = 0x8000,<br>VDI_ERR_PARITY_FIFO = 0x10000,<br>VDI_ERR_PARITY_WEIGHTS = 0x20000,<br>VDI_ERR_PARITY_GAMMA = 0x40000,<br>VDI_ERR_PARITY_HIST = 0x80000,<br>VDI_ERR_START_AFTER_START = 0x100000,<br>VDI_ERR_CORE_ID = 0x200000,<br>FRAME_COUNT_READ_FAIL = 0x400000,<br>HISTOGRAMS_MISMATCH = 0x800000,<br>CCFT_DECISIONS_MISMATCH = 0x1000000,<br>HEADER_VERIFICATION_ERROR = 0x2000000,<br>VDI_ERR_SHUTDOWN = 0x4000000,<br>SHOT_OUT_OF_SYNC = 0x8000000 |
| Camera2_VideoErrorFlags | 480 | 60  | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | TBD. Error Flags 2-8 are for systems with multiple cameras.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Camera3_VideoErrorFlags | 544 | 68  | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | TBD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Camera4_VideoErrorFlags | 608 | 76  | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | TBD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Camera5_VideoErrorFlags | 672 | 84  | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | TBD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Camera6_VideoErrorFlags | 736 | 92  | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | TBD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Camera7_VideoErrorFlags | 800 | 100 | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | TBD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Camera8_VideoErrorFlags | 864 | 108 | 64 | uint |  | 1 | 0 | 0 | 1.84E+19 | bitwise | TBD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Total                   | 928 | 116 |    |      |  |   |   |   |          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## CPYR AGENT\_VEH\_STATE\_MSG (0x20)

CPYR AGENT\_VEH\_STATE\_MSG(0x20)-STNumber(53)-STVersion(1) **The DASP Data in the CPYR agent Vehicle State Message shall be as shown below.**

| Member Name                  | Type                       | Array Size | Description                                                                                                                                               | Range                   | Units |
|------------------------------|----------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------|
| vehStateInfo                 | CPYR agent_vehicle_state_t | n/a        | This contains information about the Ego vehicle state with 64 bit timestamp.                                                                              | n/a                     | n/a   |
| CPYR agent_sync_timestamp_ms | uint32_t                   | n/a        | The time when the CPYR agent received the latest VisionQ4 Clock Sync pulse.                                                                               | 0 to 2 <sup>32</sup> -1 | ms    |
| global_real_time             | uint32_t                   | n/a        | Global real time from the vehicle bus data to be populated and sent every SPI Transmission. This will be used in the EDR in the VisionQ. (This is not for | 0 to 2 <sup>32</sup> -1 | s     |

|                   |          |     |                                                                                                                 |                         |     |
|-------------------|----------|-----|-----------------------------------------------------------------------------------------------------------------|-------------------------|-----|
|                   |          |     | is the date/time data received from the vehicle bus. Date/time data is stored with the EDR.)                    |                         |     |
| SPI_Rolling_Count | uint16_t | n/a | Rolling counter to be incremented by one on every message transmission                                          | 0 to 2 <sup>16</sup> -1 | n/a |
| WheelSlipEvent    | uint8_t  | n/a | a Current wheel slip event:<br>0 = No Slip<br>1 = Brake Slip Control Active<br>2 = Traction Slip Control Active | 0 to 2                  | n/a |

## VISIONCO\_BOOT\_STATUS\_MSG (0x10)

VISIONCO\_BOOT\_STATUS\_MSG(0x10)-STNumber52)-STVersion(1) VISIONCO\_BOOT\_STATUS\_MSG (0x10)

| Message Field                         | Byte | Start bit | Array Size(Bytes) | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|------|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zero byte                             | 0    | 0         | 1                 | Constant zero value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Protocol Version                      | 1    | 0         | 1                 | The version of the message structure and data fields. Current protocol version is 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Reserved0                             | 2    | 0         | 2                 | Reserved fields value is 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Test results                          | 4    | 0         | 2                 | It is a bit map where each bit represents single test result:<br>0 – OK, 1 – Failure.<br>The bit values are as follows-<br>Bit 0 = VisionCo Over Temperature<br>Bit 1 = BIST<br>Bit 2 = PLL init finished<br>Bit 3 = Parity Check<br>Bit 4 = Flash CRC of Boot<br>Bit 5 = DDR init finished<br>Bit 6 = DDR Temperature<br>Bit 7 = DDR Test<br>Bit 8 = Flash CRC of Application<br>Bit 9 = Flash CRC of FFS<br>Bit 10 = Unsupported VisionCo version<br>Bit 11 = Authentication failure of the secure SW signature.<br>Bit 12-15 = Reserved |
| Board project code ('s internal code) | 6    | 0         | 1                 | Internal ' s code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Board revision (based on GPIO)        | 7    | 0         | 1                 | Based on GPIO.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Boot version                          | 8    | 0         | 4                 | 4-bytes field:<br>byte 0 (MSB) – first number of version;<br>byte 1 – second number of version;<br>byte 2 – third number of version;<br>byte 3 – reserved;                                                                                                                                                                                                                                                                                                                                                                                 |
| Application version                   | 12   | 0         | 4                 | Example: Boot version 7.1.23 will be represented as the following byte sequence: byte#8 = 0x07, byte#9 = 0x01, byte#10 = 0x17, byte#11 = 0x00                                                                                                                                                                                                                                                                                                                                                                                              |
| VisionCo Type                         | 16   | 0         | 4                 | -Not Implemented                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| VisionCo ID                           | 20   | 0         | 4                 | -Serial number of the specific VisionCo4 chip.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| DDR manufacturer code                 | 24   | 0         | 1                 | LPDDR4 Manufacturer ID according to JEDEC standard No. 209.<br>For example: 0x08 – Winbond; 0xFF – Micron.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DDR size in MB                        | 25   | 0         | 3                 | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                                                                                                                             |    |   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------|----|---|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flash manufacturer code                                                                                                                     | 28 | 0 | 1  | Each flash memory device has a manufacturer ID (which is assigned by JEDEC). For example: 0x01 – Spansion; 0x20 – Micron; Numonyx – 0x89.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Flash size in MB                                                                                                                            | 29 | 0 | 3  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| VisionCo Temperature Sensor 1                                                                                                               | 32 | 0 | 1  | The temperature measurement provided by two VisionCo4 internal sensors in degree Celsius.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| VisionCo Temperature Sensor 2                                                                                                               | 33 | 0 | 1  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| LPDDR4 Temperature and resh Status                                                                                                          | 34 | 0 | 1  | LPDDR4 Temperature and resh Status 0 – temperature is 85°C; default resh.<br>1 – SDRAM low temperature operating limit exceeded.<br>2 – SDRAM high temperature operating limit exceeded.<br>3 – temperature is much greater than 85°C; 0.25x resh, with derating.<br>4 – temperature is much less than 85°C; default resh. 5 – temperature is less than 85°C; default resh.<br>6 – temperature is greater than 85°C; 0.5x resh.<br>7 – temperature is greater than 85°C; 0.25x resh, no derating                                                                                                                                                                                                                                                      |
| Reserved1                                                                                                                                   | 35 | 0 | 1  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Flash 2 manufacturer code                                                                                                                   | 36 | 0 | 1  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Flash 2 size in MB                                                                                                                          | 37 | 0 | 3  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Boot stage completed (The sequential order is: 1. Boot Authentication, 2. BIST, 3.DRAM initialization, 4. DRAM test, 5. Application loading | 40 | 0 | 1  | 0 – reserved (illegal value).<br>1 – BIST results.<br>2 – Boot authentication result.<br>3 – DRAM initialization & training result.<br>4 – DRAM test result.<br>5 – Application authentication and loading result.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Failure information                                                                                                                         | 41 | 0 | 1  | Reports only 0' s until TBD' s are defined.<br>The failure information format is dependent on the boot stage and the failure type.<br>1. BIST result.<br>Reports failures of the following types:<br>- temperature issue;<br>- VDI memory parity error;<br>- BIST fault;<br>Format: TBD<br><br>2. Boot authentication result.<br>pass / fail<br><br>3. DRAM initialization & training result.<br>- Memory controller initialization issue;<br>- Memory PHY initialization issue;<br>- DRAM training failure;<br>Format: TBD<br><br>4. DRAM test result.<br>- temperature issue;<br>- DRAM test failure;<br>Format: TBD<br><br>5. Application authentication and loading result.<br>- authentication pass / fail;<br>- loading failure;<br>Format: TBD |
| Padding (0x00)                                                                                                                              | 42 | 0 | 79 | Padding by zeroes up to 128 bytes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### 1.1.1 Init\_Params\_Vision (0x92)

Init\_Params\_Vision (0x92) - ST Number 54) - ST Version (1)  
Init\_Params\_Vision (0x92)

## Feature Stream Messages

Algorunner Signals needed:

1. Bus: BT\_TPP\_SPP
  - Stream Number (60) ST Version (1) Length 48 Bytes
2. Bus: BT\_TPP\_TAD
  - Stream Number (61) ST Version (1) Length 460 Bytes
3. Bus: BT\_TPP\_AutoHitch
  - Stream Number (62) ST Version (1) Length 64 Bytes
4. Bus: BT\_TPP\_DriveHistory
  - Stream Number (63) ST Version (1) Length 40 Bytes
5. Bus: BT\_TPP\_Localization\_Results
  - Stream Number (64) ST Version (1) - Length 32 Bytes

## CPUload PDU signal

### CPUload\_PDU (0x100) - ST Number (82) - ST Version (1)

This message provides information used for calculating CPU load.

Message CPUload\_PDU (0x100) shall be implemented as follows:

| Signal Name   | Description | Start bit | Length [Bit] | Value Type | Factor | Offset | Minimum | Maximum | Unit |
|---------------|-------------|-----------|--------------|------------|--------|--------|---------|---------|------|
| headerSize    | header      | 0         | 8            | NA         | 1      | 0      | 0       | 255     | N/A  |
| recordVersion | header      | 8         | 8            | NA         | 1      | 0      | 0       | 255     | N/A  |
| sourceTxCnt   | header      | 16        | 16           | NA         | 1      | 0      | 0       | 65535   | N/A  |



|                |                                              |     |    |    |   |   |   |          |     |
|----------------|----------------------------------------------|-----|----|----|---|---|---|----------|-----|
| sourceTxTime   | header                                       | 32  | 32 | NA | 1 | 0 | 0 | (2^32)-1 | ms  |
| reservedSrc1   | header                                       | 64  | 8  | NA | 1 | 0 | 0 | 255      | N A |
| reservedSrc2   | header                                       | 72  | 8  | NA | 1 | 0 | 0 | 255      | N A |
| reservedSrc3   | header                                       | 80  | 8  | NA | 1 | 0 | 0 | 255      | N A |
| streamRefIndex | header                                       | 88  | 32 | NA | 1 | 0 | 0 | (2^32)-1 | N A |
| streamDataLen  | header                                       | 120 | 16 | NA | 1 | 0 | 0 | 65535    | N A |
| streamTxCnt    | header                                       | 136 | 8  | NA | 1 | 0 | 0 | 255      | N A |
| streamNumber   | header                                       | 144 | 8  | NA | 1 | 0 | 0 | 255      | N A |
| streamVersion  | header.                                      | 152 | 8  | NA | 1 | 0 | 0 | 255      | N A |
| streamChunks   | header                                       | 160 | 8  | NA | 1 | 0 | 0 | 255      | N A |
| streamChunkIdx | header                                       | 168 | 8  | NA | 1 | 0 | 0 | 255      | N A |
| reservedStr3   | header                                       | 176 | 8  | NA | 1 | 0 | 0 | 255      | N A |
| timestamp      | Rte_DT_Idt__CpuLoadData                      | 184 | 32 | NA | 1 | 0 | 0 | (2^32)-1 | ms  |
| CpuLoad_Min    | Rte_DT_Idt__CpuLoadData,<br>Minimum CPU Load | 216 | 32 | NA | 1 | 0 | 0 | (2^32)-1 | N A |

|                 |                                           |     |    |    |   |   |   |                      |     |
|-----------------|-------------------------------------------|-----|----|----|---|---|---|----------------------|-----|
| CpuLoad_Max     | Rte_DT_Idt__CpuLoadData, Maximum CPU load | 248 | 32 | NA | 1 | 0 | 0 | (2 <sup>32</sup> )-1 | N/A |
| CpuLoad_Avg     | Rte_DT_Idt__CpuLoadData, Average CPU load | 280 | 32 | NA | 1 | 0 | 0 | (2 <sup>32</sup> )-1 | N/A |
| CpuLoad_Current | Rte_DT_Idt__CpuLoadData, Current CPU load | 312 | 32 | NA | 1 | 0 | 0 | (2 <sup>32</sup> )-1 | N/A |

## \_Execution\_Time\_Pdu

### \_Execution\_Time\_Pdu (0x101) - ST Number (83) - ST Version (1)

This message provides information used for calculating Execution Time Pdu  
 Message \_Execution\_Time\_Pdu (0x101) shall be implemented as follows:

| Signal Name   | Description | Start bit | Length [Bit] | Value Type | Factor | Offset | Minimum | Maximum              | Unit |
|---------------|-------------|-----------|--------------|------------|--------|--------|---------|----------------------|------|
| headerSize    | header      | 0         | 8            | NA         | 1      | 0      | 0       | 255                  | N/A  |
| recordVersion | header      | 8         | 8            | NA         | 1      | 0      | 0       | 255                  | N/A  |
| sourceTxCnt   | header      | 16        | 16           | NA         | 1      | 0      | 0       | 65535                | N/A  |
| sourceTxTime  | header      | 32        | 32           | NA         | 1      | 0      | 0       | (2 <sup>32</sup> )-1 | ms   |
| reservedSrc1  | header      | 64        | 8            | NA         | 1      | 0      | 0       | 255                  | NA   |
| reservedSrc2  | header      | 72        | 8            | NA         | 1      | 0      | 0       | 255                  | NA   |

|                   |                                                                                                    |     |    |    |   |   |   |              |    |
|-------------------|----------------------------------------------------------------------------------------------------|-----|----|----|---|---|---|--------------|----|
| reservedSrc3      | header                                                                                             | 80  | 8  | NA | 1 | 0 | 0 | 255          | NA |
| streamRefIndex    | header                                                                                             | 88  | 32 | NA | 1 | 0 | 0 | $(2^{32})-1$ | NA |
| streamDataLen     | header                                                                                             | 120 | 16 | NA | 1 | 0 | 0 | 65535        | NA |
| streamTxCnt       | header                                                                                             | 136 | 8  | NA | 1 | 0 | 0 | 255          | NA |
| streamNumber      | header                                                                                             | 144 | 8  | NA | 1 | 0 | 0 | 255          | NA |
| streamVersion     | header.                                                                                            | 152 | 8  | NA | 1 | 0 | 0 | 255          | NA |
| streamChunks      | header                                                                                             | 160 | 8  | NA | 1 | 0 | 0 | 255          | NA |
| streamChunkIndex  | header                                                                                             | 168 | 8  | NA | 1 | 0 | 0 | 255          | NA |
| reservedStr3      | header                                                                                             | 176 | 8  | NA | 1 | 0 | 0 | 255          | NA |
| Min_Idle_Task     | reports the shortest idle time seen. Immediately before the micro-controller enters the idle task. | 184 | 32 | NA | 1 | 0 | 0 | $(2^{32})-1$ | ms |
| Current_Idle_Task | reports the idle time seen. during micro-controller enters the idle task.                          | 216 | 32 | NA | 1 | 0 | 0 | $(2^{32})-1$ | ms |

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|                           |                                                                                              |            |           |           |          |          |          |                           |           |
|---------------------------|----------------------------------------------------------------------------------------------|------------|-----------|-----------|----------|----------|----------|---------------------------|-----------|
| <b>OS_Task_Execution</b>  | <b>updates the maximum execution time the different OS tasks.</b>                            | <b>248</b> | <b>32</b> | <b>NA</b> | <b>1</b> | <b>0</b> | <b>0</b> | <b>(2<sup>32</sup>)-1</b> | <b>ms</b> |
| <b>R_Execution</b>        | <b>updates the maximum execution time the different features runnables.</b>                  | <b>280</b> | <b>32</b> | <b>NA</b> | <b>1</b> | <b>0</b> | <b>0</b> | <b>(2<sup>32</sup>)-1</b> | <b>ms</b> |
| <b>Idle_Task_Miss_Cnt</b> | <b>records the current execution time for all OS Tasks when the Min_Idel_Task is updated</b> | <b>312</b> | <b>16</b> | <b>NA</b> | <b>1</b> | <b>0</b> | <b>0</b> | <b>65535</b>              | <b>ms</b> |